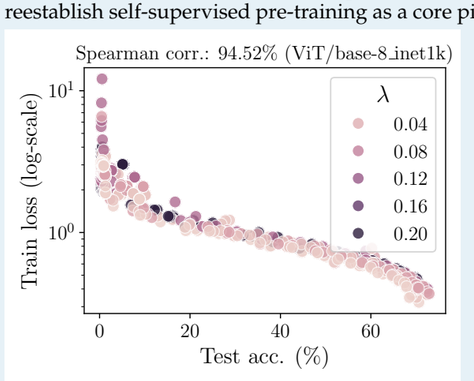
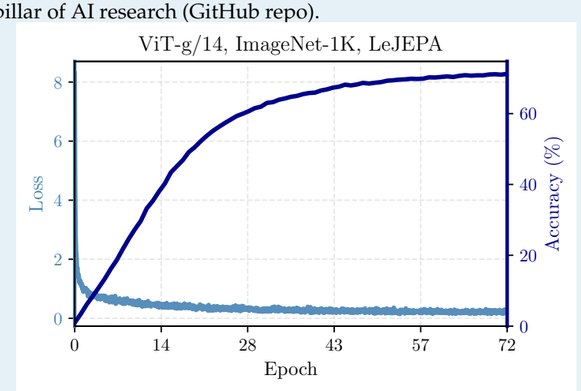
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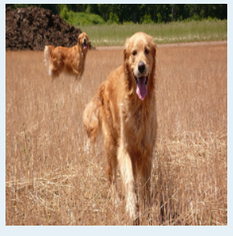
LeJEPA: Provable and Scalable Self-Supervised Learning Without the Heuristics

Randall Balestriero[1,2],* Yann LeCun[3,2],*
[1] Brown University [3] New York University (NYU) [2] Meta-FAIR
* Equal contribution

Learning manipulable representations of the world and its dynamics is central to AI. Joint-Embedding Predictive Architectures (JEPAs) offer a promising blueprint, but lack of practical guidance and theory has led to ad-hoc R&D. We present a comprehensive theory of JEPAs and instantiate it in **LeJEPA**, a lean, scalable, and theoretically grounded training objective. First, we identify the isotropic Gaussian as the optimal distribution that JEPAs' embeddings should follow to minimize downstream prediction risk. Second, we introduce a novel objective–**Sketched Isotropic Gaussian Regularization** (SIGReg)–to constrain embeddings to reach that ideal distribution. Combining the JEPA predictive loss with SIGReg yields LeJEPA with numerous theoretical and practical benefits: (i) single trade-off hyperparameter, (ii) linear time and memory complexity, (iii) stability across hyper-parameters, architectures (ResNets, ViTs, ConvNets) and domains, (iv) heuristics-free, e.g., no stop-gradient, no teacher–student, no hyper-parameter schedulers, and (v) distributed training-friendly implementation requiring only ≈50 lines of code. Our empirical validation covers 10+ datasets, 60+ architectures, all with varying scales and domains. As an example, using imagenet-1k for pretraining and linear evaluation with frozen backbone, LeJEPA reaches 79% with a ViT-H/14. We hope that the simplicity and theory-friendly ecosystem offered by LeJEPA will reestablish self-supervised pre-training as a core pillar of AI research (GitHub repo).









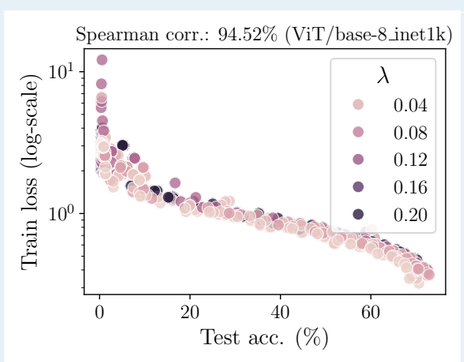
Method	Full FT 1-sh	Full FT Full	Frozen 1-sh	Frozen Full
LeJEPA (in-domain)				
ConvNeXt-V2 Nano	**29.42**	82.72	28.74	76.52
ResNet-34	24.27	**83.28**	**31.08**	**78.17**
Frontier (transfer)				
DINOv2 ViT-S/16	21.05	78.34	27.68	67.62
DINOv3 ViT-S/16	24.71	81.60	30.17	71.38

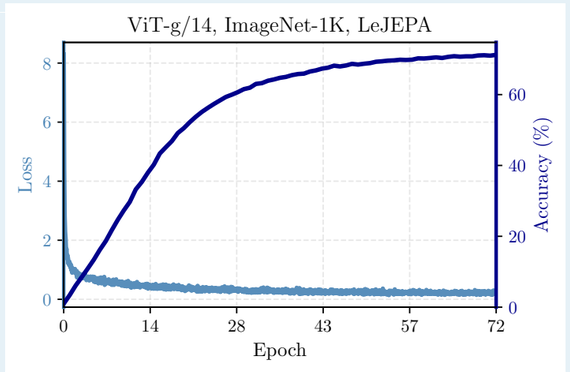
Figure 1. LeJEPA overview. Top-left: Training loss exhibits strong correlation with downstream linear probe performance on ImageNet-1k (ViT-base), providing the first practical loss for model selection without supervised probing. **Top-right:** Training stability without heuristics even on 1.8B ViT-g models, stable training loss. **Bottom-left:** PCA features from ImageNet-1k pretrained LeJEPA ViT-Large demonstrate clear semantic relationships. **Bottom-right:** Galaxy10 in-domain results showcasing LeJEPA's in-domain pretraining consistently outperforms state-of-the-art frontier foundation models transfer learning (DINOv2/v3 trained on natural images) across data regimes from 1-shot to full supervision. This demonstrates that *domain-specific SSL beats generic transfer learning*, even against massive-scale frontier models, when the framework scales effortlessly to any domain, model, and data scale.

LeJEPA：无须启发式技巧、可证明且可扩展的自监督学习

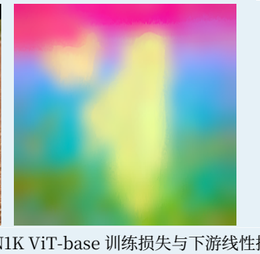
Randall Balestriero[1,2],* Yann LeCun[3,2],*[1] 布朗大学 [3] 纽约大学（NYU）[2] Meta-FAIR * 同等贡献

学习可操控的世界及动力学表征是 AI 核心。JEPA 提供有前景蓝图，但缺少实用指导和理论，导致临时拼凑式研发。我们提出完整理论，并实例化为精简、可扩展、有理论依据的 LeJEPA 目标。首先确定各向同性高斯是最小化下游风险的最优嵌入分布；其次提出 SIGReg，约束嵌入达到理想分布。将预测损失与 SIGReg 结合，得到：（i）仅一个权衡超参数；（ii）线性时间、内存；（iii）跨超参数、架构（ResNet、ViT、ConvNet）和领域稳定；（iv）无需停止梯度、教师–学生、超参数调度等启发式；（v）只需 ≈50 行代码，支持分布式训练。实验覆盖 10+ 数据集、 60+ 架构及不同规模、领域。例如 ImageNet-1k 预训练、冻结骨干线性评估，ViT-H/14 达到 79%。希望这一简单、便于理论研究的生态，使自监督预训练重新成为 AI 研究核心支柱（GitHub 仓库）。









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图 1. LeJEPA 概览。左上：IN1K ViT-base 训练损失与下游线性探测强相关，首次提供无需监督探测的实用模型选择损失。右上：1.8B ViT-g 也无需启发式，损失稳定。左下：IN1K ViT-Large PCA 特征显示清晰语义关系。右下：Galaxy10 域内预训练在从单样本到全监督各数据规模，均优于自然图像训练的 DINOv2/v3 迁移。框架可轻松扩展至任意领域、模型、规模时，领域自监督胜过通用迁移，即使面对巨大基础模型。

1 Introduction

Learning manipulable representations of the world and its dynamics is a long-standing question in AI, with roots dating back centuries ago [Von Helmholtz, 1867, Tolman, 1948, Gregory, 1980, Sutton, 1991, Friston, 2010]. Across domains, e.g., image recognition, robotics, physics, space exploration, the unifying question is *how to learn an organized and actionable high-dimensional embedding space from observations?* Using Deep Networks–parameterized nonlinear operators f_θ–to map observations to embeddings is a standard first piece of that puzzle [LeCun et al., 2015, Goodfellow et al., 2016]. The second, less standardized, piece of that puzzle is *how to train* f_θ. Joint-Embedding Predictive Architectures (JEPAs) suggest training f_θ by maximizing predictive agreement between the embeddings of semantically related *views* [Bromley et al., 1993, LeCun, 2022, Balestriero et al., 2023]. Views can come in two forms: transformations or corruptions. They can involve masking, cropping, blurring, temporal or spatial translations, geometric or photometric transformations, viewpoint changes, views from different sensor modalities, etc. The supervised forms involve human-produced components such as image-caption pairs, text-code pairs, etc [Tian et al., 2020]. In any case, views are expected to share some degree of semantic relationship to allow the prediction task to align f_θ's embeddings towards the underlying knowledge present in the data.

Alas, JEPA's prediction task admits failure modes, such as representation collapse, where f_θ maps all inputs to nearly identical embeddings (*complete collapse*) or to a low-dimensional subspace (*dimensional collapse*) [Jing et al., 2021][Jing et al., 2021, Cosentino et al., 2022, Balestriero and LeCun, 2022]. To mitigate such shortcut solutions, state-of-the-art recipes rely on heuristics–stop-gradient [Chen et al., 2020a], asymmetric view generation [Wang et al., 2022], teacher–student networks with carefully tuned EMA schedules [Caron et al., 2021, Tian et al., 2021], explicit normalization and whitening layers [Ermolov et al., 2021, Chen et al., 2021]–and a delicate balance of hyperparameters. As a result, today's JEPA training is brittle and most research has shifted toward scaling data [Vo et al., 2024], models [Fan et al., 2025] and even post-training Rodas et al. [2025] while leaving the theoretical foundations of JEPAs largely unexplored.

Our study proposes to break that cycle by questioning some of the fundamental design principles underpinning JEPAs. That introspection will start by asking *what are the necessary conditions that JEPAs should abide by?* Those minimal conditions will then act as *axioms* for us to design a novel and lean JEPA. We identify two axioms: (i) solving the prediction task while (ii) enforcing an isotropic Gaussian distribution of the embeddings (Section 3). While (i) follows standard practice [Balestriero and LeCun, 2022], we introduce in Section 4 a novel distribution matching objective–Sketched Isotropic Gaussian Regularization (SIGReg)–to enforce (ii). The use of SIGReg not only removes the need for the numerous heuristics previously employed to prevent representation collapse, but SIGReg also exhibits favorable scaling properties as its *memory and computational complexity is linear in dimension and sample size*. Crucially, SIGReg's isotropic Gaussian enforcement solves the collapsed shortcut solution and provably minimizes the model's expected risk over the space of downstream tasks to be encountered post-training. The resulting JEPA solution–coined Latent-Euclidean JEPA (LeJEPA)–is introduced in Section 5. Beyond theoretical optimality, LeJEPA offers numerous benefits such as (i) provable statistical guarantees, (ii) removal of heuristics such as teacher-student networks, (iii) linear memory and computational complexity, and most importantly (iv) a unified design with a single trade-off parameter that works out of the box across datasets, architectures and scales (see Section 6). We summarize our contributions below.

Contribution 1: We prove the optimal embedding distribution for foundation models. We establish that the isotropic Gaussian uniquely minimizes downstream prediction risk across broad task families. In Section 3, we derive this result rigorously for both linear (Section 3.1) and nonlinear probes (Section 3.2), providing the first principled answer to what distribution f_θ's embeddings should follow. This theoretical result transforms JEPA design from heuristic exploration to targeted optimization.
Contribution 2: We introduce SIGReg, a distribution matching objective that uniquely combines provable correctness with computational efficiency at scale. We present *Sketched Isotropic Gaussian Regularization* (SIGReg), a novel objective that enforces distributional alignment via random projections and characteristic-function matching (Section 4 and Figure 2). SIGReg provides statistical guarantees (Sections 4.1 and 4.2) while achieving linear complexity and bounded gradients—a combination that existing distribution matching methods do not offer. Critically, its projection-based construction defeats the curse of dimensionality (Section 4.3), making it both theoretically sound and practically efficient for high-dimensional embeddings.
Contribution 3: We design LeJEPA, a statistically optimal JEPA that eliminates collapse by construction. By combining JEPA's predictive objective with SIGReg targeting the isotropic Gaussian, we introduce *LeJEPA*—Latent-Euclidean JEPA (Section 5). LeJEPA requires only a single hyperparameter, eliminates representational collapse without stop-gradients or teacher-student architectures, and transfers across architectures and datasets without hyperparameter tuning. This demonstrates that principled

1 引言

学习可操控的世界及动力学表征是 AI 长期问题，思想可追溯数百年 [Von Helmholtz，1867；Tolman，1948；Gregory，1980；Sutton，1991；Friston，2010]。图像识别、机器人、物理、太空探索的共同问题是：如何从观测学到有组织、能支持行动的高维嵌入空间？用深网络，即参数化非线性算子 f_θ映射观测为嵌入，是标准第一步 [LeCun 等，2015；Goodfellow 等，2016]。第二步是如何训练 f_θ，尚未标准化。JEPA 建议最大化语义相关视图嵌入间的预测一致性来训练 f_θ [Bromley 等，1993；LeCun，2022；Balestriero 等，2023]。视图可来自变换或破坏，包括遮蔽、裁剪、模糊、时空平移、几何／光度变换、视角变化及不同传感模态等。监督形式含人工构造的图文对、文本–代码对等 [Tian 等，2020]。视图应共享一定语义关系，使预测将 f_θ的嵌入对齐到数据内在知识。

然而，JEPA 预测任务存在失效模式，如表征坍塌：f_θ 把所有输入映射到几乎相同嵌入（完全坍塌）或低维子空间（维度坍塌） [Jing 等，2021][Jing 等，2021；Cosentino 等，2022；Balestriero 和 LeCun，2022]。为避免捷径解，先进方案依赖启发式设计：停止梯度[Chen 等，2020a]，、非对称视图生成 [Wang 等，2022]，、仔细调 EMA 的教师–学生网络 [Caron 等，2021；Tian 等，2021]，、显式归一化与白化层 [Ermolov 等，2021；Chen 等，2021]，以及微妙的超参数平衡。因此当前 JEPA 训练脆弱，研究转向扩大数据 [Vo 等，2024]，、模型 [Fan 等，2025]，甚至后训练 Rodas 等 [2025]，其理论基础仍大体未被探索。

本研究提出，通过重新审视 JEPA 背后的一些基本设计原则，打破这一循环。我们首先自问：JEPA 必须满足哪些条件？这些最小条件随后将作为公理，指导我们设计一种新颖而精简的 JEPA。我们确定了两条公理：（i）解决预测任务，同时（ii）约束嵌入服从各向同性高斯分布。（第 3 节）。（i）遵循常规 [Balestriero 和 LeCun，2022]，；第 4 节提出新分布匹配目标 SIGReg 来约束（ii）。它消除许多防坍塌启发式，同时内存和计算对维度、样本数呈线性扩展。关键是各向同性高斯约束排除坍塌捷径，并可证明最小化模型在训练后将遇到的下游任务空间中的期望风险。第 5 节介绍所得潜在欧氏 JEPA（LeJEPA）。除理论最优外，还有：（i）可证明统计保证；（ii）去除教师–学生等启发式；（iii）线性内存与计算；（iv）统一设计、仅一个权衡参数，可直接用于不同数据集、架构、规模（第 6 节）。下面总结贡献。

贡献 1：证明基础模型的最优嵌入分布。各向同性高斯在广泛任务族中唯一最小化下游预测风险。第 3 节分别对线性（3.1）及非线性探测器（3.2）严格推导，首次原则性回答 f_θ嵌入应遵循何种分布，使 JEPA 设计从启发式探索变为针对性优化。贡献 2：SIGReg 同时实现可证明正确与大规模计算效率。这一新目标通过随机投影和特征函数匹配来约束分布对齐（第 4 节、图 2）。它具有统计保证（4.1、4.2）、线性复杂度和有界梯度，现有分布匹配方法不能同时提供这些性质。关键是投影构造克服维度灾难（4.3），使高维嵌入在理论和实践上均高效。

贡献 3：构造统计最优、排除坍塌的 LeJEPA。预测目标结合各向同性高斯 SIGReg，形成潜在欧氏 JEPA（第 5 节）。仅一个超参数，无停止梯度或教师–学生，也无表征坍塌，无需调参跨架构、数据迁移。说明有原则的

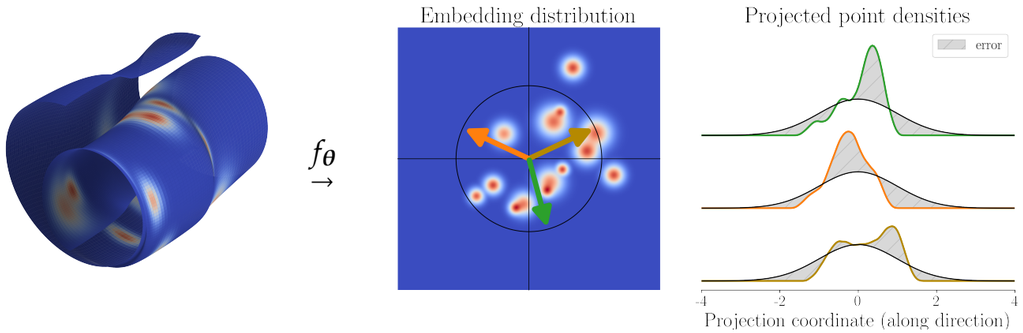


Figure 2. Sketched Isotropic Gaussian Regularization (SIGReg): Given some arbitrary input data with density p_x with support that may or may not lie on a manifold (**left**), a Deep network (DN) encoder (f_θ) produces embeddings $\boldsymbol{z} = f_\theta(\boldsymbol{x})$ with some distribution $\boldsymbol{z} \sim p_z$ (**middle**). Our proposed Backward Cramér-Wold Statistics (Section 4) objective pushes p_z to match a target distribution p_t by projecting the embeddings along $1d$ directions (**middle, arrows**) and enforcing that the univariate densities (**right, colored lines**) match the distribution of p_t, projected along the same directions. Any popular statistical test (provided in Section 4.2) can assess the goodness-of-fit–in practice we argue for characteristic function tests (Section 4.2). By using SIGReg with p_t isotropic Gaussian (**right, black lines**), we introduce a lean and provably optimal (Section 3) JEPA, coined LeJEPA, free of numerous heuristics and able to produce competitive performances (Sections 5 and 6).

theory directly yields practical simplicity.

Contribution 4: We validate LeJEPA at scale across diverse architectures and establish in-domain pretraining as viable. Our experiments (Section 6) span ViTs, ConvNeXts, ResNets, MaxViTs, and Swin Transformers at scales approaching 1 billion parameters, where LeJEPA matches or exceeds state-of-the-art methods while maintaining training simplicity and robustness. Critically, on domain-specific datasets (Galaxy10, Food101), LeJEPA outperforms DINOv2-based transfer learning when pretrained directly on target data. This challenges the transfer learning paradigm and demonstrates that principled SSL can unlock effective in-domain pretraining—previously considered impractical for small datasets.

2 Background and Notations

We start by introducing some of the notations we will be using throughout our manuscript (Section 2.1), followed by a review of JEPAs (Section 2.2), and existing literature studying their design (Section 2.3).

2.1 Notations and Definitions

Data. We are in possession of a dataset of shape $(N, V, D) \in \mathbb{N}^{*3}$ where N is the number of samples, V is the number of views, and D is the dimension. One entry of this dataset is accessed via $\boldsymbol{x}_{n,v,d}$. Those dimensions are often interpreted as follows: (**N**) is the number of independent samples, e.g., different images or different videos, (**V**) is the number of *views*, e.g., data-augmentations for images, frames for videos, and (**D**) is the dimension of each $\boldsymbol{x}_{n,v}$, e.g., number of RGB pixels for images. In many cases the ordering over V is given by *time*–but in some cases, e.g., data-augmentation of an image, ordering becomes irrelevant. Our study does not require any particular choice to organize one's dataset into a (N, V, D) tensor–*and none of our theory and implementation assumes a particular design decision for that tensor*. However, we will rely on the following two properties, (*independence*) the samples $\boldsymbol{x}_n, \boldsymbol{x}_{n'}$ have been obtained independently from each other $\forall n \neq n'$, and (*identically distributed*) the sampling process was identical among $\boldsymbol{x}_n, \forall n$.

Deep Networks. Today's AI solutions rely on *Deep (Neural) Networks* (DNs), which are compositions of a large number of parameterized linear and nonlinear operators. We denote the DN's mapping as $f_\theta : \mathbb{R}^D \to \mathbb{R}^K$ with K the dimension of the embedding space. The internals of f_θ are designed by the researcher to incorporate as much prior knowledge about the data as possible. The details of f_θ are irrelevant to our study–as we will see the proposed LeJEPA works out-of-the-box on any f_θ. In any case, all the *learnable parameters* are gathered in the vector $\boldsymbol{\theta} \in \mathbb{R}^P$, with P counting the total number of parameters. A central challenge in AI research is to design the right architecture and training objective so that $\boldsymbol{\theta}$ can be learned from gradient descent to ultimately produce a useful system, or foundation model, f_θ.

JEPAs. A foundation model is any system, e.g., a DN, able to solve numerous downstream tasks without requiring any change in its internal parameters $\boldsymbol{\theta}$. This is in sharp contrast with a supervised model that only considers its training task. JEPAs have formally been introduced by LeCun [2022] as a vehicle to produce foundation models. The core building blocks of JEPAs rely on numerous well-established techniques such as siamese networks [Bromley et al., 1993] and predictive coding [Helmholtz et al., 1867, Bruner and Postman, 1949]. While the exact blueprint of

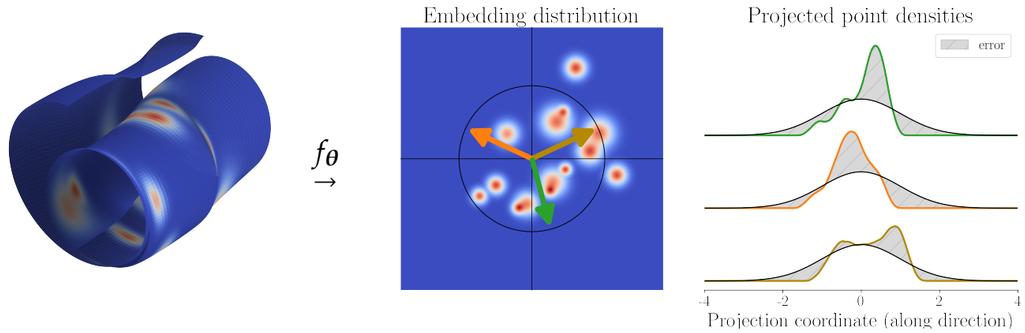


图 2. 草图化各向同性高斯正则化（SIGReg）。任意输入密度 p_x 的支撑集可能位于流形，也可能不在（左）。深网络编码器 f_θ生成分布 $\boldsymbol{z} \sim p_z$ 的嵌入 $\boldsymbol{z} = f_\theta(\boldsymbol{x})$（中）。反向 Cramér–Wold 统计目标（第 4 节）沿 $1d$ 个方向投影（中，箭头），使一维密度（右，彩线）匹配 p_t的同向投影，从而推动 p_z 匹配 p_t。常用统计检验（第 4.2 节）可评估拟合优度，我们主张特征函数检验。将 SIGReg 目标设为 p_t 各向同性高斯（右，黑线），得到精简且可证明最优的 LeJEPA（第 3 节），无需许多启发式，性能具有竞争力（第 5、6 节）。

理论直接带来了实践上的简洁。

贡献 4：跨架构大规模验证，并确立域内预训练可行。第 6 节覆盖 ViT、ConvNeXt、ResNet、MaxViT、Swin，规模接近十亿参数，性能匹配或超过先进方法，同时简单稳健。在 Galaxy10、Food101 等专用域直接预训练，超过 DINOv2 迁移。这挑战迁移范式，表明有原则的 SSL 可使小数据集上曾被认为不实用的域内预训练有效。

2 背景与记号

我们首先介绍全文所用的部分记号（第 2.1 节），随后回顾 JEPA（第 2.2 节）以及研究其设计的现有文献（第 2.3 节）。

2.1 记号与定义

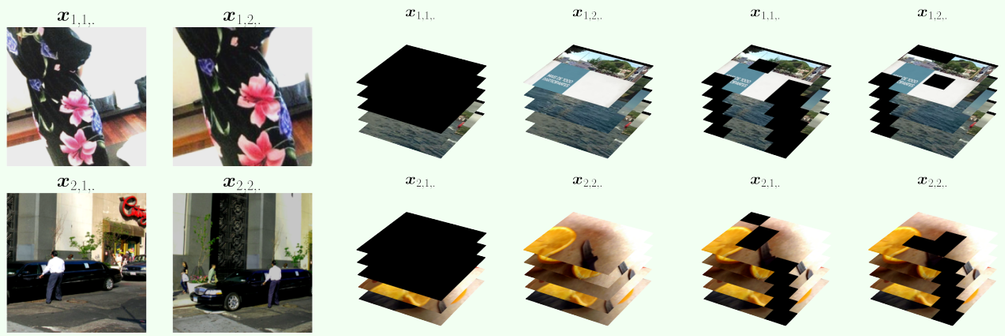
数据。数据集形状$(N, V, D) \in \mathbb{N}^{*3}$， N 样本数、 V 视图数、 D 维度。元素用$\boldsymbol{x}_{n,v,d}$访问。通常 N 为独立样本数（不同图像／视频），V 为视图数（图像增强／视频帧），D 为每个$\boldsymbol{x}_{n,v}$的维度（如 RGB 像素数量）。 V 常按时间排序，但图像增强等情况中，顺序变得无关。数据整理为(N, V, D)张量不需特定方式，理论、实现也不假设特定设计。但需：（独立）$\boldsymbol{x}_n, \boldsymbol{x}_{n'}$ 样本彼此独立取得$\forall n \neq n'$；（同分布） $\boldsymbol{x}_n, \forall n$采样过程相同。

深网络。当前 AI 依赖由大量参数化线性与非线性算子组成的深神经网络（DN）。映射记为 f_θ： $\mathbb{R}^D \to \mathbb{R}^K$， K是嵌入维度。研究者设计 f_θ 内部，尽量融入数据先验；f_θ 细节与本文无关，因为 LeJEPA 可直接用于任意 f_θ。全部可学习参数组成向量 $\boldsymbol{\theta} \in \mathbb{R}^P$， P 为总参数数。核心挑战是选择架构与训练目标，使梯度下降学得 $\boldsymbol{\theta}$，产生有用系统或基础模型 f_θ。

JEPA。基础模型是无需修改内部参数 $\boldsymbol{\theta}$，即可解决多个下游任务的系统（如深网络），不同于仅考虑训练任务的监督模型。LeCun [2022] 正式提出 JEPA 作为产生基础模型的载体，核心部件基于成熟的孪生网络 [Bromley 等， 1993]和预测编码 [Helmholtz 等，1867; Bruner 和 Postman， 1949]。尽管具体蓝图

Definition 1: JEPA

$$\text{JEPA}(\boldsymbol{x}) \iff \text{Enc}\left(\boldsymbol{x}_{n,t+1,.}\right) \text{ is predictable from Enc}\left(\boldsymbol{x}_{n,t,.}\right), \forall n,t \text{ and Enc}\left(\boldsymbol{x}_{.,.,.}\right) \text{ is not degenerate.} \quad (1)$$



JEPAs varies greatly between use-cases, they all rely on two core principles: (i) being able to predict the embedding of a view $\boldsymbol{x}_{n,v}$ from the embedding of another view $\boldsymbol{x}_{n,v'}, v' \neq v$, all while (ii) ensuring that the embeddings do not become degenerate. Concretely, once a JEPA is designed and trained, it should be able to solve numerous downstream tasks in zero or few shots. The JEPA objective function, along with some examples for $\boldsymbol{x}$, is provided in Equation (1). The *predictability* criterion can be done by directly comparing the embeddings of the partial views $Enc(\boldsymbol{x}_{n,v,.})$ and $Enc(\boldsymbol{x}_{n,v',.})$ with a metric, e.g., ℓ_p. In some cases, an additional DN coined *Pred*, is employed to compare $Pred(Enc(\boldsymbol{x}_{n,v,.}))$ against $Enc(\boldsymbol{x}_{n,v',.})$–which is only justified when there exists an asymmetry between the information content of the different views, e.g., by conditioning the predictions on observed actions from robotics data [Khazatsky et al., 2024].

2.2 The Need for Reliable Pretraining

The JEPA's prediction task is designed based on a priori knowledge of the data. Its design is often quite natural since it is relatively intuitive to form $\boldsymbol{x}$ so that its views share the relevant information content one hope to capture. On the other hand, the design of the "anti-collapse" criterion is much closer to a game of Whac-A-Mole. Today's designs rely on many different under-specified safeguards which are carefully combined in the hope that degenerate shortcut solutions are avoided during training. Such mechanisms include (i) feature whitening [Ermolov et al., 2021, Bardes et al., 2021], (ii) negative samples [Chen et al., 2020a, He et al., 2020], and (iii) asymmetric views and teacher-student networks with stop-gradient [Caron et al., 2021, Assran et al., 2023]. Those mechanisms all suffer from at least two of the following limitations: (i) under-specification, i.e., the criteria can be minimized while embeddings are in a degenerate configuration, (ii) quadratic time and memory complexity with mini-batch size and/or embedding dimension, (iii) sensitivity to data distribution, hyperparameters, architecture, and (iv) lack of theoretical understanding and guarantees.

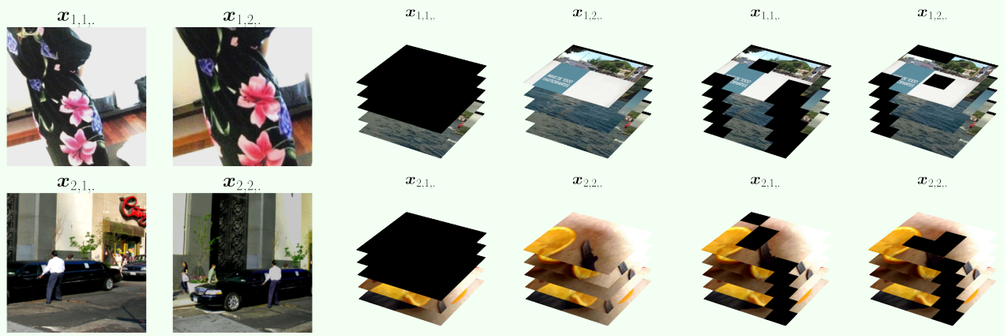
2.3 The Need for Actionable Theory

For decades, the two major solutions for AI were supervised learning [LeCun et al., 2015] and learning by reconstruction [Rumelhart et al., 1986]–sometimes combined together, e.g., for semi-supervised learning [Kingma et al., 2014]. In supervised learning, the labels both ensure that semantically similar samples are close to each other in embedding space while preventing complete representation collapse. In particular, it is possible to measure the amount of collapse in supervised learning as a function of the number of classes [Papyan et al., 2020]. The reconstruction objective is similarly well suited to prevent representation collapse as the original input must be recovered from the embeddings, i.e., the embeddings must be as informative about the input as possible–up to some optional denoising tasks that users can setup as part of the training [Vincent et al., 2010].

Because supervised and reconstruction-based learning have been widely studied for decades, there exists a large body of work to explain and inform practical designs–as well as studying their limitations in producing foundation models [Balestriero and LeCun, 2024, Van Assel et al., 2025]. This is not the case for the more recent JEPAs where empirical advances quickly outpace anyone hoping to delve into their inner workings. This dynamic led the community to focus on post-hoc theoretical justification of already found solutions [Liu et al., 2021, Shwartz Ziv

定义 1：JEPA

$$\text{JEPA}(\boldsymbol{x}) \iff \text{Enc}\left(\boldsymbol{x}_{n,t+1,.}\right) \text{ is predictable from Enc}\left(\boldsymbol{x}_{n,t,.}\right), \forall n,t \text{ and Enc}\left(\boldsymbol{x}_{.,.,.}\right) \text{ is not degenerate.} \quad (1)$$



JEPA 虽在不同应用中差异很大，都依赖两个核心原则：（i）从视图$\boldsymbol{x}_{n,v'}, v' \neq v$的嵌入预测另一视图$\boldsymbol{x}_{n,v}$ 的嵌入；同时（ii）确保嵌入不退化。设计并训练后，应能零样本或少样本解决许多下游任务。式（1）给出目标及 $\boldsymbol{x}$的例子。可直接用如 l_p的度量比较部分视图嵌入 $Enc(\boldsymbol{x}_{n,v,.})$和 $Enc(\boldsymbol{x}_{n,v',.})$。有时加入称为 Pred 的深网络，比较 $Pred(Enc(\boldsymbol{x}_{n,v,.}))$与 $Enc(\boldsymbol{x}_{n,v',.})$；只有视图的信息内容非对称时，这才有依据，例如以机器人数据中的观测动作为预测条件 [Khazatsky 等， 2024]。

2.2 可靠预训练的必要性

JEPA 预测任务依据数据先验设计，通常自然：构造 $\boldsymbol{x}$ 使视图共享希望捕获的信息较直观。但防坍塌设计更像打地鼠，仔细组合许多规定不足的防护，希望避免退化捷径。包括（i）白化 [Ermolov 等，2021；Bardes 等， 2021]，；（ii）负样本 [Chen 等，2020a；He 等， 2020]，；（iii）非对称视图和停止梯度的教师–学生 [Caron 等，2021；Assran 等， 2023]。它们至少受到以下两个限制：（i）约束不充分，即这些准则可以在嵌入处于退化配置时被最小化；（ii）时间和内存复杂度随小批量大小和／或嵌入维度呈二次增长；（iii）对数据分布、超参数和架构敏感；（iv）缺少理论理解与保证。

2.3 可指导实践的理论的必要性

数十年来，AI 主要依靠监督学习 [LeCun 等， 2015] 及重建学习 [Rumelhart 等， 1986]，有时结合，如半监督 [Kingma 等，2014]。监督标签使语义相似样本嵌入接近，同时防止完全坍塌，甚至能根据类别数测量坍塌程度 [Papyan 等， 2020]。重建也自然防坍塌，因为必须从嵌入恢复原输入，所以要尽可能保留输入信息；用户可额外设置去噪任务 [Vincent 等， 2010]。

监督与重建研究数十年，已有大量工作解释并指导实际设计，也研究产生基础模型的局限 [Balestriero 和 LeCun，2024；Van Assel 等，2025]。较新 JEPA 并非如此，经验进展远超内部机理探索，社区转向为已有解事后给理论依据 [Liu 等，2021；Shwartz Ziv

and LeCun, 2024, Shwartz-Ziv et al., 2022, Zhang et al., 2023]. In most cases, those studies involve the *Mutual Information (MI)* [Shannon, 1948, Cover, 1999] whose different bounds recover established methods [Gutmann and Hyvärinen, 2010, Ma and Collins, 2018, Oord et al., 2018, Poole et al., 2019, Hjelm et al., 2018, McAllester and Stratos, 2020]. Because existing studies focus on explaining and interpreting already developed JEPAs, too little principled guidance and innovation has been brought forward. Instead, most of the recent empirical advances take the form of collecting larger dataset, scaling up pre-existing training recipes [Goyal et al., 2019, Chen et al., 2020b, Oquab et al., 2023, Fan et al., 2025], and deriving novel data curation processes [Vo et al., 2024, Kerdreux et al., 2025].

In contrast, our goal in the following Sections 3 to 5 will be to derive a novel JEPA solution from first principles, i.e., whose design relies on proved necessary conditions for optimality, and with a pretraining recipe that can finally reconcile exploratory research, scalability, and state-of-the-art performances.

3 Latent Euclidean: Embeddings Should be Isotropic Gaussian

We address a fundamental question: *which distribution should* $\text{Enc}(\boldsymbol{x})$ *follow to minimize empirical risk on any downstream task?* We prove that the isotropic Gaussian is the unique optimal distribution for both linear (Section 3.1) and nonlinear probing (Section 3.2), with geometric intuition provided in Section 3.3. This theoretical result establishes the necessary design principle for our JEPA; Section 4 then provides the practical implementation to achieve it.

3.1 Linear Probing

We begin by identifying the optimal distribution for $f_{\boldsymbol{\theta}}$'s embeddings by analyzing linear probes–one of the most popular methods for frozen encoder evaluation. Specifically, we ask: *which distribution for* $f_{\boldsymbol{\theta}}(\boldsymbol{x})$ *would be most favorable for solving arbitrary downstream tasks, i.e., for any realization of targets* $\boldsymbol{y}$*?*

Denote as $\boldsymbol{Z} \in \mathbb{R}^{N\times K}$ the matrix of N embeddings, each K-dimensional, from $f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)$. The *unknown* corresponding labels are denoted as $\boldsymbol{y} \in \mathbb{R}^N$. Without loss of generality, we consider univariate targets; the following analysis extends to multivariate targets. The linear probe minimizes the following least square problem [Bishop and Nasrabadi, 2006]

$$\hat{\beta} = \arg\min_{\beta\in\mathbb{R}^K} \|\boldsymbol{y} - \boldsymbol{Z}\beta\|_2^2 + \lambda\|\beta\|_2^2, \qquad \text{(OLS)}$$

where $\hat{\beta}$ is the optimal probe parameters, and $\lambda \geq 0$ is an hyperparameter controlling the Tikhonov regularizer strength [Bishop, 1995, Golub et al., 1999]. Despite not knowing $\boldsymbol{y}$, it is possible to describe the bias and variance of the estimator $\hat{\beta}$ as a function of the distribution of $\boldsymbol{Z}$. Consider two embeddings with identical column spans $\boldsymbol{Z}_{\text{aniso}}, \boldsymbol{Z}_{\text{iso}}$. $\boldsymbol{Z}_{\text{aniso}}$'s covariance matrix eigenvalues are given by $\{\lambda_k\}_{k=1}^K$ with at least two distinct values, while $\boldsymbol{Z}_{\text{iso}}$'s covariance matrix eigenvalues are all equal to $\frac{1}{K}\sum_{k=1}^K \lambda_k$. Hence, the two candidate embeddings $\boldsymbol{Z}_{\text{aniso}}, \boldsymbol{Z}_{\text{iso}}$ capture the same intrinsic features and have same energy, but different geometries.

Lemma 1: Anisotropy amplifies bias

Whenever $\lambda_K > \lambda_1$, there always exists a downstream task ($\boldsymbol{y}$) for which $\boldsymbol{Z}_{\text{aniso}}$ produces a higher bias estimator than $\boldsymbol{Z}_{\text{iso}}$ for $\lambda > 0$. (Proof in Section B.1.)

Lemma 2: Anisotropy amplifies variance

With $\lambda = 0$, the total variance of $\hat{\beta}$ (OLS) is minimized for $\boldsymbol{Z}_{\text{iso}}$ with $\text{tr}(\text{Var}(\hat{\boldsymbol{\beta}}_{\text{aniso}})) > \text{tr}(\text{Var}(\hat{\boldsymbol{\beta}}_{\text{iso}}))$. (Proof in Section B.2.)

From the above lemmas. 1 and 2 we obtain that the distribution of features must be isotropic. We now move to nonlinear probing where the standard Gaussian will emerge as the unique optimum.

3.2 Nonlinear Probing

To allow for more flexible evaluation of the pretrained encoder $f_{\boldsymbol{\theta}}$, it has become increasingly common to work with a nonlinear probe. We analyze two widely-used nonlinear methods: radius-based k-NN [Taunk et al., 2019, Sun and Huang, 2010, Zhang et al., 2017, Abu Alfeilat et al., 2019] for its simplicity and kernel methods [Nadaraya, 1964, Watson, 1964] for their theoretical tractability.

As in Section 3.1, we ask ourselves which distribution of embeddings would be preferable for a foundation model. We first define our prediction function. The training data consists of the N embeddings along with their training labels $\{(\boldsymbol{z}_n, \boldsymbol{y}_n)\}_{n=1}^N$. The prediction, using radius-based k-NN for a query vector $\boldsymbol{q}$ is formed as

$$\widehat{\boldsymbol{y}}(\boldsymbol{q}) := \frac{1}{|\mathcal{N}_{r_0}(\boldsymbol{q})|}\sum_{n\in\mathcal{N}_{r_0}(\boldsymbol{q})} \boldsymbol{y}_n, \qquad \text{(kNN)}$$

where $\mathcal{N}_{r_0}(\boldsymbol{q}) = \{n : \|\boldsymbol{z}_n - \boldsymbol{q}\| \leq r_0\}$. The specific choice of radius r_0 controls how many neighbors predictions are averaged to form the query's prediction. The kernel's prediction at a query $\boldsymbol{q} \in \mathbb{R}^K$ is given by

$$\widehat{\boldsymbol{y}}(\boldsymbol{q}) \triangleq \frac{\sum_{n=1}^N K_h(\boldsymbol{q} - \boldsymbol{z}_n)\boldsymbol{y}_n}{\sum_{n=1}^N K_h(\boldsymbol{q} - \boldsymbol{z}_n)}. \qquad \text{(Kernel)}$$

We search over all distributions of Z subject to a fixed total variance constraint, e.g., $\text{Tr}(\text{Cov}(\boldsymbol{Z})) = \kappa_1$ or $\|\text{Cov}(\boldsymbol{Z})\|_F = \kappa_2$. The specific value of κ does not affect the optimal dis-

和 LeCun，2024；Shwartz-Ziv 等，2022；Zhang 等，2023]）。多数研究涉及互信息（MI）[Shannon，1948；Cover，1999]，其不同界可恢复既有方法 [Gutmann 和 Hyvärinen，2010；Ma 和 Collins，2018；Oord 等，2018；Poole 等，2019；Hjelm 等，2018；McAllester 和 Stratos，2020]。现有研究侧重解释已开发的 JEPA，提出的原则性指导与创新不足。近期经验进展多是收集更大数据、扩大既有训练方案 [Goyal 等，2019；Chen 等，2020b；Oquab 等，2023；Fan 等，2025]，，或新数据筛选流程 [Vo 等，2024；Kerdreux 等，2025]。

相比之下，随后第 3 至第 5 节的目标，是从第一性原理推导出一种新的 JEPA 方案：其设计依据是经过证明的最优性必要条件，其预训练方案最终能够兼顾探索性研究、可扩展性与最先进的性能。

3 潜在欧氏空间：嵌入应服从各向同性高斯分布

我们讨论一个根本问题：$\text{Enc}(\boldsymbol{x})$ 应服从什么分布，才能使任意下游任务的经验风险最小？我们证明，对于线性探测（第 3.1 节）和非线性探测（第 3.2 节），各向同性高斯分布都是唯一最优分布；第 3.3 节给出几何直觉。这一理论结果确立了我们的 JEPA 必须遵循的设计原则；随后第 4 节给出实现该原则的实际方法。

3.1 线性探测

我们首先分析线性探测，确定 $f_{\boldsymbol{\theta}}$ 的嵌入的最优分布；线性探测是评估冻结编码器最常用的方法之一。具体而言，我们问：对于任意目标 $\boldsymbol{y}$ 的实现，$f_{\boldsymbol{\theta}}(\boldsymbol{x})$ 采用什么分布最有利于解决任意下游任务？

将来自 $f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)$ 的 N 个嵌入组成的矩阵记为 $\boldsymbol{Z} \in \mathbb{R}^{N\times K}$，每个嵌入为 K 维。其未知的对应标签记为 $\boldsymbol{y} \in \mathbb{R}^N$。不失一般性，我们考虑一元目标；以下分析可推广到多元目标。线性探测器最小化下面的最小二乘问题 [Bishop and Nasrabadi,2006]:

$$\hat{\beta} = \arg\min_{\beta\in\mathbb{R}^K} \|\boldsymbol{y} - \boldsymbol{Z}\beta\|_2^2 + \lambda\|\beta\|_2^2, \qquad \text{(OLS)}$$

其中 $\hat{}\beta$ 为最优探测器参数，$\lambda \geq 0$ 是控制 Tikhonov 正则化强度的超参数 [Bishop, 1995, Golub et al., 1999]。尽管即使不知道 $\boldsymbol{y}$，也能将估计器 $\hat{}\beta$ 的偏差和方差写为 $\boldsymbol{Z}$ 分布的函数。两个嵌入列空间相同 $\boldsymbol{Z}_{\text{aniso}}, \boldsymbol{Z}_{\text{iso}}$，$\boldsymbol{Z}_{\text{aniso}}$ 协方差特征值为 $\{\lambda_k\}_{k=1}^K$，至少两个不同；$\boldsymbol{Z}_{\text{iso}}$ 特征值均为 $\frac{1}{K}\sum_{k=1}^K \lambda_k$。因此 $\boldsymbol{Z}_{\text{aniso}}, \boldsymbol{Z}_{\text{iso}}$ 包含相同内在特征、能量相同，但几何不同。

–

引理 1：各向异性放大偏差

只要 $\lambda_K > \lambda_1$，就总存在某个下游任务（$\boldsymbol{y}$），使 $\boldsymbol{Z}$aniso 在 $\lambda > 0$ 情况下得到的估计量偏差高于 $\boldsymbol{Z}$iso。（证明见附录 B.1。）

引理 2：各向异性放大方差

当 $\lambda = 0$ 时，$\hat{}\beta$ (OLS) 的总方差在 $\boldsymbol{Z}$iso 下最小，且 $\text{tr}(\text{Var}(\hat{}\boldsymbol{\beta}\text{aniso})) > \text{tr}(\text{Var}(\hat{}\boldsymbol{\beta}\text{iso}))$。（证明见附录 B.2。）

由上述引理 1 和引理 2 可得，特征分布必须是各向同性的。下面转向非线性探测，标准高斯分布将在此成为唯一最优解。

3.2 非线性探测

灵活评估预训练编码器 $f_{\boldsymbol{\theta}}$，越来越常用非线性探测器。分析两种：简单的半径 kNN [Taunk 等，2019；Sun 和 Huang，2010；Zhang 等，2017；Abu Alfeilat 等，2019]；理论易处理的核方法 [Nadaraya，1964；Watson，1964]。

与第 3.1 节一样，我们考察基础模型的嵌入应采用哪种分布。首先定义预测函数。训练数据由 N 个嵌入及其训练标签 $\{(\boldsymbol{z}_n, \boldsymbol{y}_n)\}_{n=1}^N$ 构成。对于查询向量 $\boldsymbol{q}$，采用基于半径的 k 近邻方法，预测为

$$\widehat{\boldsymbol{y}}(\boldsymbol{q}) := \frac{1}{|\mathcal{N}_{r_0}(\boldsymbol{q})|}\sum_{n\in\mathcal{N}_{r_0}(\boldsymbol{q})} \boldsymbol{y}_n, \qquad \text{(kNN)}$$

其中 $\mathcal{N}_{r_0}(\boldsymbol{q}) = \{n : \|\boldsymbol{z}_n - \boldsymbol{q}\| \leq r_0\}$。半径 r_0 的具体选择决定了对多少个邻居的预测取平均，以形成查询点的预测。在查询点 $\boldsymbol{q} \in \mathbb{R}^K$ 处，核预测为

$$\widehat{\boldsymbol{y}}(\boldsymbol{q}) \triangleq \frac{\sum_{n=1}^N K_h(\boldsymbol{q} - \boldsymbol{z}_n)\boldsymbol{y}_n}{\sum_{n=1}^N K_h(\boldsymbol{q} - \boldsymbol{z}_n)}. \qquad \text{(Kernel)}$$

我们在所有满足固定总方差约束的 Z 分布中搜索，例如 $\text{Tr}(\text{Cov}(\boldsymbol{Z})) = \kappa_1$ 或 $\|\text{Cov}(\boldsymbol{Z})\|_F = \kappa_2$。$\kappa$ 的具体取值不影响最优分布

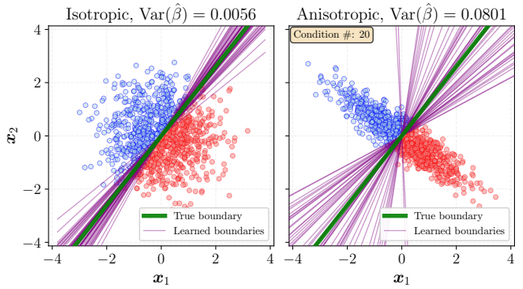


Figure 3. Illustration of lemma. 2 showcasing how anisotropic (**right**) embeddings lead to higher variance estimator compared to isotropic embeddings (**left**). We sample 100 training points for the 2-class classification task and fit a logistic regression–repeating the process over numerous training set sample. Each sampling results in a decision boundary (**purple**).

tribution shape. Following the same type of derivations as done in the linear regime–with the exception of some additional regularity conditions–we are able to precisely identify the isotropic Gaussian as the unique optimum to minimize bias as formalized below.

Theorem 1: isotropic Gaussian Optimality

The integrated square bias (ISB) over query points is given by

$$\text{ISB}_{k\text{-NN}} = \frac{r_0^4}{(K+2)^2}\tau_s^2 J(p) + O(r_0^4), \quad \text{(k-NN)}$$

$$\text{ISB}_{\text{kernel}} \leq \left(\frac{h^2\mu_2(K)}{2}\right)^2 \left(2B^2 + 8L^2 J(p)\right) + o(h^4), \quad \text{(kernel)}$$

and among distributions with a scalar-based covariance constraint, the isotropic Gaussian is the unique minimizer of the integrated square bias. (Proof in Sections B.4 and B.7.)

Numerous additional details and discussions on the regularity assumptions we employed are provided in Section A. Together, these results establish the isotropic Gaussian distribution as the optimal design to minimize the worst-case risk of a foundation model across downstream tasks.

3.3 Geometric and Practical Insights

We now empirically validate that the isotropic Gaussian is optimal when no information about downstream tasks is available. We focus on linear probing (Section 3.1), where all considered distributions have the same total variance.

When employing a linear probe, an anisotropic distribution increases both bias (with Tikhonov regularization) and variance. Examining bias first (lemma. 1), we present in Figure 18 visualizations for both continuous regression and discrete classification tasks. We observe that the cosine similarity between estimated and ground-truth parameters equals 1 only for isotropic distributions, degrading for anisotropic cases regardless of sample size or regularization strength. Regarding variance (lemma. 2), we show in Figure 3 that learned parameters vary significantly more across training sets when the covariance is anisotropic (right) compared to isotropic (left)—even when using logistic regression instead of OLS. Figure 17 further illustrates this effect, showing the distribution of learned β parameters across different training samples for both cases. The anisotropic distribution clearly produces higher-variance estimators.

These theoretical and empirical results establish our design principle for LeJEPA: *embeddings $f_\theta(x)$ should follow an isotropic Gaussian distribution to minimize worst-case risk across downstream tasks encountered post-training*. Section 4 introduces a novel regularizer to achieve this distribution.

4 SIGReg: Reliable Isotropic Gaussian Regularization in High-Dimension

Having established the isotropic Gaussian as the optimal embedding distribution (Section 3), we now introduce *Sketched Isotropic Gaussian Regularization* (SIGReg)–a distribution matching objective that is simultaneously (i) *differentiable*, (ii) *scalable*, (iii) *provable*, and (iv) *interpretable*. SIGReg builds on three key innovations. First, we formulate distribution matching as a statistical test under the null hypothesis $P_\theta = Q$ (Section 4.1). Second, we identify a test that guarantees bounded gradients and curvature while maintaining linear complexity and efficient multi-GPU scaling (Section 4.2). Third, SIGReg bypasses the curse of dimensionality, eliminating collapsed shortcut solutions entirely (Section 4.3).

4.1 Hypothesis Testing as a Judge

Asking for $f_\theta(x)$'s distribution P_θ to match a target distribution Q is typically done by creating various measures of distance or divergence, and estimating them in high-dimension. We propose a different starting point grounded in statistics. Consider the hypothesis testing framework [Fisher, 1928, Neyman and Pearson, 1933] given by

$$H_0 : P_\theta = Q \quad \text{vs.} \quad H_1 : P_\theta \neq Q, \tag{2}$$

with H_0 being referred to as the *null hypothesis*. That is, we are asking in Equation (2) if there is enough empirical evidence to reject the null. To answer that question, one (i) employs a *test-statistic*, i.e., a single scalar value summarizing the evidence from the empirical samples, (ii) determines a critical value τ_α for the test-statistic based on the probability α of Type I error, i.e., of mistakenly rejecting a true null hypothesis, (iii) compares the test-statistic to

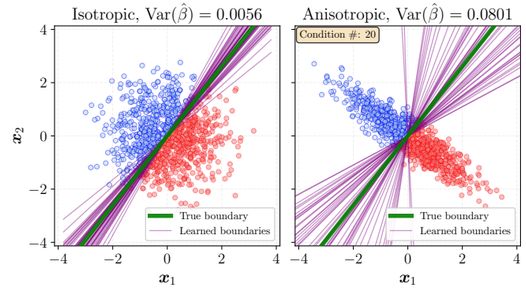


图 3. 引理 2 的示意图：相比各向同性嵌入（左），各向异性嵌入（右）使估计量具有更高的方差。我们为二分类任务采样 100 个训练点并拟合逻辑回归，对多个训练集样本重复此过程。每次采样均产生一条决策边界（紫色）。

的形状。除增加一些正则性条件外，沿用线性情形中的同类推导，我们能精确确定：各向同性高斯分布是使偏差最小的唯一最优解。形式化结论如下。

定理 1：各向同性高斯最优性

对查询点积分的平方偏差（ISB）为

$$\text{ISB}_{k\text{-NN}} = \frac{r_0^4}{(K+2)^2}\tau_s^2 J(p) + O(r_0^4), \quad \text{(k-NN)}$$

$$\text{ISB}_{\text{kernel}} \leq \left(\frac{h^2\mu_2(K)}{2}\right)^2 \left(2B^2 + 8L^2 J(p)\right) + o(h^4), \quad \text{(kernel)}$$

且在协方差满足基于标量约束的分布中，各向同性高斯分布是使积分平方偏差最小的唯一分布。（证明见附录 B.4 和 B.7。）

有关所采用正则性假设的更多细节与讨论，见附录 A。这些结果共同确立了各向同性高斯分布的最优性：它能使基础模型在各类下游任务中的最坏情况风险最小。

3.3 几何与实践直觉

下面通过实验验证：当没有关于下游任务的信息时，各向同性高斯分布最优。我们重点考察线性探测（第 3.1 节），其中所有被比较的分布均具有相同总方差。

使用线性探测器时，各向异性分布会同时增大偏差（采用 Tikhonov 正则化时）和方差。首先考察偏差（引理 1）：图 18 展示连续回归任务和离散分类任务的可视化结果。我们观察到，估计值与真实值之间的余弦相似度参数只对各向同性等于 1，各向异性时无论样本量、正则强度均下降。方差方面（引理 2），图 3 中各向异性（右）跨训练集参数变化明显大于各向同性（左），即使用逻辑回归代替 OLS。图 17 展示不同样本学得 β 分布，各向异性明显产生更高方差。

这些理论与经验结果确立了 LeJEPA 的设计原则：嵌入 $f_\theta(x)$ 应服从各向同性高斯分布，以使训练后遇到的各类下游任务的最坏情况风险最小。第 4 节介绍一种实现该分布的新正则化方法。

4 SIGReg：高维空间中可靠的各向同性高斯正则化

第 3 节确立各向同性高斯最优后，引入 SIGReg：同时可微、可扩展、可证明、可解释的分布匹配目标。三项创新：以零假设 $P_\theta = Q$ 下的统计检验表述匹配（4.1）；选择梯度和曲率有界、线性复杂度、多 GPU 高效的检验（4.2）；绕过维度灾难，完全排除坍塌捷径（4.3）。

4.1 以假设检验作为判据

要求 $f_\theta(x)$ 的分布 P_θ 与目标分布 Q 匹配，通常通过构造各种距离或散度度量，并在高维空间中估计它们来实现。我们提出一个以统计学为基础的不同起点。考虑以下假设检验框架 [Fisher, 1928, Neyman and Pearson, 1933]：

$$H_0 : P_\theta = Q \quad \text{vs.} \quad H_1 : P_\theta \neq Q, \tag{2}$$

H_0 称零假设，式（2）询问是否有足够经验证据拒绝它。做法：（i）检验统计量，用单一标量总结样本证据；（ii）根据 I 类错误概率 α（误拒真实零假设）确定临界值 τ_α；（iii）比较统计量与

the critical value τ_α; if the test-statistic exceeds τ_α, reject the null hypothesis. If the null is not rejected, we can only claim that *there is not sufficient empirical evidence against* $P_{\boldsymbol{\theta}} = Q$.

As it stands, Equation (2) remains impractical in large dimension as existing tests have at least quadratic complexity with the number of samples considered (more details in Section F). We thus propose to derive a sketching strategy by decomposing Equation (2) into simpler univariate tests. Denoting the push-forward distributions $P^{(\boldsymbol{a})}_{\boldsymbol{\theta}} \triangleq (\boldsymbol{a}^\top)_\# P_{\boldsymbol{\theta}}$ and $Q^{(\boldsymbol{a})} \triangleq (\boldsymbol{a}^\top)_\# Q$, we can define the following *directional* univariate test

$$H_0(\boldsymbol{a}) : P^{(\boldsymbol{a})}_{\boldsymbol{\theta}} = Q^{(\boldsymbol{a})} \quad \text{vs.} \quad H_1(\boldsymbol{a}) : P^{(\boldsymbol{a})}_{\boldsymbol{\theta}} \neq Q^{(\boldsymbol{a})}, \tag{3}$$

for a given directional unit-norm vector $\boldsymbol{a} \in \mathcal{S}^{K-1}$. The corresponding *directional test-statistic* of Equation (3) is computed as $T(\{\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N)$. Examples of tests T will be provided in the later Section 4.2. Repeating that process over a set of M directions $\mathbb{A} \triangleq \{\boldsymbol{a}_1, \dots, \boldsymbol{a}_M\}$ and aggregating the individual values lead to the following *global test-statistic*

$$T_{\mathbb{A}}(\{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \triangleq \max_{\boldsymbol{a} \in \mathbb{A}} T(\{\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N). \tag{4}$$

We now provide a formal statement asserting the consistency of Equation (4) to test the original multivariate null hypothesis from Equation (2). Our result leverages the well-known union-intersection principle [Roy, 1953], and a slightly modified Cramér-Wold theorem. We denote by $\overset{d}{=}$ equality in distribution.

Lemma 3: Hyperspherical Cramér-Wold

Let X, Y be $\mathbb{R}^d$-valued random vectors, then

$$\langle \boldsymbol{u}, X\rangle \overset{d}{=} \langle \boldsymbol{u}, Y\rangle, \forall \boldsymbol{u} \in \mathbb{S}^{d-1} \iff X \overset{d}{=} Y.$$

Convergence in distribution also holds. (Proof in Section B.8.)

Theorem 2: Sufficiency of directional tests

Equation (4) is a valid statistical test for Equation (3) as

$$P = Q \implies \limsup_{n\to\infty} \Pr\left(T_{\mathbb{A}}(\{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \geq \tau_\alpha\right) \leq \alpha, \textbf{(level)}$$

$$P \neq Q \implies \limsup_{n\to\infty} \Pr\left(T_{\mathbb{A}}(\{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \geq \tau_\alpha\right) = 1, \textbf{(power)}$$

(Proof in Section B.9.)

The assumptions required in the proof of thm. 2 hold for classical consistent univariate tests T such as the ones presented in the following Section 4.2.

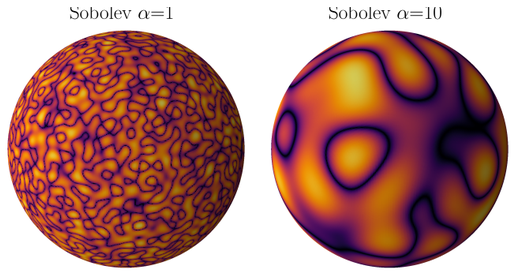


Figure 4. Examples of distributions living on the surface of the sphere with varying Sobolev smoothness coefficients α. As per thm. 5, the greater α is, the more global will be the impact of SIGReg for a given number of directions M. Practically, this represents the distribution of the encoder's output. Because the target density (isotropic Gaussian) is smooth, the α coeffcients of the embedding will quickly grow hereby making SIGReg (def. 2) immune to the curse of dimensionality.

4.2 SIGReg: Sketching the Epps-Pulley Test is Stable and Scalable

Our proposed regularizer–coined Sketched Isotropic Gaussian Regularization (SIGReg)–follows directly from thm. 2 using any statistical test T targeted towards the isotropic Gaussian, illustrated in Figures 2 and 5, and formalized below.

Definition 2: SIGReg (PyTorch code in algorithm 1)

SIGReg sketches a statistical test T towards isotropic Gaussian

$$\text{SIGReg}_T(\mathbb{A}, \{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \triangleq \frac{1}{|\mathbb{A}|} \sum_{\boldsymbol{a} \in \mathbb{A}} T(\{\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N), \tag{SIGReg}$$

where we recommend the Epps-Pulley test (Section 4.2.3) for T.

We replace the maximum over $\boldsymbol{a} \in \mathbb{A}$ in thm. 2 by an average in (SIGReg) to avoid sparse gradient over the directions in $\mathbb{A}$. We now delve on the choice of T for which we compare well-known candidate tests in the field of statistics that are categorized into (i) moment based (Section 4.2.1), (ii) CDF based (Section 4.2.2), and (iii) CF based (Section 4.2.3) statistics–ultimately justifying our choice of the Epps-Pulley statistic.

4.2.1 Moments are Unstable and Insufficient

The first family of statistics we consider are moment-based. Taking the standard Gaussian as an instanciation for the moments, we can define the Jarque-Bera [Jarque and Bera, 1980] test that compares the third and fourth moments,

临界值 τ_α；若检验统计量超过 τ_α，则拒绝零假设。若未拒绝零假设，我们只能声称：没有足够的经验证据反对 $P_{\boldsymbol{\theta}} = Q$。

目前，式（2）在高维情形下仍不实用，因为现有检验的复杂度随所考虑的样本数至少呈二次增长（详见附录 F）。因此，我们提出将式（2）分解为更简单的一元检验，从而推导一种随机压缩策略。

将推前分布记为 $P^{(\boldsymbol{a})}_{\boldsymbol{\theta}} \triangleq (\boldsymbol{a}^\top)_\# P_{\boldsymbol{\theta}}$ 和 $Q^{(\boldsymbol{a})} \triangleq (\boldsymbol{a}^\top)_\# Q$，即可定义以下方向性一元检验：

$$H_0(\boldsymbol{a}) : P^{(\boldsymbol{a})}_{\boldsymbol{\theta}} = Q^{(\boldsymbol{a})} \quad \text{vs.} \quad H_1(\boldsymbol{a}) : P^{(\boldsymbol{a})}_{\boldsymbol{\theta}} \neq Q^{(\boldsymbol{a})}, \tag{3}$$

给定单位方向向量 $\boldsymbol{a} \in \mathcal{S}^{K-1}$，式（3）方向统计量为 $T(\{\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N)$，候选检验 T 见 4.2。对 M 方向集合 $\mathbb{A} \triangleq \{\boldsymbol{a}_1, \dots, \boldsymbol{a}_M\}$重复并汇总，得全局统计量：

$$T_{\mathbb{A}}(\{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \triangleq \max_{\boldsymbol{a} \in \mathbb{A}} T(\{\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N). \tag{4}$$

下面给出形式化结论，说明式（4）在检验式（2）中原始多元零假设时具有一致性。该结果利用了熟知的并交原则 [Roy，1953], 以及稍加修改的 Cramér–Wold 定理。我们用 $\overset{d}{=}$ 表示依分布相等。

引理 3：超球面 Cramér–Wold 定理

设 X, Y 是取值于 $\mathbb{R}^d$ 的随机向量，则

$$\langle \boldsymbol{u}, X\rangle \overset{d}{=} \langle \boldsymbol{u}, Y\rangle, \forall \boldsymbol{u} \in \mathbb{S}^{d-1} \iff X \overset{d}{=} Y.$$

依分布收敛也成立。（证明见附录 B.8。）

定理 2：方向检验的充分性

式（4）是式（3）的有效统计检验，因为

$$P = Q \implies \limsup_{n\to\infty} \Pr\left(T_{\mathbb{A}}(\{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \geq \tau_\alpha\right) \leq \alpha, \textbf{(level)}$$

$$P \neq Q \implies \limsup_{n\to\infty} \Pr\left(T_{\mathbb{A}}(\{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \geq \tau_\alpha\right) = 1, \textbf{(power)}$$

(Proof in Section B.9.)

定理 2 证明所需的假设，对于经典的一致一元检验 T 均成立，例如随后第 4.2 节介绍的检验。

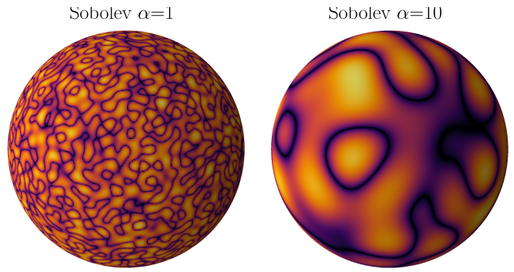


图 4. 球面分布，不同 Sobolev 平滑系数 α。定理 5 中，α 越大，固定 M方向的 SIGReg 全局影响越强。实际为编码器输出分布；目标高斯平滑，嵌入 α 系数迅速增大，使 SIGReg（定义 2）不受维度灾难影响。

4.2 SIGReg：Epps–Pulley 检验的随机压缩既稳定又可扩展

我们提出的正则化方法称为随机压缩各向同性高斯正则化（SIGReg），它直接来自定理 2，可采用任意以各向同性高斯分布为目标的统计检验 T，示意见图 2 和图 5，形式化定义如下。

定义 2：SIGReg（PyTorch 代码见算法 1）

SIGReg 对面向各向同性高斯分布的统计检验 T 进行随机压缩。

$$\text{SIGReg}_T(\mathbb{A}, \{f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N) \triangleq \frac{1}{|\mathbb{A}|} \sum_{\boldsymbol{a} \in \mathbb{A}} T(\{\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}_n)\}_{n=1}^N), \tag{SIGReg}$$

其中，对于 T，我们推荐 Epps–Pulley 检验（第 4.2.3 节）。

SIGReg 将定理 2 对 $\boldsymbol{a} \in \mathbb{A}$ 的最大值改为平均，避免 $\mathbb{A}$方向稀疏梯度。接着比较统计学常用 T 检验：矩（4.2.1）、CDF（4.2.2）、CF（4.2.3），最终论证 Epps–Pulley 选择。

4.2.1 矩不稳定且不充分

我们考虑的第一类统计量基于矩。以标准高斯分布的矩为例，可定义 Jarque–Bera [Jarque and Bera,1980] 检验，比较三阶矩与四阶矩，

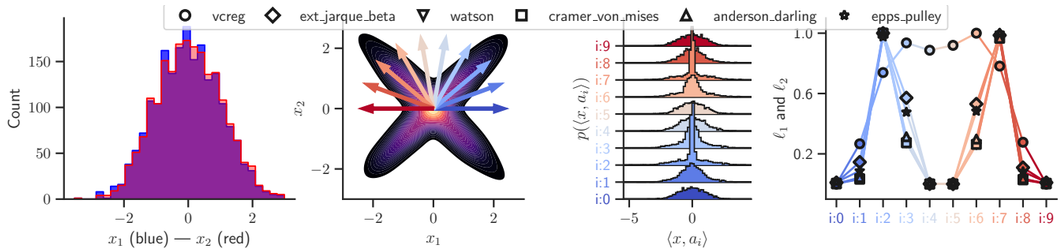


Figure 5. Constructed data density with "X" distribution whose marginals are standard Gaussian and whose covariance is identity (**left densities**). Applying $M = 10$ projections on the half circle directions produces 10 univariate distributions that can be compared against a standard Gaussian (**left**) using any preferred statistic from Section 4.2. The appropriate direction is able to capture the degenerate distribution of the data hereby creating a spike in the statistic value.

i.e., skewness and kurtosis, as

$$\text{JB}(\boldsymbol{u}) \triangleq \frac{N}{6}\left(\widehat{\text{skew}}(\boldsymbol{u})^2 + \left(\frac{\widehat{\text{kurt}}(\boldsymbol{u}) - 3}{2}\right)^2\right), \quad \text{(Jarque-Bera)}$$

where $\widehat{\text{skew}}$ is the skewness computed from the data as $\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^3}{\hat{\sigma}^3}$ and $\widehat{\text{kurt}}$ is the kurtosis $\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^4}{\hat{\sigma}^4}$. Typically, the (Jarque-Bera) test is used to see if a density follows a Gaussian distribution of any mean and variance–hence it only looks at moments 3 and 4. In our case we aim for a standard Gaussian test and thus add the usual statistics on the first two moments, leading to the extended test

$$\text{EJB}(\boldsymbol{u}) \triangleq \frac{N\hat{\mu}(\boldsymbol{u})^2}{\hat{\sigma}(\boldsymbol{u})^2} + \frac{(N-1)\left(\hat{\sigma}(\boldsymbol{u})^2 - 1\right)^2}{2} + \text{JB}(\boldsymbol{u}). \quad \text{(Extended Jarque-Bera)}$$

The (Extended Jarque-Bera) acts as a moment matching problem over the first four moments. Such moment matching methods have proven powerful not only for statistical tests but also as mean to learn parametric and nonparametric models of data.

The Stability and Identifiability Conundrum. We now explain why moment-based tests–albeit powerful–will not be suited for LeJEPA. The kth of a distribution P is denoted as $m_k(P)$. The first observation is that well-behaved distributions abiding the Carleman's condition $\sum_{k=1}^{\infty} m_{2k}(Q)^{-1/(2k)} = \infty$ [Carleman, 1926], such as the Gaussian, or for distributions with finite interval [Hausdorff, 1923] are uniquely determined by their moments. However, using a finite number of moments creates the following non-identifiability issue which well-known in statistics and often used as a motivation to use *all* moments [Lehmann and Romano, 2005].

Theorem 3: Insufficiency of K Moments

Minimizing the following objective with $c_k > 0, \forall k$

$$\sum_{k=1}^{K} c_k \left(m_k\left(P_{\boldsymbol{\theta}}^{(\boldsymbol{a})}\right) - m_k\left(Q^{(\boldsymbol{a})}\right)\right)^2,$$

for finite K does not imply $P_{\boldsymbol{\theta}}^{(\boldsymbol{a})} = Q^{(\boldsymbol{a})}$. (Proof in Section B.11.)

Hence thm. 3 prescribes us with the guideline to employ as large K as possible to remove collapsed shortcut solution by making sure our distribution matching is accurate. Yet, doing so leads to unstable gradient-based training due to the gradient norm scaling as $O(k)$, and the variance of Monte Carlo gradient estimates growing as $O(k^2 m_{2(k-1)})$ for the k-th moment since $\left\|\nabla_\theta m_k(P_{\boldsymbol{\theta}}^{(\boldsymbol{a})})\right\| = \left\|\mathbb{E}\left[k(\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}))^{k-1}\boldsymbol{a}^\top J_{f_{\boldsymbol{\theta}}}(\boldsymbol{x})\right]\right\|$, with $J_{f_{\boldsymbol{\theta}}}(\boldsymbol{x}) \in \mathbb{R}^{K\times P}$ the Jacobian matrix–hereby creating an impractical situation where training stability and identifiability can not be achieved simultaneously.

4.2.2 Cumulative Density Functions are Impractical

The second family of tests acts upon the CDF. Because those tests require sorting, let's denote the kth order-statistics of N samples by $x_{k:N}$. Two highly standard tests are quadratic Empirical Density Function statistics with different weighting known as Cramér-von Mises [Cramér, 1928, Von Mises, 1981] and Anderson Darling [Anderson and Darling, 1952], and given by

$$T_w = N\int_{-\infty}^{\infty}\left(F_N(x) - F(x)\right)^2 w(x)dF(x)$$
$$w(x) = 1, \quad \text{(Cramér-von Mises)}$$
$$w(x) = [F(x)(1-F(x))]^{-1}, \quad \text{(Anderson-Darling)}$$

where $w(x)$ is a weighting function. Adding the U^2 statistics on top of Equation (Cramér-von Mises) recovers the

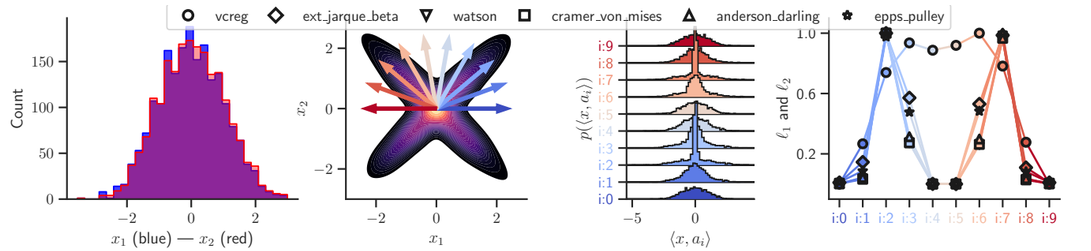


图 5. 构造的"X"形数据密度，其边缘分布为标准高斯分布，协方差矩阵为单位矩阵（左侧密度图）。沿半圆上的方向进行 $M = 10$ 次投影，得到 10 个一元分布；可使用第 4.2 节中任意统计量，将其与标准高斯分布（左侧）比较。合适的方向能够捕捉数据分布的退化现象，从而使统计量的值出现峰值。

即偏度与峰度，为

$$\text{JB}(\boldsymbol{u}) \triangleq \frac{N}{6}\left(\widehat{\text{skew}}(\boldsymbol{u})^2 + \left(\frac{\widehat{\text{kurt}}(\boldsymbol{u}) - 3}{2}\right)^2\right), \quad \text{(Jarque-Bera)}$$

$\widehat{\text{skew}}$ 其中 skew 是偏度，数据计算为 $\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^3\hat{\sigma}^3$，$\widehat{\text{kurt}}$ 是峰度 $\frac{1}{n}\sum_{i=1}^{n}(x_i-\hat{\mu})^4\hat{\sigma}^4$。通常 Jarque–Bera 检验任意均值、方差的高斯，因此仅考察三、四阶矩。我们要检验标准高斯，加入前两阶矩常用统计，得到扩展检验：

$$\text{EJB}(\boldsymbol{u}) \triangleq \frac{N\hat{\mu}(\boldsymbol{u})^2}{\hat{\sigma}(\boldsymbol{u})^2} + \frac{(N-1)\left(\hat{\sigma}(\boldsymbol{u})^2 - 1\right)^2}{2} + \text{JB}(\boldsymbol{u}). \quad \text{(Extended Jarque-Bera)}$$

（扩展 Jarque–Bera）检验相当于前四阶矩的匹配问题。这类矩匹配方法不仅用于统计检验，也被证明是学习数据参数模型和非参数模型的有力工具。

稳定性与可辨识性难题。矩检验虽强，却不适合 LeJEPA。分布 P的 k^{th} 记作 $m_k(P)$。满足 Carleman 条件 $\sum_{k=1}^{\infty} m_{2k}(Q)^{-1/(2k)} = \infty$ [Carleman， 1926], 的良好分布（如高斯），或有限区间分布 [Hausdorff，1923]，由矩唯一确定。但有限数量矩产生统计学熟知的不可辨识性，通常因此要求全部矩 [Lehmann 和 Romano， 2005]。

定理 3：K 阶矩的不充分性

在 $c_k > 0, \forall k$ 条件下最小化以下目标：

$$\sum_{k=1}^{K} c_k \left(m_k\left(P_{\boldsymbol{\theta}}^{(\boldsymbol{a})}\right) - m_k\left(Q^{(\boldsymbol{a})}\right)\right)^2,$$

对于有限 K，并不推出 $P(\boldsymbol{a})_{\boldsymbol{\theta}} = Q_{(\boldsymbol{a})}$。（证明见附录 B.11。）

因此定理 3 建议尽可能增大 K，确保分布匹配准确、排除坍塌捷径。但这使梯度训练不稳定：梯度范数按 $O(k)$ 增长，第 k阶矩的蒙特卡洛梯度方差按 $O(k^2 m_{2(k-1)})$增长，因为 $\left\|\nabla_\theta m_k(P_{\boldsymbol{\theta}}^{(\boldsymbol{a})})\right\| = \left\|\mathbb{E}\left[k(\boldsymbol{a}^\top f_{\boldsymbol{\theta}}(\boldsymbol{x}))^{k-1}\boldsymbol{a}^\top J_{f_{\boldsymbol{\theta}}}(\boldsymbol{x})\right]\right\|$，其中 $J_{f_{\boldsymbol{\theta}}}(\boldsymbol{x}) \in \mathbb{R}^{K\times P}$为雅可比矩阵。因此稳定性与可辨识性无法同时达到。

4.2.2 累积分布函数不实用

第二类检验作用于累积分布函数（CDF）。由于这些检验需要排序，我们将 N 个样本的 k^{th} 阶顺序统计量记为 $x_{k:N}$。两种十分常用的检验是采用不同权重的二次型经验分布函数统计量，分别称为 Cramér–von Mises [Cramér, 1928, Von Mises,1981] 和 Anderson–Darling [Anderson and Darling, 1952],，其表达式为

$$T_w = N\int_{-\infty}^{\infty}\left(F_N(x) - F(x)\right)^2 w(x)dF(x)$$
$$w(x) = 1, \quad \text{(Cramér-von Mises)}$$
$$w(x) = [F(x)(1-F(x))]^{-1}, \quad \text{(Anderson-Darling)}$$

其中 $w(x)$ 为权重函数。在式（Cramér–von Mises）基础上加入 U^2 统计量，可得到

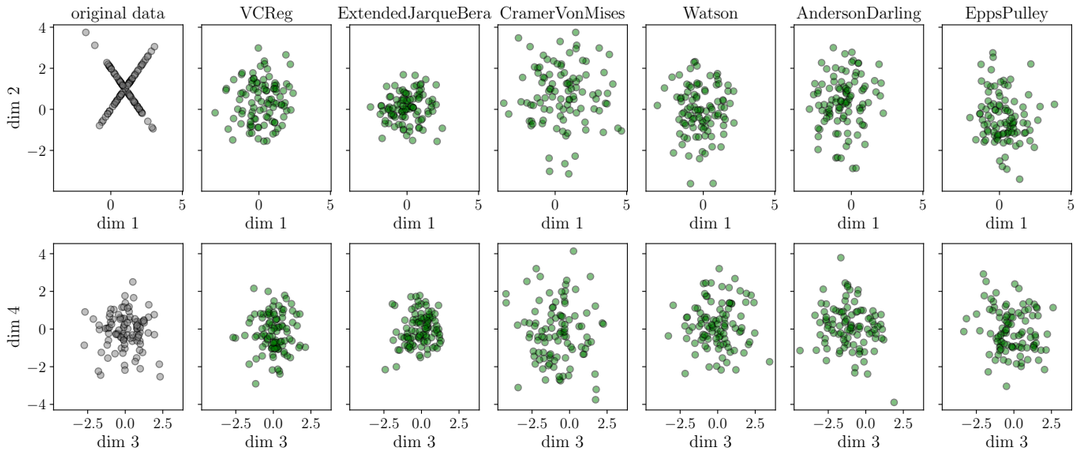


Figure 6. $N = 100$ samples are drawn from a 1024-dimensional standard Gaussian, and the first 2 coordinates are altered to produce the "X" distribution from Figure 5 (**left-most column**). For each statistic (**all other columns**), we perform gradient descent on the samples to minimize their value, at each iteration step with sample $M = 10$ random directions to evaluate SIGReg (recall def. 2). We obtain that albeit this is a high-dimensional distribution with limited number of samples, SIGReg is able to capture the degenerate subspace and adapt the data accordingly to match an isotropic Gaussian distribution. Additional figures with varying dimensions and number of 1d projections are provided in Figure 16.

Watson test [Watson, 1961]

$$U^2 = T_w - N\left(\bar{F} - \frac{1}{2}\right)^2. \qquad \text{(Watson)}$$

We do not consider the Kolmogorov-Smirnov test [Kolmogorov, 1933] as it employs the ℓ_∞-norm instead of the ℓ_2-norm hereby producing sparse gradients. Another common test is the Shapiro-Wilk test [Shapiro and Wilk, 1965] which we found to be unstable in practice–details are provided in Section E.

Lack of Scalability and Differentiability. CDF-based tests require sorting that have been highly optimized, e.g., with the $O(N \log(N))$ Quicksort algorithm [Hoare, 1962] but that nonetheless breaks the embarrassingly parallel nature of SGD–especially on multi-GPU [Tanasic et al., 2013, Maltenberger et al., 2022] due to synchronization requirements. Moreover, these tests involve non-differentiable operations (sorting and order statistics), making them unsuitable for gradient-based optimization without relaxations [Cuturi et al., 2019, Grover et al., 2019, Petersen et al., 2022]. While there exists intricate sketching solutions [Dunning and Ertl, 2019, Masson et al., 2019, Dunning, 2021], each of those solutions introduce numerous additional hyper-parameters–going against our first motivation for LeJEPA.

4.2.3 Characteristic Functions are Stable, Scalable and Identifiable

The third family of tests is concerned with Empirical Characteristic Functions (ECF) which are the Fourier transform of the density function. The Epps–Pulley test [Epps and Pulley, 1983] is one of the most popular test and simply compares in weighted ℓ_2-norm the ECF of the data against a target CF

$$EP = N \int_{-\infty}^{\infty} \left|\hat{\phi}_X(t) - \phi(t)\right|^2 w(t)dt. \qquad \text{(Epps–Pulley)}$$

The first crucial observation is that the ECF being defined as $\hat{\phi}_X(t) = \frac{1}{n}\sum_{j=1}^{n} e^{itX_j}$ is naturally differentiable and easily computed in distributed settings via efficient `all_reduce` operations, as the ECF is a simple average of complex exponentials. The weight function is typically Gaussian, such as $w(t) = e^{-t^2/\sigma^2}$ with σ commonly set to 1.

Other tests, e.g., based on the Entropy [Székely and Rizzo, 2005] are not considered here as they require numerous additional design choices for the univariate Entropy estimation [Silverman, 2018, Beirlant et al., 1997], e.g., using kernels [Joe, 1989], or M-estimators [Miller, 2003].

Epps-Pulley has bounded loss, gradient and curvature. We now consider the remaining two families of tests: moment-based and CF-based. First, recall that moments are polynomial in the data and with extreme growth rate

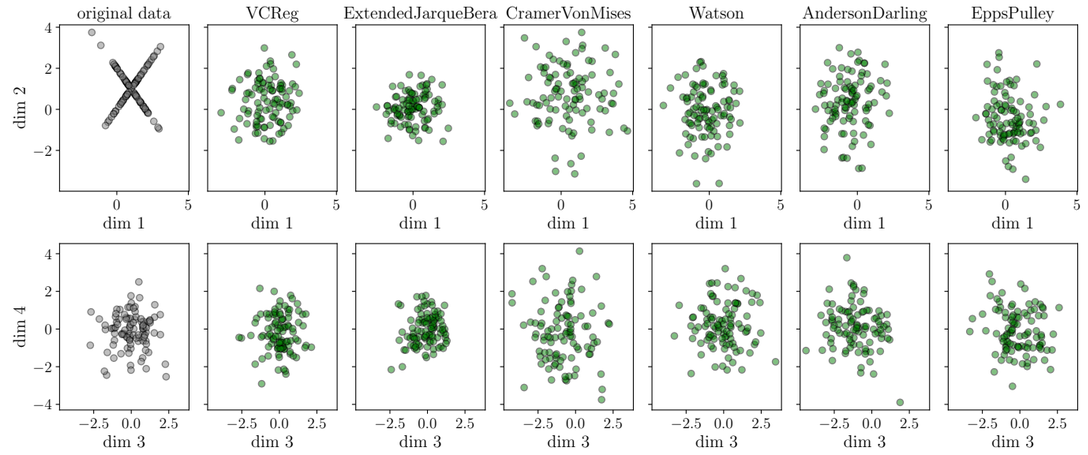


图 6. 从 1024 维标准高斯抽 $N = 100$ 样本，修改前两坐标产生图 5 的 X 分布（最左列）。各统计量（其余列）对样本梯度下降，每步采 $M = 10$ 随机方向计算 SIGReg（定义 2）。高维且样本有限，仍可发现退化子空间，调整数据匹配各向同性高斯。其他维度和一维投影数见图 16。

Watson 检验 [Watson，1961]

$$U^2 = T_w - N\left(\bar{F} - \frac{1}{2}\right)^2. \qquad \text{(Watson)}$$

我们不考虑 Kolmogorov–Smirnov 检验 [Kolmogorov，1933]，因为它采用 l_∞ 范数而非 l_2 范数，会产生稀疏梯度。另一种常用检验是 Shapiro–Wilk 检验 [Shapiro and Wilk,1965]，但我们发现它在实践中不稳定，详见附录 E。

缺少扩展性与可微性。CDF 检验需要排序，虽有高度优化的 $O(N \log(N))$快速排序 [Hoare， 1962]，同步要求仍破坏 SGD 的易并行性，尤其多 GPU [Tanasic 等，2013；Maltenberger 等， 2022] 。排序与顺序统计量不可微，不做松弛就不适合梯度优化 [Cuturi 等，2019；Grover 等，2019；Petersen 等， 2022]。复杂草图方案虽存在[Dunning 和 Ertl，2019；Masson 等，2019；Dunning，2021]，，却引入许多额外超参数，违背 LeJEPA 初衷。

4.2.3 特征函数稳定、可扩展且可识别

第三类检验关注经验特征函数（ECF），即密度函数的傅里叶变换。Epps–Pulley 检验 [Epps and Pulley，1983] 是最常用的检验之一，它直接用带权 l_2 范数比较数据的 ECF 与目标特征函数：

$$EP = N \int_{-\infty}^{\infty} \left|\hat{\phi}_X(t) - \phi(t)\right|^2 w(t)dt. \qquad \text{(Epps–Pulley)}$$

第一个关键观察是：经验特征函数（ECF）定义为 $\hat{\phi}_X(t) = \frac{1}{n}\sum_{j=1}^{n} e^{itX_j}$，天然可微，并且由于它只是复指数的简单平均，可通过高效的 all_reduce 操作轻松进行分布式计算。权重函数通常选用高斯函数，例如 $w(t) = e^{-t^2/\sigma^2}$，其中 σ 通常设为 1。

本文不考虑其他检验，例如基于熵的检验 [Székely and Rizzo，2005]，因为它们在估计一元分布的熵时，需要额外作出许多设计选择 [Silverman, 2018, Beirlant etal., 1997]，，例如采用核方法 [Joe, 1989]，或 M 估计量 [Miller, 2003]。

Epps–Pulley 的损失、梯度与曲率均有界。下面考虑剩余两类检验：基于矩和基于特征函数的检验。首先回顾：矩是数据的多项式，增长速率极高，

Algorithm 1. SIGReg with Epps-Pulley statistic with DDP support and $O(N)$ time and memory complexity. x is a (N, K) tensor, num_slices is $|\mathbb{A}|$ in def. 2, 'global_step' is used for sync. sampling across GPUs and can be omited for single-GPU training. An optimized implementation with caching is also provided in our official codebase, computation times provided in Table 6.

```
def SIGReg(x, global_step, num_slices=256):
    # slice sampling -- synced across devices --
    dev = dict(device=x.device)
    g = torch.Generator(**dev)
    g.manual_seed(global_step)
    proj_shape = (x.size(1), num_slices)
    A = torch.randn(proj_shape,  generator=g, **dev)
    A /= A.norm(p=2, dim=0)
    # -- Epps-Pulley stat. see Sec. 4.3 for alt. --
    # integration points
    t = torch.linspace(-5, 5, 17, **dev)
    # theoretical CF for N(0, 1) and Gauss. window
    exp_f = torch.exp(-0.5 * t**2)
    # empirical CF -- gathered across devices --
    x_t = (x @ A).unsqueeze(2) * t  # (N, M, T)
    ecf = (1j * x_t).exp().mean(0)
    ecf = all_reduce(ecf, op="AVG")
    # weighted L2 distance
    err = (ecf - exp_f).abs().square().mul(exp_f)
    N = x.size(0) * world_size
    T = torch.trapz(err, t, dim=1) * N
    return T
```

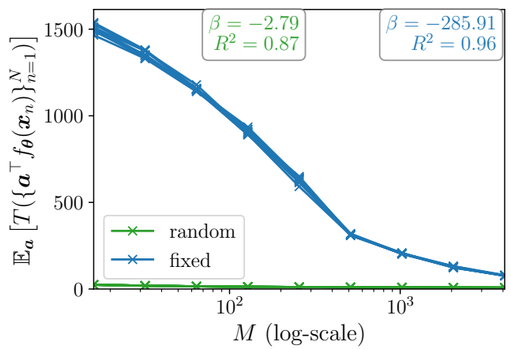


Figure 7. Expected directional statistic at the end of training (**y-axis**) for varying M (number of directions used at each training step, **x-axis**). The M directions are either resampled (**green**) or kept fixed (**blue**) at each training step. While for fixed directions we benefit from thm. 5 bound where increasing M reduces the overall expected loss, being able to resample at every step provides significant coverage boost for free.

for higher moment–assuming they even exist. Even for well-behaved distributions, raising values to a power of k can quickly lead to exploding gradients. This comes in sharp contrast with the ECF which is always bounded and with bounded gradients for any input distribution for the projected samples $z_i = \boldsymbol{a}^\top f_\theta(\boldsymbol{x}_n)$, $n = 1, \ldots, N$.

Theorem 4: Stability of Epps-Pulley Test

(Epps–Pulley) satisfies for samples $z_1, \ldots, z_N$

$$\left|\frac{\partial EP(\mathbf{a})}{\partial z_i}\right| \le \frac{4\sigma^2}{N}, \quad \left|\frac{\partial^2 EP(\mathbf{a})}{\partial z_i^2}\right| \le \frac{C\sqrt{\pi}\sigma^3}{2N},$$

with constant C, and bandwidth σ. (Proof in Section B.12.)

By the chain rule, thm. 4 directly gives $\|\nabla_\theta EP(\mathbf{a})\| \le \frac{4\sigma^2}{N}\sum_{i=1}^N \|\mathbf{a}^\top \nabla_\theta f_\theta(\mathbf{x}_i)\|$, providing stable gradients. The limitations of moment-based and CDF-based tests coupled with thm. 4 justifies our choice of the (Epps–Pulley): (i) DDP-friendly and scalable, (ii) uniformly bounded gradients and curvature regardless of input distribution, and (iii) hyper-parameter free implementation. Lastly, we highlight that *our implementation has a linear memory and computational complexity of $O(N)$, with N the minibatch size*. The implementation of SIGReg using that statistical test is provided in algorithm 1, along with computation times of the forward-backward pass in Table 6.

As a last step before introducing LeJEPA, we ought to study the requirements on the number of directions ($|\mathbb{A}|$) for (2) to be effective in high-dimension.

4.3 How SIGReg Beats the Curse of Dimensionality

This last section seeks to characterize how many slices in $\mathbb{A}$ one must sample for (SIGReg) to be an effective statistical test. That design is crucial if we hope for LeJEPA to successfully converge towards isotropic Gaussian embeddings.

Smoothness Beats the Curse of Dimensionality

Our first argument arguing for a favorable scaling of $|\mathbb{A}|$ with the embedding dimension K relies on the smoothness of $P_{\boldsymbol{\theta}}$ as measured by its Sobolev regularity α [Adams and Fournier, 2003]. We formalize below a bound on the directional test from Equation (3) over all possible directions $\boldsymbol{a}$ when the test statistic is minimized over $|\mathbb{A}| = M$ directions. While we provide bounds on the expected discrepancy over random directions $\boldsymbol{a}$ when the EP test is satisfied (equals zero) on a finite set of directions, the provided proof includes the case of moment-based and CDF-based tests as well.

算法 1. 支持 DDP、采用 Epps–Pulley 统计量的 SIGReg，其时间和内存复杂度均为 $O(N)$。x 是形状为 (N, K) 的张量，num_slices 对应定义 2 中的 $|\mathbb{A}|$；'global_step' 用于在各 GPU 间同步采样，单 GPU 训练时可以省略。官方代码库还提供了带缓存的优化实现，计算耗时见表 6。

```
def SIGReg(x, global_step, num_slices=256):# slice sampling --synced across devices --dev =dict(device=x.device)g =torch.Generator(**dev)g.manual_seed(global_step)proj_shape =(x.size(1), num_slices)A =torch.randn(proj_shape, generator=g, **dev)A/=A.norm(p=2, dim=0)# --Epps-Pulley stat. see Sec. 4.3 for alt. --#integrationpointst =torch.linspace(-5,5,17, **dev)#theoreticalCFforN(0,1)and Gauss.windowexp_f =torch.exp(-0.5 *t**2)#empiricalCF --gathered across devices --x_t =(x@A).unsqueeze(2) *t#(N, M, T)ecf =(1j *x_t).exp().mean(0)ecf =all_reduce(ecf, op="AVG")#weighted L2 distanceerr =(ecf -exp_f).abs().square().mul(exp_f)N =x.size(0) *world_sizeT =torch.trapz(err, t, dim=1) *Nreturn T
```

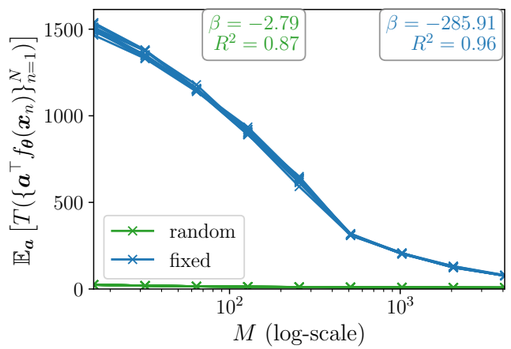


图 7. 训练结束时的期望方向统计量（纵轴）随 M（每个训练步骤使用的方向数，横轴）的变化。M 个方向在每个训练步骤中重新采样（绿色），或始终固定（蓝色）。对于固定方向，定理 5 的界说明，增大 M 能降低总体期望损失；而每步重新采样则能无须额外代价地显著提高覆盖范围。

对于更高阶矩——即使假设它们存在，也会出现问题。即便分布性质良好，将数值提升到 k 次幂也很容易引发梯度爆炸。相比之下，对投影样本 $z_i = \boldsymbol{a}^\top f_\theta(\boldsymbol{x}_n)$，$n = 1, \ldots, N$ 的任意输入分布，ECF 始终有界，其梯度也有界。

定理 4：Epps–Pulley 检验的稳定性

对于样本 $z_1, \ldots, z_N$，（Epps–Pulley）满足

$$\left|\frac{\partial EP(\mathbf{a})}{\partial z_i}\right| \le \frac{4\sigma^2}{N}, \quad \left|\frac{\partial^2 EP(\mathbf{a})}{\partial z_i^2}\right| \le \frac{C\sqrt{\pi}\sigma^3}{2N},$$

其中常数为 C，带宽为 σ。（证明见附录 B.12。）

链式法则和定理 4 直接给出 $\|\nabla_\theta EP(\mathrm{a})\| \le \frac{4\sigma^2}{N}\sum_{i=1}^N \|\mathrm{a}^\top \nabla_\theta f_\theta(\mathrm{x}_i)\|$，确保梯度稳定。矩、CDF 检验的局限与定理 4 支持选 Epps–Pulley：（i）适合 DDP、可扩展；（ii）不论输入分布，梯度与曲率一致有界；（iii）无需超参数。实现内存和计算为线性 $O(N)$，N 为小批量。使用该检验的 SIGReg 实现为：

见算法 1，前向与反向传播的计算耗时见表 6。

在介绍 LeJEPA 前的最后一步，我们需要研究在高维空间中使式（2）有效所需的方向数（$|\mathbb{A}|$）。

4.3 SIGReg 如何克服维数灾难

本节最后讨论：必须从 $\mathbb{A}$ 中采样多少个切片，才能使（SIGReg）成为有效的统计检验。如果希望 LeJEPA 成功收敛到各向同性高斯嵌入，这一设计至关重要。

光滑性克服维数灾难

第一论据用 $P_{\boldsymbol{\theta}}$ 的 Sobolev 正则性 α [Adams 和 Fournier， 2003]说明 $|\mathbb{A}|$随维度 K 有利扩展。下面给出在$|\mathbb{A}| = M$ 方向最小化检验时，式（3）对全部 $\boldsymbol{a}$ 方向的界。有限方向 EP 为零时，约束随机方向 $\boldsymbol{a}$ 期望差异；证明也包括矩与 CDF 检验。

Theorem 5: Unified Error Bounds

Let $p_{\boldsymbol{\theta}} \in H^{\alpha}(\mathbb{R}^K)$, $\boldsymbol{a} \sim \mathcal{U}(\mathcal{S}^{K-1})$, and (Epps–Pulley)= 0, i.e., $P_{\theta}^{(\mathbf{a})} = Q^{(\mathbf{a})}, \forall \boldsymbol{a} \in \mathbb{A}$, then

$$\mathbb{E}_{\boldsymbol{a}}\left[\int_{\mathbb{R}} \left|\varphi_a(t) - \varphi_{\mathcal{N}}(t)\right|^2 dt\right] \le C(K,\alpha)|\mathbb{A}|^{-2\alpha/(K-1)} \times \int_0^{\infty} \left\|\varphi_{\cdot}(r) - \varphi_{\mathcal{N}}(r)\right\|^2_{H^{\alpha}(\mathcal{S}^{K-1})} dr,$$

(Proof in Section B.10.)

As $|\mathbb{A}| \to \infty$, the bound decays as $|\mathbb{A}|^{-2\alpha/(K-1)}$, showing that $|\mathbb{A}| = O(K)$ directions suffice for ϵ-approximation when α is large. Some examples of embedding densities with varying α are provided in Figure 4. The following statement characterizes how the M directions actually constrain the entire space as a function of α. The constant $C(K,\alpha) = \frac{2^{2\alpha}\pi^{(K-1)/2}\Gamma\left(\alpha+\frac{K-1}{2}\right)}{(K-1)\Gamma(\alpha)\Gamma\left(\frac{K-1}{2}\right)}$ is visualized in Figure 15 (left) depicting how α and $|\mathbb{A}|$ interact. In words, we obtain that thanks to the natural smoothness of DN–either stemming from the architecture or the implicit and explicit regularizers used during training–applying SIGReg on $|\mathbb{A}|$ directions can be sufficient to tightly constrain the entire space. We note that considering the worst case over $\boldsymbol{a}$ or using low-discrepancy sequences for $\boldsymbol{a}$ does not impact the asymptotic bounds, details provided in Section D.

SGD Beats the Curse of Dimensionality

Our second argument leverages the iterative nature of DN training. Although we may use only $|\mathbb{A}|$ to be a few hundreds, the cumulative number of sampled directions grows linearly with training time. This resampling effect (illustrated in Figure 7, bottom) enables rapid convergence. Even small $|\mathbb{A}|$ achieves tight distributional matching compared to keeping the set $\mathbb{A}$ fixed throughout minibatches (recall thm. 5). Our experiments show that even with $|\mathbb{A}|$ as low as 16 can easily outperform a fixed set with $|\mathbb{A}|$ of order of thousands thanks to the compounding effect of resampling at each minibatch.

Empirical Validation on Synthetic Data

We conclude this section with a controlled experiment applying (SIGReg) with gradient-based training to produce isotropic embeddings. In this setup, we directly consider embeddings $\boldsymbol{Z}$ which we will differentiate and optimized to minimize (SIGReg). By directly optimizing the embeddings we are able to observe the impact of the loss without any possible constraint and regularization that would come from the architecture. We sample N i.i.d. samples $\boldsymbol{x}_n$ in a D-dimensional space. This sampling is based on an isotropic Gaussian distribution–but the first two dimensions are again set to the adversarial "X" shape. That is, among the D dimensions, only two must be transformed as all the other ones already obey the isotropic Gaussian target distribution. We then make the samples $\boldsymbol{x}_n$ differentiable and optimize then to minimize the value of the different statistical tests compute on M random directions. Those directions are resampled after each gradient step–which follows the procedure we will employ in LeJEPA. We present the results in Figure 6 demonstrating that even in challenging case, i.e., $D = 512$ and $M = 16$, SIGReg is able to detect the two degenerate dimensions and unfold them back to how they should look like under the target distribution.

Algorithm 2. LeJEPA implementation–works out-of-the-box on any dataset, with DDP, with any backbone, e.g., torchvision or timm. For non-ViT architectures (e.g., ResNet), set global_views = all_views. We use bs for the minibatch size, SIGReg is from algorithm 1.

```
def LeJEPA(global_views, all_views, lambd):
    """global_views and all_views are lists of
       tensors, lambd is a scalar"""

    # embedding of global views
    g_emb = forward(torch.cat(glob_views))
    # embedding of local views
    # if resnet: skip with a_emb=g_emb
    a_emb = forward(torch.cat(all_views))

    # LeJEPA loss
    centers = g_emb.view(-1, bs, K).mean(0)
    a_emb = a_emb.view(-1, bs, K)
    sim = (centers - a_emb).square().mean()
    sigreg = mean(SIGReg(emb, global_step) for emb
        in a_emb)
    return (1-lambd)*sim + lambd*sigreg
```

5 LeJEPA: Stable and Scalable Implementation

Having established that isotropic Gaussians are the optimal embedding distribution for foundation models (Section 3) and introduced SIGReg to achieve this distribution (def. 2), we now present the complete LeJEPA framework. We first evaluate candidate statistical tests (Sections 4.2.1 and 4.2.2) and identify characteristic function-based tests as optimal for gradient-based training (Section 4.2.3). The full LeJEPA implementation follows in Section 5.1.

5.1 LeJEPA: SIGReg + Prediction Loss

We now discuss the implementation of LeJEPA starting with SIGReg and followed by the prediction and total losses.

The SIGReg Loss. We chose (Epps–Pulley) for its provable boundedness (thm. 4) and its scalability. Its implementation follows exactly the equation except for the integrate which is estimated using a quadrature approximation. We

定理 5：统一误差界

令 $p_{\boldsymbol{\theta}} \in H^{\alpha}(\mathbb{R}^K)$、 $\boldsymbol{a} \sim \mathcal{U}(\mathcal{S}^{K-1})$ 和 (Epps–Pulley)= 0，即 $P(\mathrm{a})_{\theta} = Q(\mathrm{a}), \forall \boldsymbol{a} \in \mathbb{A}$，则

$$\mathbb{E}_{\boldsymbol{a}}\left[\int_{\mathbb{R}} \left|\varphi_a(t) - \varphi_{\mathcal{N}}(t)\right|^2 dt\right] \le C(K,\alpha)|\mathbb{A}|^{-2\alpha/(K-1)} \times \int_0^{\infty} \left\|\varphi_{\cdot}(r) - \varphi_{\mathcal{N}}(r)\right\|^2_{H^{\alpha}(\mathcal{S}^{K-1})} dr,$$

（证明见附录 B.10。）

当 $|\mathbb{A}| \to \infty$，界按 $|\mathbb{A}|^{-2\alpha/(K-1)}$衰减，说明 α 较大时，$|\mathbb{A}| = O(K)$个方向足以实现 ϵ近似。图 4 展示不同 α 密度。下述结论刻画 M 个方向如何随 α约束整体空间。常数$C(K,\alpha) = \frac{2^{2\alpha}\pi^{(K-1)/2}\Gamma\left(\alpha+\frac{K-1}{2}\right)}{(K-1)\Gamma(\alpha)\Gamma\left(\frac{K-1}{2}\right)}$见图 15 左，体现 α 、 $|\mathbb{A}|$的相互作用。深网络的自然平滑性（来自架构或隐式、显式正则化），使 $|\mathbb{A}|$个 SIGReg 方向即可紧约束全空间。对 $\boldsymbol{a}$ 取最坏情况，或为 $\boldsymbol{a}$ 使用低差异序列，不影响渐近界，见第 D 节。

SGD 克服维数灾难

第二论据利用深网络迭代训练。 $|\mathbb{A}|$ 虽仅几百，累计抽样方向随时间线性增加。重采效果（图 7 下）可快速收敛。即使小 $|\mathbb{A}|$，匹配也比每小批量保持 $\mathbb{A}$固定更紧（定理 5）。实验中，每批重采的累积效应，使 $|\mathbb{A}|$低至 16 也轻易超过固定数千 $|\mathbb{A}|$ 方向。

合成数据上的实验验证

以受控实验结束：梯度训练 SIGReg 产生各向同性嵌入。直接对 $\boldsymbol{Z}$ 求导、优化，最小化 SIGReg，以排除架构的约束和正则化，单独观察损失作用。在 D维抽取 N 个独立同分布 $\boldsymbol{x}_n$ 样本，来自各向同性高斯，但前

算法 2. LeJEPA 实现：可直接用于任意数据集、DDP 和任意骨干网络，例如 torchvision 或 timm。对于非 ViT 架构（例如 ResNet），设置 global_views = all_views。bs 表示小批量大小，SIGReg 来自算法 1。

```
def LeJEPA( global_views , all_views , lambd ) : """ global_vie

    # embedding of g l o b a l views g_emb  =forward ( t

    # LeJEPA loss centers  =g_emb . view ( -1 , bs , K) .
```

两个维度设为对抗性 X 形。 D 维中仅两维需变换，其余已符合各向同性高斯。让样本$\boldsymbol{x}_n$ 可微，最小化在 M 个随机M 方向上计算的各检验；每次梯度步后重采方向，与 LeJEPA 相同。图 6 显示，即便 $D = 512$、 $M = 16$这样的困难情况，SIGReg 仍检测两退化维并展开至目标分布应有形态。

5 LeJEPA：稳定且可扩展的实现

第 3 节确定各向同性高斯最优、定义 2 提出 SIGReg 后，介绍完整 LeJEPA。先评估候选检验（4.2.1、4.2.2），确定特征函数检验最适合梯度训练（4.2.3），完整实现见 5.1。

5.1 LeJEPA：SIGReg +预测损失

下面讨论 LeJEPA 的实现，先介绍 SIGReg，再介绍预测损失和总损失。

SIGReg 损失。我们选用（Epps–Pulley）检验，是因为它具有可证明的有界性（定理 4）和可扩展性。实现严格遵循公式，唯一的区别是积分通过求积近似来估计。我们

find that the simple trapezoidal quadrature rule is sufficient even with as few knots as 17, as ablated in Figure 20. In particular, we leverage the symmetry of the integrand to double the number of knots for free, see the official code. On the other hand, the use of minibatches introduces a bias vanishing at rate $O(1/N)$, as formalized below.

Theorem 6: Vanishing gradient bias

The expectation of (Epps–Pulley) satisfies

$$\mathbb{E}\left[\widehat{L}_n(\theta)\right] = L(\theta) + \frac{1}{N}\int_{\mathbb{R}} w_s(t)\left(1 - |\varphi_P(t)|^2\right)dt,$$

therefore both the loss and its derivative have a bias of order $O(1/n)$. (Proof in Section B.13.)

Hence, the gradients we obtain from using (Epps–Pulley) are biased by an explicit $O(1/N)$ term. We found this bias to be minimal and not a concern even for minibatches as small as 16. Unbiased alternatives include using U-statistic debiasing of $|\phi_\theta|^2$ or sample splitting, which we do not explore in this study. Our final implementation of the SIGReg term with Epps-Pulley statistic is provided in algorithm 1.

The Prediction Loss. To standardize notations, we adopt the DINO [Caron et al., 2021] setup of generating V_g global views and V_l local views, leading to a total of $V = V_g + V_l$ views. We set the first $1, \ldots, V_g$ indices of each $z_{n,v}$ as the global views. For the cases without local views, simply set $V_l = 0$. The prediction loss is then given by having all views predict the global views as

$$\mathcal{L}_{\text{pred}}(\{z_{n,v}\}_{v=1}^{V}) = \frac{1}{V_g}\sum_{v=1}^{V_g}\frac{1}{V}\sum_{v'=1}^{V}\|z_{n,v} - z_{n,v'}\|_2^2 \quad (5)$$

$$= \frac{1}{V}\sum_{v'=1}^{V}\left\|\frac{1}{V_g}\sum_{v=1}^{V_g} z_{n,v} - z_{n,v'}\right\|_2^2 \quad (6)$$

$$\triangleq \frac{1}{V}\sum_{v'=1}^{V}\left\|\mu_n - z_{n,v'}\right\|_2^2, \quad (7)$$

where we denote $\mu_n \triangleq \frac{1}{V_g}\sum_{v=1}^{V_g} z_{n,v}$, the Equation (5) to Equation (6) derivations are detailed in Section B.6.

LeJEPA Loss. The final total loss simply combines the above prediction loss along with SIGReg on each views as per

$$\mathcal{L}_{\text{LeJEPA}}(\{x_{n,v}\}_{n,v=1}^{B,V}) = \frac{\lambda}{V}\sum_{v=1}^{V}\text{SIGReg}(\{\{z_{n,v}\}_{n=1}^{B}\}) + \frac{1-\lambda}{B}\sum_{n=1}^{B}\mathcal{L}_{\text{pred}}^{(V_g)}(\{z_{n,v}\}_{v=1}^{V}). \quad \text{(LeJEPA)}$$

We present (LeJEPA)'s implementation in algorithm 2. Altogether, the entire implementation–besides the usual model definitions, optimizers, and data loaders–only takes a few dozens lines in PyTorch (algorithms 1 and 2). The absence of prototypes, stop-gradients, and teacher-student networks makes (LeJEPA) appealing as it only contains one hyperparameter, λ, balancing the trade-off between the prediction and isotropic Gaussian terms.

5.2 Relation to Prior Work

Prior to presenting our experiments (Section 6), we conclude by discussing how our proposed LeJEPA and SIGReg objective relate to existing frameworks in the literature.

While there is no existing solution employing such slicing and distribution matching for JEPAs, there exists similar pipelines for generative models and optimal transport. Notably, the Sliced Score Matching [Song et al., 2020] proposes to leverage univariate slicing of the space to ease the estimation of a density for generative models. In a similar vein, the sliced Wasserstein distance [Bonneel et al., 2015, Nguyen and Ho, 2023] uses such strategy to speed up and improve optimal transport. Furthermore, when the integral of the (Epps–Pulley) test is computed exactly, as opposed to our quadrature, each slice loss value recovers the kernel MMD [Sriperumbudur et al., 2010, Gretton et al., 2012, Chwialkowski et al., 2016] measuring the distance between two distributions–albeit with a quadratic complexity. Lastly, it is possible to recover some existing SSL frameworks in the limit by employing LeJEPA with a particular test–instead of the preferred (Epps–Pulley). For example, Setting $T(\{x_n\}_{n=1}^{B}) = \text{mean}(\{x_n\}_{n=1}^{B})^2 + (\text{std}(\{x_n\}_{n=1}^{B}) - 1)^2$ and using that T with SIGReg in LeJEPA recovers the VICReg SSL method in the limit of large number of slices. In fact, SIGReg will enforce in expectation that $\mathbb{E}[\mathbf{Z}] = \mathbf{0}$ and $\text{Cov}(\mathbf{Z}) = \mathbf{I}_d$, where $\mathbf{I}_d$ denotes the $d \times d$ identity matrix–derivations provided in Section B.14. And since our invariance term is simply the ℓ_2 distance between the views' embeddings, LeJEPA recovers VICReg for this degenerate statistical test. Based on thm. 3, we however strongly advocate against such a setting as it would lead to shortcut solutions–a phenomenon already observed in VICReg.

6 LeJEPA: Empirical Validation

We now use the LeJEPA implementation described in Section 5.1 to demonstrate its effectiveness through comprehensive experiments. We show that LeJEPA: (i) trains reliably across diverse architectures and datasets (Section 6.1), (ii) provides an informative training loss for model selection (Section 6.2), (iii) outperforms frontier vision models on small-scale in-domain pretraining (Section 6.3), (iv) scales successfully to nearly 1 billion parameters on ImageNet-1k (Section 6.4), and (v) learns rich

我们发现，即使节点数少至 17，简单的梯形求积法也已足够，相关消融实验见图 20。具体而言，我们利用被积函数的对称性，无须额外代价即可使节点数翻倍，详见官方代码。另一方面，使用小批量会引入偏差，该偏差以 $O(1/N)$ 的速率消失，下面给出形式化结论。

定理 6：趋于消失的梯度偏差

（Epps–Pulley）的期望满足

$$\mathbb{E}\left[\widehat{L}_n(\theta)\right] = L(\theta) + \frac{1}{N}\int_{\mathbb{R}} w_s(t)\left(1 - |\varphi_P(t)|^2\right)dt,$$

因此，损失及其导数的偏差均为 $O(1/n)$ 阶。（证明见附录 B.13。）

Epps–Pulley 梯度含显式 $O(1/N)$偏差。偏差极小，批量低至 16 也无问题。可用 $|\phi_\theta|^2$ 的 U 统计去偏或样本拆分取得无偏，但本文不探索。最终实现见算法 1。

预测损失。采用 DINO [Caron 等， 2021] 设定统一符号： V_g 全局视图、 V_l 局部视图，共 $V = V_g + V_l$。每个 $z_{n,v}$ 前 $1, \ldots, V_g$ 索引为全局；无局部时设 $V_l = 0$。让全部视图预测全局，损失为：

$$\mathcal{L}_{\text{pred}}(\{z_{n,v}\}_{v=1}^{V}) = \frac{1}{V_g}\sum_{v=1}^{V_g}\frac{1}{V}\sum_{v'=1}^{V}\|z_{n,v} - z_{n,v'}\|_2^2 \quad (5)$$

$$= \frac{1}{V}\sum_{v'=1}^{V}\left\|\frac{1}{V_g}\sum_{v=1}^{V_g} z_{n,v} - z_{n,v'}\right\|_2^2 \quad (6)$$

$$\triangleq \frac{1}{V}\sum_{v'=1}^{V}\left\|\mu_n - z_{n,v'}\right\|_2^2, \quad (7)$$

这里记 $\mu_n \triangleq \frac{1}{V_g}\sum_{v=1}^{V_g} z_{n,v}$，从式（5）到式（6）的推导详见附录 B.6。

LeJEPA 损失。最终总损失只需将上述预测损失与各视图上的 SIGReg 相结合：

$$\mathcal{L}_{\text{LeJEPA}}(\{x_{n,v}\}_{n,v=1}^{B,V}) = \frac{\lambda}{V}\sum_{v=1}^{V}\text{SIGReg}(\{\{z_{n,v}\}_{n=1}^{B}\}) + \frac{1-\lambda}{B}\sum_{n=1}^{B}\mathcal{L}_{\text{pred}}^{(V_g)}(\{z_{n,v}\}_{v=1}^{V}). \quad \text{(LeJEPA)}$$

算法 2 给出实现。除常规模型、优化器、加载器外，全部只需几十行 PyTorch（算法 1、2）。无需原型、停止梯度、教师–学生，仅超参数 λ平衡预测与各向同性高斯，因而很有吸引力。

5.2 与先前工作的关系

在介绍实验（第 6 节）之前，我们最后讨论所提出的 LeJEPA 和 SIGReg 目标与文献中现有框架之间的关系。

JEPA 尚无此类切片与分布匹配方案，但生成模型和最优传输已有类似流程。切片分数匹配 [Song 等， 2020] 通过一维切片简化生成密度估计；切片 Wasserstein 距离 [Bonneel 等， 2015；Nguyen 和 Ho， 2023]类似地加速并改善最优传输。若 Epps–Pulley 积分精确计算，而非采用我们的求积，每切片损失等价于测量分布距离的核 MMD [Sriperumbudur 等， 2010；Gretton 等， 2012；Chwialkowski 等， 2016]，但复杂度为二次方。LeJEPA 改用特定检验（而非推荐的 Epps–Pulley），可在极限恢复某些 SSL 框架。设 $T(\{x_n\}_{n=1}^{B}) = \text{mean}(\{x_n\}_{n=1}^{B})^2 + (\text{std}(\{x_n\}_{n=1}^{B}) - 1)^2$，把该 T 用于 SIGReg，大量切片极限下恢复 VICReg。SIGReg 将在期望中约束 $\mathbb{E}[\text{Z}] = \mathbf{0}$ 和 $\text{Cov(Z)} = \text{I}_d$，$\text{I}_d$ 为 $d \times d$ 单位矩阵（推导 B.14）。不变项只是视图嵌入的 l_2 距离，因此退化检验恢复 VICReg。但基于定理 3，我们强烈反对这一设置，因为会产生捷径，VICReg 中已观察到该现象。

6 LeJEPA：实验验证

采用 5.1 实现全面实验，展示：（i）跨架构、数据可靠训练（6.1）；（ii）损失提供模型选择信息（6.2）；（iii）小规模域内预训练超过先进模型（6.3）；（iv）IN1K 成功扩展到近十亿参数（6.4）；（v）学到丰富的

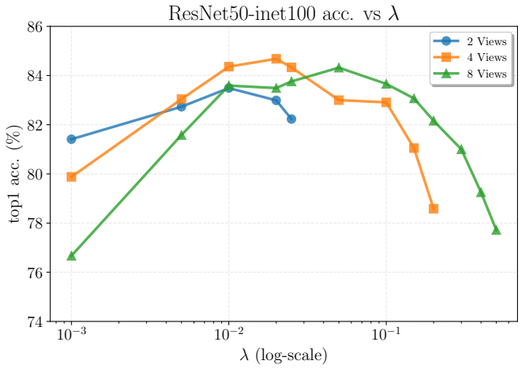


Figure 8. Inet100 with 400 pretraining epochs and resnet50 backbone. We depict linear probe performances as a function of λ and the number of views V (recall (LeJEPA)). We observe that performances are stable over λ–with **peak performance obtain by slightly adjust λ proportionally to the number of views**. The corresponding performance values are provided in Table 7.

semantic segmentation features without explicit supervision.

6.1 LeJEPA's Stability Across Hyper-Parameters and Architectures

We now demonstrate LeJEPA's stability across hyperparameters, architectures, and experimental setups. Additional cross-domain stability results are presented in Section 6.3.

Stability across standard hyperparameters. We begin by evaluating LeJEPA on ImageNet-100 and ImageNet-1K. On ImageNet-100, we train a ResNet-50 and vary the number of views and the loss weighting λ (Figure 8). Performance remains stable across both dimensions, leading us to recommend $\lambda = 0.05$ as a robust default. On ImageNet-1K, we train a ViT-Large/14 and explore batch size, as well as the number of global (V_g) and local (V_l) views (Table 1b). We find that the configuration commonly used in prior work ($V_g = 2, V_l = 8$) transfers well to LeJEPA. Notably, LeJEPA achieves competitive performance with batch sizes as small as 128 on ImageNet-1K (Table 1c), suggesting reduced memory requirements compared to existing methods. *We thus recommend to use $\lambda = 0.05$, $V_g = 2$, $V_l = 8$, and batch size ≥ 128 as starting points.*

Stability across Epps-Pulley hyperparameters. We next examine hyperparameters specific to LeJEPA: the number of slices $|\mathcal{A}|$ in SIGReg, the integration domain for the Epps-Pulley test (Epps–Pulley), and the number of quadrature points for numerical integration. Table 1a shows ablations on ImageNet-1K with ViT-Large/14. Both the integration domain and number of quadrature points have negligible impact on performance. This is expected: since the characteristic function is accurate at zero, the moments of the distribution are well-characterized even with a modest integration range. The number of slices $|\mathcal{A}|$ has a modest effect—while more slices slightly improve performance, even 512 slices yield competitive results. *We thus recommend to use 17 integration points, an integration domain of $[-5, 5]$, and 1024 slices as starting points.*

Table 1. ViT/Large-14, on inet1k pretraining for 100 epochs and evaluated with frozen backbone linear probing (top1 accuracy, %).**LeJEPA's performance is stable across all its hyperparameters** and while some may slightly improve performance, e.g., the number of slices $|\mathbb{A}|$ and the projector sizes, none of the choices lead to a catastrophic collapse.

(a) (Epps–Pulley) **parameters**

integration	num_slices	config/bstat_n_points		
		5	17	41
$[-1, 1]$	512	71.82	72.13	72.04
	2048	72.88	72.30	72.69
$[-3, 3]$	512	73.95	74.16	74.04
	2048	75.02	74.68	74.77
$[-5, 5]$	512	73.71	74.21	74.15
	2048	74.50	74.80	74.77

(b) Number of local/global views

# global_views (V_g) # views ($V = V_g + V_l$)	1	2	4
4	53.06	72.26	–
6	58.65	73.07	73.68
8	64.46	74.24	73.94
10	68.97	74.06	75.08

(c) Mini-batch size

batch_size	128	256	512	1024
	72.20	74.15	74.72	74.07

(d) Embedding/Projector dim.

num_slices	1024		4096	
emb. dim. proj. dim.	512	2048	512	2048
64	75.29	75.32	75.50	75.65
128	74.77	75.09	75.26	75.47
256	74.56	74.66	75.08	75.02
512	73.94	74.11	74.81	74.65
1024	73.65	73.94	74.71	74.79

(e) Register tokens

reg_tokens num_slices	0	1	2	4	8
1024	75.14	75.18	75.08	75.34	75.23
4096	75.61	75.58	75.67	75.63	75.84

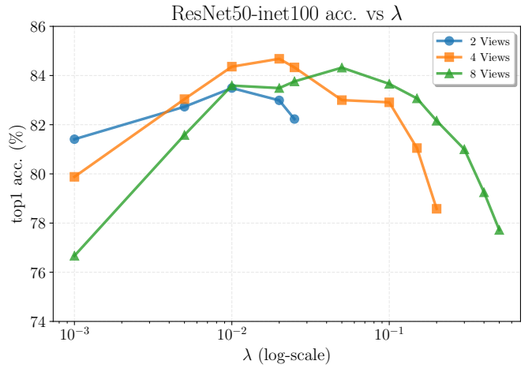


图 8. 在 Inet100 上使用 resnet50 骨干网络预训练 400 个轮次。我们绘制线性探测性能随 λ 和视图数 V 的变化（参见（LeJEPA））。性能在不同 λ 下保持稳定；按视图数比例对 λ 略作调整即可达到峰值性能。对应性能数值见表 7。

无需显式监督的语义分割特征。

6.1 LeJEPA 在不同超参数和架构下的稳定性

下面展示 LeJEPA 在不同超参数、架构和实验设置下的稳定性。更多跨领域稳定性结果见第 6.3 节。

常规超参数稳定性。先评估 ImageNet-100/1K。IN100 的 ResNet-50 改变视图数和损失权重 λ （图 8），性能均稳定，推荐 $\lambda = 0.05$。IN1K 的 ViT-Large/14 探索批量及全局V_g、局部V_l视图数（表 1b）；此前常用配置 $V_g = 2, V_l = 8$可直接迁移。批量仅 128 也有竞争力（表 1c），说明内存需求降低。因此建议从 $\lambda = 0.05$、$V_g = 2$、 $V_l = 8$及批量 ≥ 128 开始。

Epps–Pulley 超参数稳定性。考察 SIGReg 切片数 $|\mathcal{A}|$ 、积分域及求积点数。表 1a 在 IN1K ViT-Large/14 消融；积分域、点数影响可忽略。因为特征函数在零处精确，

表 1. ViT/Large-14 在 inet1k 上预训练 100 个轮次，随后冻结骨干网络进行线性探测评估（top-1 准确率，%）。LeJEPA 的性能对所有超参数均保持稳定。虽然某些选择可略微提高性能，例如切片数 $|\mathbb{A}|$ 和投影器尺寸，但没有任何选择导致灾难性的坍塌。

（a）（Epps–Pulley）参数

积分	切片数_	config/bstat_n_points		
		5	17	41
$[-1, 1]$	512	71.82	72.13	72.04
	2048	72.88	72.30	72.69
$[-3, 3]$	512	73.95	74.16	74.04
	2048	75.02	74.68	74.77
$[-5, 5]$	512	73.71	74.21	74.15
	2048	74.50	74.80	74.77

（b）局部／全局视图数

全局_视图数（V_g） 视图数（$V = V_g + V_l$）	1	2	4
4	53.06	72.26	–
6	58.65	73.07	73.68
8	64.46	74.24	73.94
10	68.97	74.06	75.08

（c）小批量大小

批量 _size	128	256	512	1024
	72.20	74.15	74.72	74.07

（d）嵌入／投影器维度

num_切片	1024		4096	
嵌入维度 投影器维度	512	2048	512	2048
64	75.29	75.32	75.50	75.65
128	74.77	75.09	75.26	75.47
256	74.56	74.66	75.08	75.02
512	73.94	74.11	74.81	74.65
1024	73.65	73.94	74.71	74.79

（e）寄存器词元

寄存器_词元 切片数_	0	1	2	4	8
1024	75.14	75.18	75.08	75.34	75.23
4096	75.61	75.58	75.67	75.63	75.84

即使积分区间不大，分布的各阶矩也能得到良好刻画。切片数 $|\mathcal{A}|$ 的影响较小：更多切片会略微改善性能，但即便只有 512 个切片也能取得有竞争力的结果。因此，我们建议以 17 个积分点、积分区域 $[-5, 5]$ 和 1024 个切片作为起始设置。

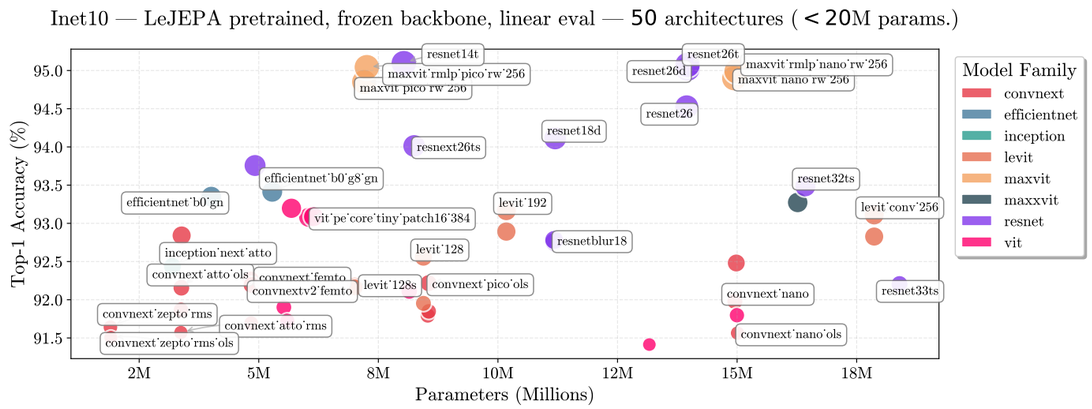


Figure 9. INet10 pretraining and frozen backbone linear evaluation across 50 timm models using LeJEPA out of the box. We cross-validate the learning rate and weight-decay. While there is a small variation between the best and worst performing model, we clearly see that **across** 50 **models spanning** 8 **families, LeJEPA is able to produce non-trivial representations able to solve the downstream task at SOTA levels**.

Stability across architectures. A key advantage of LeJEPA over recent methods (e.g., IJEPA, DINOv2) is its architecture-agnostic design. While most modern self-supervised methods are tailored to Vision Transformers, LeJEPA works across diverse architecture families without modification. To validate this claim, we pretrain approximately 50 architectures from 8 different families on ImageNet-10, selecting all models in the timm library with fewer than 20M parameters. All models are able to learn high-quality representations reaching between 91.5% to 95% top 1 accuracy with frozen backbone linear probing. It seems that models performing well in supervised learning setups are also the ones to favor for LeJEPA, such as resnets and ViTs. *We thus recommend to use standard architectures such as ResNets and ViTs over specialized models like EfficientNet as stating point.*

Removal of popular heuristics. In addition to providing reliable performance across models and datasets, LeJEPA's provable construction enables us to *remove* many heuristics traditionally used to prevent collapse. First, prior work has shown both empirically and theoretically that predictors in image JEPA (without asymmetric information) and teacher-student architectures serve primarily to prevent collapse [Grill et al., 2020, Jing et al., 2021, Tian et al., 2021, Caron et al., 2021, Chen et al., 2021]. Removing these components produces *collapsed* encoders, i.e., with performances at chance-level. Thanks to LeJEPA's SIGReg loss, we can remove both the predictor and teacher-student architecture without suffering from collapse, as shown in Table 4. While a teacher-student configuration does provide a small performance boost for ViT models—consistent with observations in supervised learning via Stochastic Weight Averaging [Izmailov et al., 2019]—it is not necessary to prevent collapse. In our setup, we apply SWA on the encoder producing μ in Equation (6). Second, recent work demonstrated that register tokens are needed to prevent training instabilities in vision models [Oquab et al., 2023, Siméoni et al., 2025, Darcet et al., 2023]. We show in Table 1 that such instabilities likely stem from poorly conditioned training objectives. In contrast, LeJEPA *does not* require register tokens and achieves stable performance with or without them. *We thus recommend training without a predictor or register tokens, and optionally applying SWA with ViTs for a possible performance gain.*

Experiment Details 1

We strive for **simplicity** and thus adopt a unified pretraining pipeline. The following parameters apply to *all* experiments and figures unless stated otherwise in the corresponding caption and come from Section 6.1:

- LeJEPA's implementation is given in algorithm 2 with hyperparameter λ
- All backbones are from timm and all optimizers/schedulers are from PyTorch without modifications
- We employ eight views ($V = 8$) containing two global views ($V_g = 2$) with resolution 224x224 and 96x96 for the local views
- AdamW optimizer with lr $\in \{5e-3, 5e-4\}$ and wd $\in \{1e-1, 1e-2, 1e-5\}$–no scheduler on weight-decay, standard linear warm-up cosine-annealing for lr

6.2 LeJEPA's Training Loss is Informative of Downstream Performance

A major challenge in SSL pretraining is the lack of reliable signals conveying the quality of the learned representation. As a result, it is common to monitor a supervised

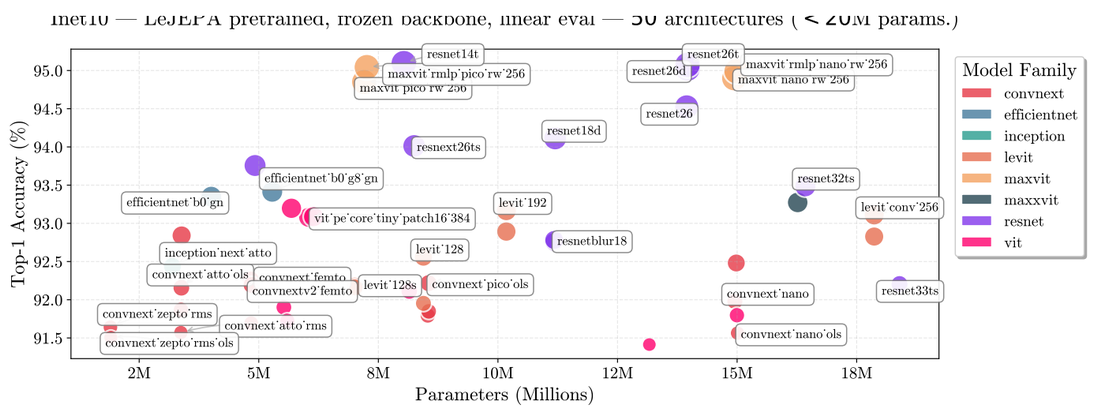


图 9. 在 INet10 上使用开箱即用的 LeJEPA，对 50 个 timm 模型进行预训练与冻结骨干网络后的线性评估。我们通过交叉验证选择学习率和权重衰减。尽管最佳模型与最差模型之间存在小幅差异，但可以清楚地看到：对于涵盖 8 个模型家族的 50 个模型，LeJEPA 都能学习出具有实际意义的表征，并以最先进的水平解决下游任务。

架构稳定性。相比 IJEPA、DINOv2，LeJEPA 不依赖特定架构。现代自监督多为 ViT 设计，LeJEPA 无需修改即可跨架构。IN10 上选 timm 中少于 20M 参数的约 50 种、8 个家族模型，全部冻结线性探测达到 91.5%–95% top-1。监督学习表现好的 ResNet、ViT 也较适合 LeJEPA，因此推荐标准 ResNet、ViT，而非 EfficientNet 等专用模型作为起点。

去除常用启发式。除跨模型、数据可靠外，可证明构造还允许移除传统防坍塌设计。此前实验与理论表明，无信息非对称的图像 JEPA 预测器，以及教师–学生，主要用于防坍塌 [Grill 等，2020；Jing 等，2021；Tian 等，2021；Caron 等，2021；Chen 等，2021]。移除通常会使编码器坍塌至随机水平。SIGReg 使预测器、教师–学生都可移除而不坍塌（表 4）。教师–学生对 ViT 有小幅提升，与监督学习中随机权重平均 [Izmailov 等，2019]，但它并非防坍塌所必需。我们对式（6）生成 μ 的编码器应用 SWA。其次，近期工作称视觉模型需寄存器令牌防训练不稳定 [Oquab 等，2023；Siméoni 等，2025；Darcet 等，2023]。表 1 显示，不稳定可能源于条件不良的训练目标。LeJEPA 无需寄存器，有无均稳定。因此推荐不使用预测器、寄存器，ViT 可选 SWA 以小幅提升。

实验细节 1

我们追求简洁，因此采用统一的预训练流程。除对应图注另有说明外，下列参数适用于所有实验和图，取自第 6.1 节：

- LeJEPA 的实现见算法 2，超参数为 λ。
- 所有骨干网络均来自 timm，所有优化器和调度器均来自 PyTorch，未作修改。
- 我们使用八个视图（$V = 8$），包括两个分辨率为 224 ×224 的全局视图（$V_g = 2$），局部视图分辨率为 96×96。
- AdamW 优化器，lr $\in \{5e-3, 5e-4\}$，wd $\in \{1e-1, 1e-2, 1e-5\}$；权重衰减不使用调度器，学习率采用标准的线性预热与余弦退火。

6.2 LeJEPA 的训练损失能反映下游性能

自监督预训练面临的一大难题，是缺少反映所学表征质量的可靠信号。因此，通常会监测有监督的

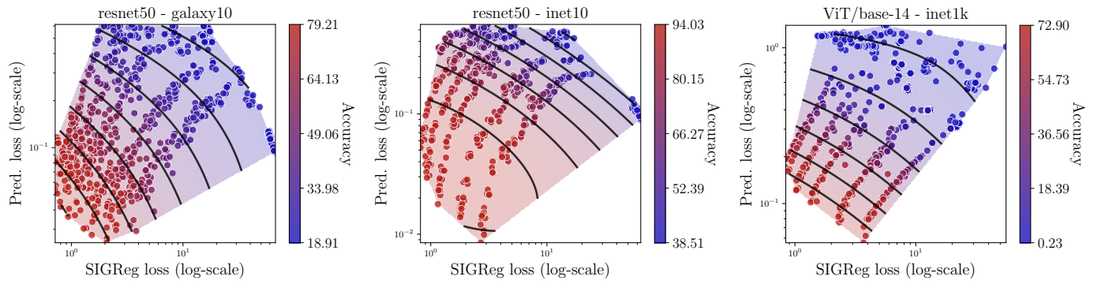


Figure 10. (SIGReg, prediction loss) 2d-plane with downstream task accuracy shown with colors from **blue** (low) to **red** (high). We clearly observe that within this plane, **there exists trade-off fronts between the two terms of LeJEPA producing similar downstream performance** corresponding to different values of λ. Yet, those fronts are linear and pointed towards the lower left corner, i.e., LeJEPA's training loss informs of downstream test performance across models and datasets (**columns**). Additional models and datasets provided in Figure 21.

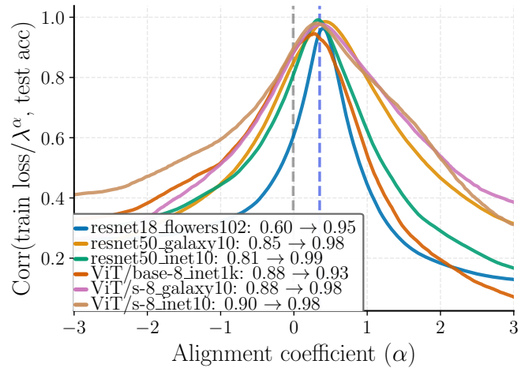


Figure 11. Spearman correlation (**y-axis)** between LeJEPA's training loss and downstream accuracy on the dataset's classification task with a frozen backbone and linear evaluation. The **x-axis** varies α in Equation (8) following our scaling law of the loss w.r.t. λ. Using $\alpha = 0$ recovers the plain training loss. We clearly observe a very high correlation already for $\alpha = 0$, which further increases up to 99% for $\alpha = 0.4$. The entire set of points is obtained across numerous hyper-parameters such as learning rate, weight decay, number of epochs, λ–demonstrating how **LeJEPA's training loss is strongly predictive of downstream performance** which can be used for label-free cross-validation.

downstream task performance, sometimes supplemented with unsupervised embedding statistics [Agrawal et al., 2022, Garrido et al., 2023, Thilak et al., 2023]. This process is highly limiting since it requires labeled data that is costly and overly specialized. This is further exacerbated in the latest JEPA models where training losses exhibit low correlation with downstream performance–and may not even decrease monotonically during training.

In contrast, we find that LeJEPA's training loss behaves much more favorably–providing us with a meaningful signal on model quality. First, we provide in Figure 10, the 2D plane spanned by the SIGReg and prediction losses where a clear trend with downstream task accuracy can be observed. More strikingly, the combined training loss (LeJEPA) with mixing coefficient λ exhibits very high Spearman correlation [Spearman, 1961], denoted as ρ_s, of about 85% with downstream accuracy–which is considered a strong signal. This strong relationship holds across datasets and architectures. As a result, a lower LeJEPA training loss reliably indicates a better downstream performance.

We can further improve this correlation through a simple scaling law based upon the trade-off weighting hyperparameter λ

$$C^{(\alpha)} = \rho_s\left(\frac{\text{train_loss}}{\lambda^\alpha}, \text{test_accuracy}\right). \tag{8}$$

By setting $\alpha \approx 0.4$, LeJEPA's training loss is able to achieve nearly 99% correlation with downstream performance across multiple datasets and models. We depict the changes in $C^{(\alpha)}$ as a function of α on multiple datasets and models in Figure 11, as well as the training LeJEPA loss against downstream performance in Figure 19. **The strong alignment between LeJEPA's training loss and model quality enables label-free SSL model selection and cross-validation**.

6.3 In-Domain LeJEPA Outperforms Frontier Model Transfer Learning

A key promise of self-supervised learning is to learn universal representations that generalize across tasks and domains. However, current frontier foundation models (e.g., DINOv2/v3, IJEPA) are pretrained on natural images forcing practitioners in specialized domains to collect large amount of labels for supervised finetuning. In fact, most frontier models can not be trained directly on those domains as the number of samples may be small and searching again for the hyper-parameters would be cum-

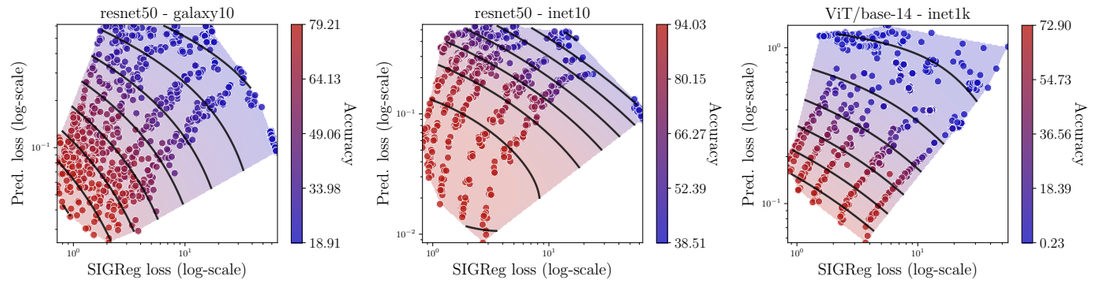


图 10. SIGReg 与预测损失的 2d平面，下游准确率由蓝（低）至红（高）。两损失存在对应不同 λ、下游性能相近的权衡前沿。前沿线性且指向左下，即训练损失可反映跨模型、数据的测试性能（列）。其他模型、数据见图 21。

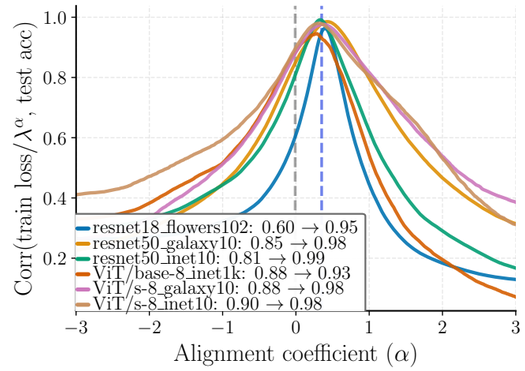


图 11. LeJEPA 训练损失与冻结线性分类准确率的 Spearman 相关（纵轴）。横轴按损失相对 λ的缩放规律，改变式（8）的 α； $\alpha = 0$ 恢复原训练损失。$\alpha = 0$时相关已很高， $\alpha = 0.4$ 时升至 99%。所有点跨学习率、权重衰减、轮数、 λ等多种超参数，说明损失能强预测下游性能，用于无标签交叉验证。

下游准确率呈清晰趋势。混合系数 λ 的 LeJEPA 总损失，与下游准确率 Spearman [Spearman， 1961], 相关 ρ_s约 85%，是强信号，跨数据、架构均成立。损失越低，可靠指示性能越好。

通过基于权衡系数超参数 λ 的简单缩放规律，可进一步提高这种相关性。

$$C^{(\alpha)} = \rho_s\left(\frac{\text{train_loss}}{\lambda^\alpha}, \text{test_accuracy}\right). \tag{8}$$

设 $\alpha \approx 0.4$，跨模型、数据的损失相关可近 99%。图 11 为 $C^{(\alpha)}$ 随 α 变化，图 19 为损失对下游性能。强一致性支持无需标签的 SSL 模型选择、交叉验证。

下游任务性能，有时还辅以无监督嵌入统计量 [Agrawal et al., 2022, Garrido et al., 2023, Thilak et al., 2023]。这一过程有很大局限，因为它需要成本高昂且过于专门化的标注数据。在最新的 JEPA 模型中，这一问题进一步加剧：训练损失与下游性能的相关性很低，甚至可能无法在训练过程中单调下降。

相比之下，我们发现 LeJEPA 的训练损失表现更好，能够提供关于模型质量的有意义信号。首先，图 10 给出由 SIGReg 损失和预测损失张成的二维平面，

6.3 领域内 LeJEPA 优于前沿模型的迁移学习

自监督希望通用表征跨任务、领域泛化。但 DINOv2/v3、IJEPA 等在自然图像预训练，专用领域需收集大量标签监督微调。大多也不能直接在这些领域训练，因为样本少，重新搜索超参数又很繁

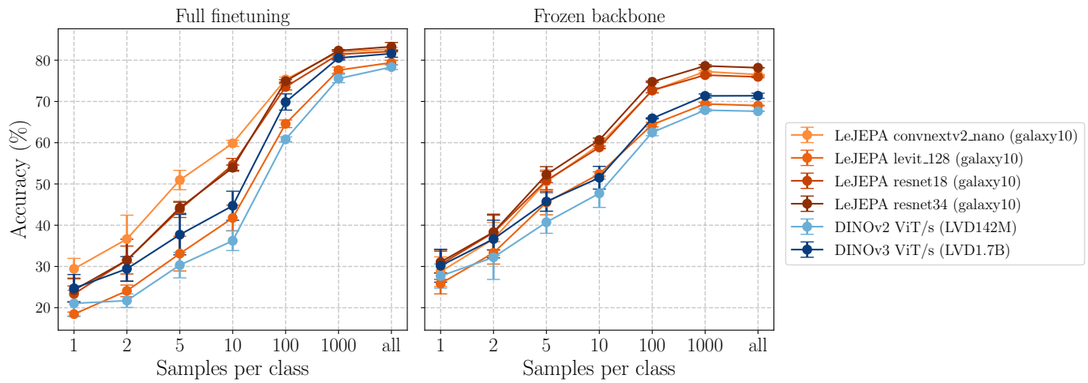


Figure 12. Small architecture in-domain (Galaxy10) LeJEPA pretraining with linear probe evaluation using frozen backbone or full finetuning (**columns**) and with varying number of samples per class (**x-axis**). We compare against state-of-the-art foundation models (DINOv2/v3, IJEPA) over 3 different random seeds. We observe that **LeJEPA enables in-domain pretraining out of the box across architectures and able to outperform frontier foundation models**. Corresponding numbers are provided in Table 3.

Table 2. Few-shot classification accuracy (percentages) on 8 datasets spanning textures, objects, and fine-grained categories. **Our LeJEPA achieves superior performance on fine-grained tasks (DTD, flowers102, food101) while requiring only 100 pretraining epochs compared to I-JEPA's 300 epochs—a 3× reduction in training time and computational resources without sacrificing downstream task performance.** This efficiency gain is particularly valuable for practical applications where training budget is limited. Bold indicates best performance within the IN-1K comparison group, all numbers are percentages.

					Dataset								
shots	model	params	pretrain	epochs	DTD	aircr.	cars	cifar10	cifar100	flowers102	food	pets	avg.
1	LeJEPA ViT-L	304M	IN-1K	100	**33.21**	9.37	3.40	51.65	27.01	48.53	17.14	46.11	29.55
	LeJEPA ConvNeXtV2-H	660M	IN-1K	100	32.15	8.07	4.28	50.95	31.48	**48.74**	**17.95**	58.98	31.58
	I-JEPA ViT-H	632M	IN-1K	300	27.71	9.86	4.33	**56.52**	30.58	44.69	14.53	53.38	30.20
	I-JEPA ViT-H + STOP	632M	IN-1K	300	26.60	**11.18**	**4.75**	56.27	**35.20**	47.17	15.75	**59.47**	32.05
	I-JEPA ViT-H (22K)	*632M*	*IN-22K*	*900*	*27.98*	*13.00*	*3.45*	*61.84*	*34.70*	*89.72*	*19.62*	*30.86*	*35.15*
10	LeJEPA ViT-L	304M	IN-1K	100	**64.72**	35.25	22.25	85.15	59.77	**92.53**	**50.90**	77.00	60.95
	LeJEPA ConvNeXtV2-H	660M	IN-1K	100	61.84	30.67	24.46	85.74	63.29	91.78	49.32	78.53	60.70
	I-JEPA ViT-H	632M	IN-1K	300	57.68	33.82	21.96	88.77	66.42	88.24	43.97	83.23	60.51
	I-JEPA ViT-H + STOP	632M	IN-1K	300	57.00	**39.77**	**25.21**	**90.09**	**70.32**	90.16	45.68	**85.13**	62.92
	I-JEPA ViT-H (22K)	*632M*	*IN-22K*	*900*	*58.74*	*43.52*	*18.27*	*94.83*	*75.23*	*98.94*	*49.06*	*67.66*	63.28
all	LeJEPA ViT-L	304M	IN-1K	100	**78.30**	57.01	57.28	96.50	83.71	**91.21**	**82.05**	89.74	79.48
	LeJEPA ConvNeXtV2-H	660M	IN-1K	100	76.60	52.99	54.88	96.15	81.34	91.11	77.64	89.76	77.56
	I-JEPA ViT-H	632M	IN-1K	300	73.32	56.61	54.47	97.54	86.42	86.47	81.02	92.11	78.50
	I-JEPA ViT-H + STOP	632M	IN-1K	300	73.87	**61.95**	**61.27**	**98.02**	**87.78**	88.08	81.72	**92.88**	80.70
	I-JEPA ViT-H (22K)	*632M*	*IN-22K*	*900*	*75.67*	*65.39*	*49.79*	*98.46*	*89.95*	*98.54*	*81.58*	*87.19*	*80.82*

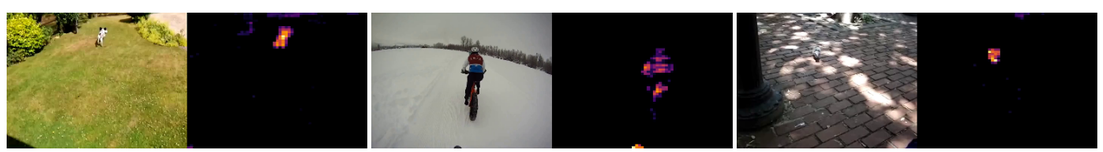


Figure 13. Emergent Object Segmentation via Last Layer Thresholding. LeJEPA naturally learns to segment and track salient objects (shown in attention maps on the right of each video) without explicit supervision. The results display impressive visual quality and strong temporal consistency across video frames (*videos provided on our project page*). This emergent capability demonstrates the rich semantic representations learned through our self-supervised approach.

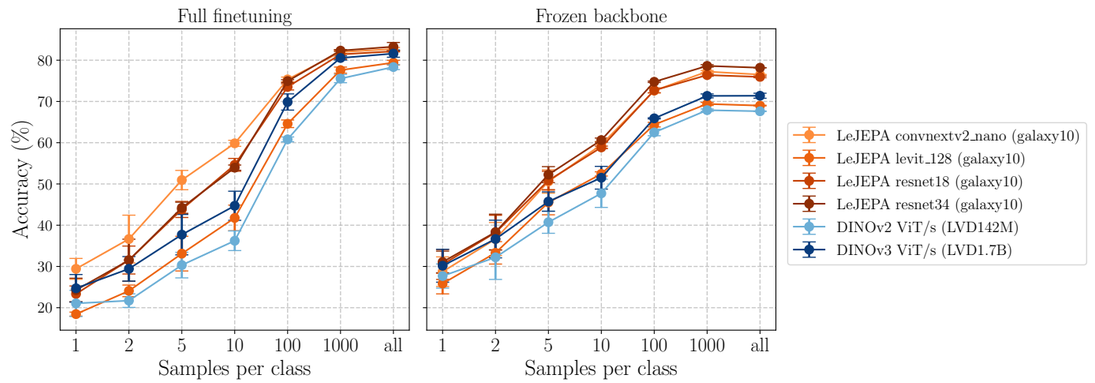


图 12. Galaxy10 小架构域内预训练，列为冻结线性探测／完整微调，横轴每类样本数。3 随机种子比较 DINOv2/v3、IJEPA。LeJEPA 跨架构直接域内训练，超过先进基础模型。数据见表 3。

表 2. 纹理、物体、细粒度类别等 8 数据集少样本分类准确率（%）。LeJEPA 在 DTD、flowers102、food101 更好，只需 100 轮预训练，而 I-JEPA 为 300，训练时间、计算减少 **3×**，不牺牲下游性能，适合预算有限应用。粗体为 IN-1K 组内最佳，全部为百分比。

					Dataset								
shots	model	params	pretrain	epochs	DTD	aircr.	cars	cifar10	cifar100	flowers102	food	pets	avg.
1	LeJEPA ViT-L	304M	IN-1K	100	**33.21**	9.37	3.40	51.65	27.01	48.53	17.14	46.11	29.55
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	I-JEPA ViT-H + STOP	632M	IN-1K	300	26.60	**11.18**	**4.75**	56.27	**35.20**	47.17	15.75	**59.47**	32.05
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	I-JEPA ViT-H	632M	IN-1K	300	73.32	56.61	54.47	97.54	86.42	86.47	81.02	92.11	78.50
	I-JEPA ViT-H + STOP	632M	IN-1K	300	73.87	**61.95**	**61.27**	**98.02**	**87.78**	88.08	81.72	**92.88**	80.70
	I-JEPA ViT-H (22K)	*632M*	*IN-22K*	*900*	*75.67*	*65.39*	*49.79*	*98.46*	*89.95*	*98.54*	*81.58*	*87.19*	*80.82*

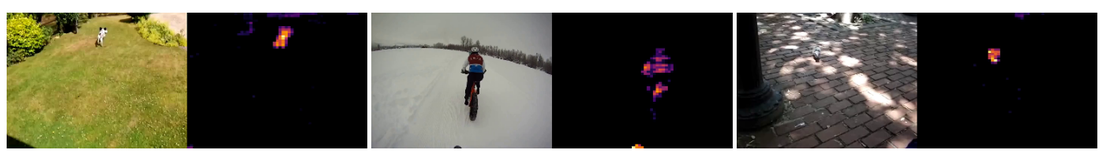


图 13. 通过对最后一层进行阈值处理而涌现的目标分割能力。LeJEPA 自然学会了分割和跟踪显著目标（各视频右侧的注意力图所示），无需显式监督。结果具有出色的视觉质量，且在视频帧之间保持较强的时间一致性（视频见项目网页）。这一涌现能力体现了自监督方法所学习到的丰富语义表征。

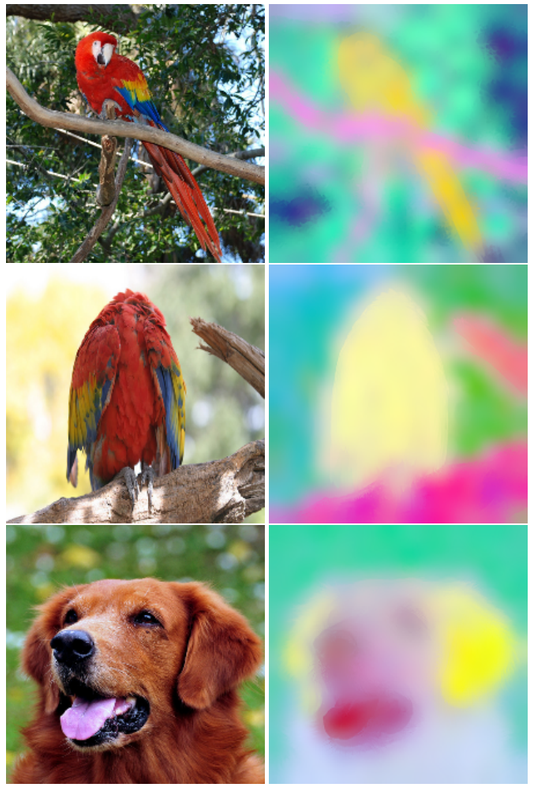


Figure 14. LeJEPA learns rich semantic representations through self-supervised learning. PCA visualization of last-layer features from LeJEPA (ViT-Large, 100 epochs on ImageNet-1K). For each image, features are independently projected to RGB using the first 3 principal components. Without any supervision, LeJEPA spontaneously develops semantically meaningful representations: notice how warm colors (red/-magenta/pink) consistently capture foreground objects (parrot bodies, dog face), while cool colors (cyan/green/yellow) represent backgrounds and foliage. This emergent object-background separation and perceptual grouping discovered the visual structure of the world purely from unlabeled data.

bersome yet necessary [Assran et al., 2022].

To demonstrate LeJEPA's versatility and ability to resolve that current pain-point, we propose to pretrain directly on a new domain without any change in the loss or the pretraining pipeline. We select the Galaxy10 dataset, a galaxy morphology classification task that differs significantly from natural images in both visual structure and statistical properties [Balestriero et al., 2025]. The dataset contains 11,000 training samples across 10 galaxy types. For LeJEPA, we use the default hyper-parameters and pretrain for 400 epochs a variety of backbones. We compare against the latest DINOv2, DINOv3 and IJEPA. We report in Figure 12 the top1 accuracy for linear probing both with frozen backbone and full-finetuning. We observe that **in-domain pretraining with LeJEPA substantially outperforms state-of-the-art frontier models (DINOv2, DINOv3) on both linear probing and full finetuning**. Additional datasets and backbones are provided in Table 5 depicting LeJEPA's ability to train in-domain, even with a dataset with 1000 samples (flowers102). Coupling this result with the stability of LeJEPA across architectures and hyper-parameters should offer a promising alternatives in domains not yet accounted for by the latest frontier models.

6.4 LeJEPA Scales Across Data and Models

We now propose to apply LeJEPA over a larger pretraining dataset, i.e., Imagenet-1k, and over larger backbones such as ViT/Large (0.3B), ConvNextV2-Huge (0.6B). For those two models, we reach an online linear probe accuracy on inet1k of 77.1% and 78.5% respectively. Beyond in-distribution performances, we also explore transfer learning. For those experiments, our baselines are IJEPA with a ViT-Huge (0.6B) which is the closest to our setup, and we also include a recent improved version of IJEPA including additional stochastic prediction tasks [Bar et al., 2023] that is coined IJEPA + STOP. For LeJEPA, we employ the same recipe as described in Section 6.1 and report transfer learning performances with frozen backbone in Table 2. We observe that we consistently outperform IJEPA while employed a smaller model and shorted training schedule. Beyond top1 accuracy, we also echo our findings from Section 6.2 about LeJEPA's training loss quality. In our setup, we observe a very stable and smooth training curve indicating a stable optimization landscape removing the need for careful hyperparameter selection (recall thm. 4). We provide an example on a ViT-gigantic (1.8B parameters) in Figure 1.

6.5 Emergent Semantic Structure in LeJEPA Representations

A hallmark of successful self-supervised learning is the emergence of semantically meaningful attention patterns

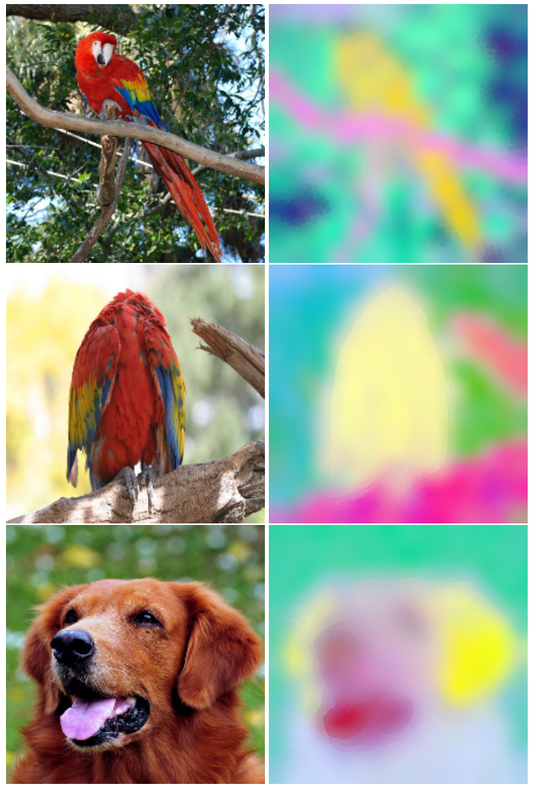


图 14. LeJEPA 自监督学到丰富语义表征。IN1K 上 ViT-Large 训练 100 轮，最后层特征 PCA。各图独立以前 3 主成分映射 RGB。无需监督，暖色（红、品红、粉）一致对应前景（鹦鹉身体、狗脸），冷色（青、绿、黄）对应背景和植物。这种涌现的前景分离与感知分组，纯从无标签数据发现世界视觉结构。

繁琐但必要 [Assran et al., 2022]。

为展示通用性及解决当前痛点，直接在新领域预训练，不改损失和流程。选择 Galaxy10 星系形态分类，其视觉结构和统计性质与自然图像很不同 [Balestriero 等，2025]，含 10 类、11,000 训练样本。默认超参数，各骨干训练 400 轮，与 DINOv2、DINOv3、IJEPA 比较。图 12 报告冻结线性探测及完整微调的 top-1。域内 LeJEPA 在两种评估均显著超过先进基础模型 DINOv2v3。表 5 提供更多数据、骨干，即使仅 1000 样本的 flowers102 也可域内训练。结合跨架构、超参数稳定性，为最新基础模型尚未覆盖的领域提供有前景选择。

6.4 LeJEPA 在数据和模型规模上的扩展

在更大 ImageNet-1k 及更大骨干 ViT/Large（0.3B）、ConvNeXtV2-Huge（0.6B）上应用 LeJEPA，在线线性探测分别为 77.1%、78.5%。也研究分布外迁移：基线为最接近设置的 IJEPA ViT-Huge（0.6B），及加入随机预测任务的改进 IJEPA [Bar 等，2023]，称 IJEPA + STOP。LeJEPA 使用第 6.1 节相同方案，表 2 报告冻结骨干迁移。更小模型、更短训练，仍一致超过 IJEPA。除 top-1 外，训练损失质量也印证 6.2：曲线稳定平滑，优化景观稳定，无需仔细选超参数（定理 4）。图 1 给出 1.8B 参数 ViT-gigantic 例子。

6.5 LeJEPA 表征中涌现的语义结构

成功的自监督学习的一个标志，是具有语义意义的注意力模式的涌现。

without explicit supervision [Caron et al., 2021]. To assess whether LeJEPA learns such structure, we visualize the attention maps of the learned representations. Following DINO [Caron et al., 2021], we apply PCA to the embeddings and visualize the first principal components, which reveal clear correspondence to object boundaries and salient regions (Figure 14). Furthermore, we explore whether these attention patterns can enable unsupervised video segmentation—a challenging task requiring temporal consistency and object understanding. By thresholding the self-attention maps of the [CLS] token, we obtain binary masks that track objects across frames without any segmentation labels during training. As shown in Figure 13, **LeJEPA's attention naturally segments foreground objects from background with remarkable temporal coherence**, suggesting that the learned representations capture both spatial semantics and temporal structure. This emergent capability demonstrates that LeJEPA's stability-focused objective does not sacrifice the semantic richness of learned features.

7 Conclusion

We have established a principled theoretical framework for JEPA-based self-supervised learning that fundamentally resolves its core pathologies. Our contributions span theory and practice: we proved that isotropic Gaussian embeddings uniquely minimize worst-case downstream risk, introduced SIGReg as a tractable and provably correct method to enforce this distribution, and demonstrated that this approach eliminates representational collapse by design–and not through ad-hoc combinations of teacher-student networks, stop-gradients, or asymmetric architectures.

We validate LeJEPA across domains and over 60 architectures including gigantic versions with 1.8B parameters. In spite of its simplicify , LeJEPA matches state-of-the-art performance while requiring fewer than 50 lines of core implementation. Critically, our approach provides what SSL has long needed: a mathematically rigorous foundation that directly informs practical algorithm design.

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无需显式监督 [Caron 等， 2021]。为检验这种结构，可视化学得注意力。按 DINO [Caron 等， 2021]，，对嵌入 PCA，前几个主成分清晰对应物体边界、显著区域（图 14）。还检验无监督视频分割，其要求时间一致性和物体理解。阈值化 [CLS] 令牌自注意力，得到无需分割标签、能跨帧追踪的二值掩码。图 13 中前景与背景自然分离，时间协调突出，说明表征捕获空间语义和时间结构。关注稳定性的目标没有牺牲特征语义丰富性。

7 结论

我们建立 JEPA 自监督的原则性理论，解决核心病态：证明各向同性高斯唯一最小化最坏下游风险，提出可计算、可证明正确的 SIGReg 约束该分布，并展示其从构造上排除表征坍塌，无需临时组合教师–学生、停止梯度、非对称架构。

我们在多个领域以及超过 60 种架构上验证 LeJEPA，其中包括拥有 18 亿参数的超大型版本。尽管十分简洁，LeJEPA 的性能仍可媲美最先进方法，而核心实现不足 50 行。关键在于，我们的方法提供了自监督学习长期以来所需要的东西：能够直接指导实际算法设计的严格数学基础。

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Appendix

A Additional Details on Nonlinear Probing

A.1 kNN Probing

To allow for more flexible evaluation of the pretrained encoder f_θ, it is standard to work with a k-NN prober [Taunk et al., 2019], both for regression and classification. We rely on the radial k-NN variation that leverages a sample-dependent k–improving performance for non uniform distributions of samples [Sun and Huang, 2010, Zhang et al., 2017, Abu Alfeilat et al., 2019].

We denote the underlying embedding density as $p_z \in C^3$ with derivatives of order up to 3 bounded, and finite Fisher information and covariance. This regularity condition is fulfilled by current encoders. The *unknown* labels come from the target function $\eta : \mathbb{R}^K \to \mathbb{R}$, assumed C^2. We handle classification tasks by setting $\eta(z) = \mathbb{P}(Y = 1 \mid z)$. The training consists of the N embeddings along with their training labels $\{(z_n, \eta(z_n))\}_{n=1}^N$, where we will denote $y_n \triangleq \eta(z_n)$. The prediction for a query vector q is formed as

$$\widehat{y}(q) := \frac{1}{y(q)} \sum_{n:\|z_n - q\| \le r_0} y_n, \tag{kNN}$$

with $y(q) \triangleq \#\{n : \|z_n - q\| \le r_0\}$ counting the number of samples within a r-radius ball around q. The radius r controls how many neighbors predictions are averaged to form the query's prediction. As per the linear probing's lemma. 1, we can characterize the bias of the estimator Equation (kNN) at a particular query point, as formalized below.

Lemma 4: k-NN Pointwise Bias

The (kNN) estimator has bias at query q given by

$$\text{Bias}(q) = \frac{r_0^2}{d+2}\left(\nabla\eta(q)^\top \nabla \log p_z(q) + \tfrac{1}{2}\Delta\eta(z)\right) + o(r_0^2),$$

where the remainder $o(r_0^2)$ is uniform in q. (Proof in Section B.3.)

To obtain the integrated bias, i.e., over the distribution of query points, we consider the following two properties. First, the distribution of query points follow the training distribution, i.e., $q \sim p_z$, second, target function η has gradient which is mean-zero and isotropic with $\mathbb{E}\left[\nabla\eta(z)\nabla\eta(z)^\top\right] = \tau_g^2 I_d$ with $\tau_g^2 \in (0, \infty)$ uniformly in z. We also have any finite scalar-constraint on the covariance of the embeddings such as $\text{Tr}(\Sigma) = c$ or $\|\Sigma\|_F = c$ for a finite constant c.

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附录

A 非线性探测的补充细节

A.1 k 近邻探测

灵活评估编码器 f_θ时，回归、分类常用 kNN 探测 [Taunk 等，2019], 。采用样本相关k的径向 kNN 变体，提高非均匀分布性能 [Sun 和 Huang，2010；Zhang 等，2017；Abu Alfeilat 等， 2019]。

嵌入密度记 $p_z \in C^3$，三阶及以下导数有界，Fisher 信息、协方差有限；当前编码器满足此正则性。未知标签来自目标函数 η： $\mathbb{R}^K \to \mathbb{R}$，假设 C^2。分类设置 $\eta(z) = \mathbb{P}(Y = 1 \mid z)$。训练包含 N 嵌入及标签$\{(z_n, \eta(z_n))\}_{n=1}^N$，记 $y_n \triangleq \eta(z_n)$。查询向量 q 预测为：

$$\widehat{y}(q) := \frac{1}{y(q)} \sum_{n:\|z_n - q\| \le r_0} y_n, \tag{kNN}$$

$y(q) \triangleq \#\{n : \|z_n - q\| \le r_0\}$计数 q周围 r半径球内样本。半径 r 控制平均多少邻居形成查询预测。类似线性引理 1，可刻画 kNN 在指定查询点的偏差：

引理 4：k 近邻的逐点偏差

（kNN）估计量在查询点 q 处的偏差为

$$\text{Bias}(q) = \frac{r_0^2}{d+2}\left(\nabla\eta(q)^\top \nabla \log p_z(q) + \tfrac{1}{2}\Delta\eta(z)\right) + o(r_0^2),$$

其中余项 $o(r_0^2)$ 对 q 一致成立。（证明见附录 B.3。）

对查询分布积分取得偏差，考虑：（1）查询分布与训练相同，即 $q \sim p_z$；（2）目标函数 η 梯度均值零、各向同性，$\mathbb{E}\left[\nabla\eta(z)\nabla\eta(z)^\top\right] = \tau_g^2 I_d$，且 $\tau_g^2 \in (0, \infty)$对 z一致。还对嵌入协方差施加任意有限标量约束，如 $\text{Tr}(\Sigma) = c$ 或 $\|\Sigma\|_F = c$，c为有限常数。

Theorem 7: k-NN isotropic Gaussian Optimality

The integrated squared bias of (kNN) satisfies

$$\mathbb{E}_z\left[\text{Bias}(z)^2\right] = \frac{r_0^4}{(K+2)^2}\tau_g^2 J(p) + O(r_0^4),$$

and the isotropic Gaussian is the unique minimizer of the integrated square bias. (Proof in Section B.4.)

As a result, we now have a unique minimizer for the optimal embedding density for both the linear and k-NN probes.

A.2 Kernel Probing

As an alternative to (kNN), it is also common to leverage kernel methods, which we consider in this section.

Consider a kernel $K : \mathbb{R}^K \to \mathbb{R}$ with the following standard properties

$$\int_{\mathbb{R}^d} K(u)du = 1, \qquad \text{(normalized)}$$
$$\int_{\mathbb{R}^d} uK(u)du = 0, \qquad \text{(symmetric)}$$
$$\int_{\mathbb{R}^d} uu^\top K(u)du = \mu_2(K) I_d, \qquad \text{(isotropic)}$$
$$R(K) \triangleq \int_{\mathbb{R}^d} K(u)^2 du < \infty, \qquad \text{(finite roughness)}$$

for some $\mu_2(K) \in (0,\infty)$, some bandwidth $h > 0$ and denoting $K_h(t) \triangleq h^{-d}K(t/h)$, we remind the reader that the Nadaraya-Watson estimator, introduced in Nadaraya [1964], Watson [1964], at a query $\boldsymbol{q} \in \mathbb{R}^d$ is

$$\widehat{\boldsymbol{y}}(\boldsymbol{q}) \triangleq \frac{\sum_{n=1}^N K_h(\boldsymbol{q}-\boldsymbol{x}_n)\boldsymbol{y}_n}{\sum_{n=1}^N K_h(\boldsymbol{q}-\boldsymbol{x}_n)}. \qquad \text{(NW)}$$

Similarly to (kNN), we will see that the performance of (NW) depends crucially on the distribution of the training points. We have access to our dataset of inputs from p_z and for each sample $\boldsymbol{z}_n$ the corresponding target is given from $\eta(\boldsymbol{z}_n) = \mathbb{E}[Y_n \mid \boldsymbol{z}_n]$. We also denote the corresponding conditional variance of the target function at that point as $v(x) = \text{Var}(Y_i \mid X_i = x)$. We follow the regularity conditions of the k-NN probing derivations and additionally assume that p has sufficiently light tails so that for each coordinate j, $\lim_{\|x\|\to\infty} p(x) = 0$ and $\lim_{\|x\|\to\infty} x_j p(x) = 0$. We first derive the pointwise bias and variance for $\widehat{\boldsymbol{y}}(\boldsymbol{q})$.

Lemma 5: Kernel Bias and Variance

For any fixed $\boldsymbol{q} \in \mathbb{R}^d$ with $p(\boldsymbol{q}) > 0$, as $h \to 0$ and $nh^d \to \infty$,

$$\text{Bias}\left[\widehat{\boldsymbol{y}}(\boldsymbol{q})\right] = \frac{h^2\mu_2(K)}{2}\Big(\Delta\boldsymbol{y}(\boldsymbol{q}) + 2\nabla\boldsymbol{y}(\boldsymbol{q})^\top \nabla \log p(\boldsymbol{q})\Big) + o(h^2),$$
$$\text{Var}\left[\widehat{\boldsymbol{y}}(\boldsymbol{q})\right] = \frac{R(K)}{nh^d}\frac{v(\boldsymbol{q})}{p(\boldsymbol{q})} + o\big((nh^d)^{-1}\big).$$

The $o(\cdot)$ terms are uniform over compact sets where p is bounded away from zero. (Proof in Section B.5.)

We now show that, under a fixed mean and total-covariance constraint on p_z, the isotropic Gaussian distribution uniquely minimizes the bias and variance of the kernel regression estimator at any test point. We restrict the smoothness class of the target function using

$$\mathcal{M}(L,B) \triangleq \Big\{m \in C^2(\mathbb{R}^d) : \|\nabla\boldsymbol{y}(\boldsymbol{q})\| \le L, \; |\Delta\boldsymbol{y}(\boldsymbol{q})| \le B, \forall \boldsymbol{q} \in \mathbb{R}^d\Big\},$$

定理 7：k 近邻中的各向同性高斯最优性

（kNN）的积分平方偏差满足

$$\mathbb{E}_z\left[\text{Bias}(z)^2\right] = \frac{r_0^4}{(K+2)^2}\tau_g^2 J(p) + O(r_0^4),$$

以及该　各向同性高斯分布是使积分平方偏差最小的唯一分布。（证明见附录 B.4。）

因此，我们现在得到一个唯一的 对于线性探测和 k 近邻探测，最优嵌入密度的唯一最小化解。 s.

A.2 核探测

除（kNN）之外，核方法也是常见选择，本节将予以讨论。考虑核 K：$\mathbb{R}^K \to \mathbb{R}$，满足以下标准性质：

$$\int_{\mathbb{R}^d} K(u)du = 1, \qquad \text{(normalized)}$$
$$\int_{\mathbb{R}^d} uK(u)du = 0, \qquad \text{(symmetric)}$$
$$\int_{\mathbb{R}^d} uu^\top K(u)du = \mu_2(K) I_d, \qquad \text{(isotropic)}$$
$$R(K) \triangleq \int_{\mathbb{R}^d} K(u)^2 du < \infty, \qquad \text{(finite roughness)}$$

对于某个 $\mu_2(K) \in (0,\infty)$、某个带宽 $\mathrm{h} > 0$，并记 $K_h(t) \triangleq \mathrm{h}^{-d}K(t/\mathrm{h})$，我们提醒读者：由 Nadaraya [1964]，和 Watson [1964]，提出的 Nadaraya–Watson 估计量，在查询点 $\boldsymbol{q} \in \mathbb{R}^d$ 处为

$$\widehat{\boldsymbol{y}}(\boldsymbol{q}) \triangleq \frac{\sum_{n=1}^N K_h(\boldsymbol{q}-\boldsymbol{x}_n)\boldsymbol{y}_n}{\sum_{n=1}^N K_h(\boldsymbol{q}-\boldsymbol{x}_n)}. \qquad \text{(NW)}$$

类似 kNN，NW 性能关键取决于训练点分布。输入来自 p_z，每样本 $\boldsymbol{z}_n$ 目标为 $\eta(\boldsymbol{z}_n) = \mathbb{E}[Y_n \mid \boldsymbol{z}_n]$，该点条件方差为$v(x) = \text{Var}(Y_i \mid X_i = x)$。沿用 kNN 推导正则条件，并假设 p 尾部足够轻，每坐标 j满足 $\lim_{\|x\|\to\infty} p(x) = 0$ 和 $\lim_{\|x\|\to\infty} x_j p(x) = 0$。先推导 $\widehat{\boldsymbol{y}}(\boldsymbol{q})$逐点偏差、方差。

引理 5：核方法的偏差与方差

对任意满足 $p(\boldsymbol{q}) > 0$ 的固定 $\boldsymbol{q} \in \mathbb{R}^d$，当 $\mathrm{h} \to 0$ 且 $n\mathrm{h}^d \to \infty$ 时，

$$\text{Bias}\left[\widehat{\boldsymbol{y}}(\boldsymbol{q})\right] = \frac{h^2\mu_2(K)}{2}\Big(\Delta\boldsymbol{y}(\boldsymbol{q}) + 2\nabla\boldsymbol{y}(\boldsymbol{q})^\top \nabla \log p(\boldsymbol{q})\Big) + o(h^2),$$
$$\text{Var}\left[\widehat{\boldsymbol{y}}(\boldsymbol{q})\right] = \frac{R(K)}{nh^d}\frac{v(\boldsymbol{q})}{p(\boldsymbol{q})} + o\big((nh^d)^{-1}\big).$$

在 p 与零保持正距离的紧集上，$o(\cdot)$ 项一致成立。（证明见附录 B.5。）

下面证明，在 p_z 的均值和总协方差约束固定时，各向同性高斯分布能唯一地最小化任意测试点处核回归估计量的偏差与方差。我们使用以下条件限制目标函数的光滑度类别：

$$\mathcal{M}(L,B) \triangleq \Big\{m \in C^2(\mathbb{R}^d) : \|\nabla\boldsymbol{y}(\boldsymbol{q})\| \le L, \; |\Delta\boldsymbol{y}(\boldsymbol{q})| \le B, \forall \boldsymbol{q} \in \mathbb{R}^d\Big\},$$

allowing us to formalize below the worst case integrated bias and the optimal density for z.

Theorem 8: Kernel isotropic Gaussian Optimality

The integrated squared bias of (NW) satisfies

$$\sup_{m\in\mathcal{M}(L,B)} \mathbb{E}_z\left[\mathrm{Bias}\left[\widehat{y}(z)\right]\right] \quad \leq \quad \left(\frac{h^2\mu_2(K)}{2}\right)^2\left(2B^2 \quad + \quad 8L^2J(p)\right) \quad + \quad o(h^4),$$

and the integrated variance is independent of p. Among all densities p on $\mathbb{R}^d$ with total-variance constrained, e.g., $\mathrm{Tr}(\Sigma) = c$, the isotropic Gaussian is the unique minimizer. (Proof in Section B.7.)

B Proofs

B.1 Proof of lemma. 1

Proof. Our proof follows standard derivations when it comes to studying the bias of an estimator. Let's consider the ridge regression problem (Tikhonov regularized least squares estimator) with close form estimator

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X} + \lambda_{\text{wd}}\mathbf{I})^{-1}\mathbf{X}^T\mathbf{Y}. \tag{9}$$

The labels are formed from the ground truth parameter β_{true} with centered error, as per $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}_{\text{true}} + \boldsymbol{\varepsilon}$ where $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$. We can now look at the bias of our estimator given by

$$\begin{aligned}
\mathrm{Bias}(\hat{\boldsymbol{\beta}}) &= \mathbb{E}[\hat{\boldsymbol{\beta}}] - \boldsymbol{\beta}_{\text{true}} \\
&= (\mathbf{X}^T\mathbf{X} + \lambda_{\text{wd}}\mathbf{I})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}_{\text{true}} - \boldsymbol{\beta}_{\text{true}} \\
&= -\lambda_{\text{wd}}(\mathbf{X}^T\mathbf{X} + \lambda_{\text{wd}}\mathbf{I})^{-1}\boldsymbol{\beta}_{\text{true}} \\
&= -\lambda_{\text{wd}}\mathbf{Q}(\boldsymbol{\Lambda} + \lambda\mathbf{I})^{-1}\mathbf{Q}^T\boldsymbol{\beta}_{\text{true}}
\end{aligned}$$

We will now compare that bias when $\boldsymbol{X}$ has isotropic and anisotropic covariance with same total variance:

$$\frac{\lambda_1 + \lambda_2 + \cdots + \lambda_p}{p} = \bar{\lambda}. \tag{10}$$

For any anisotropic covariance matrix of $\boldsymbol{X}$, denote by $\boldsymbol{q}_1$ the eigenvector with smallest eigenvalue, and let's denote by $\kappa > 0$ a positive constant. We now define

$$\boldsymbol{\beta}_{\text{true}} = \kappa \cdot \mathbf{q}_p, \tag{11}$$

leading to

$$\begin{aligned}
\|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{isotropic}} &= \frac{\lambda_{\text{wd}}}{\bar{\lambda} + \lambda_{\text{wd}}}\|\boldsymbol{\beta}_{\text{true}}\|, \\
\|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{non-isotropic}} &= \frac{\lambda_{\text{wd}}}{\lambda_p + \lambda_{\text{wd}}}\|\boldsymbol{\beta}_{\text{true}}\|
\end{aligned}$$

Since $\lambda_p < \bar{\lambda}$ (strict inequality when not isotropic):

$$\frac{\lambda_{\text{wd}}}{\lambda_p + \lambda_{\text{wd}}} > \frac{\lambda_{\text{wd}}}{\bar{\lambda} + \lambda_{\text{wd}}}$$

we obtain that

$$\|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{non-isotropic}} > \|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{isotropic}}$$

As a result, whenever the covariance matrix of $\boldsymbol{X}$ is anisotropic, there will be downstream tasks for which the estimator bias is increased compared to having isotropic covariance matrix. Anisotropic covariance structure thus amplifies regularization bias when the true parameter vector aligns unfavorably with the data's covariance structure. □

从而可形式化表述以下最坏情况积分偏差，以及 z 的最优密度。

定理 8：核方法中的各向同性高斯最优性

（NW）的积分平方偏差满足

$$\sup_{m\in\mathcal{M}(L,B)} \mathbb{E}_z\left[\mathrm{Bias}\left[\widehat{y}(z)\right]\right] \quad \leq \quad \left(\frac{h^2\mu_2(K)}{2}\right)^2\left(2B^2 \quad + \quad 8L^2J(p)\right) \quad + \quad o(h^4),$$

且积分方差与 p 无关。在 $\mathbb{R}^d$ 上所有总方差受约束的密度 p 中，例如 $\mathrm{Tr}(\Sigma) = c$，各向同性高斯分布是唯一的最小化解。（证明见附录 B.7。）

B 证明

B.1 引理 1 的证明

证明。证明遵循研究估计量偏差时的标准推导。考虑岭回归问题（Tikhonov 正则化最小二乘估计量），其闭式估计量为

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X} + \lambda_{\text{wd}}\mathbf{I})^{-1}\mathbf{X}^T\mathbf{Y}. \tag{9}$$

标签由真实参数 β_{true} 加上中心化误差形成，即 $\mathrm{Y} = \mathrm{X}\boldsymbol{\beta}_{\text{true}} + \boldsymbol{\varepsilon}$，其中 $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$。现在考察估计量的偏差，其表达式为

$$\begin{aligned}
\mathrm{Bias}(\hat{\boldsymbol{\beta}}) &= \mathbb{E}[\hat{\boldsymbol{\beta}}] - \boldsymbol{\beta}_{\text{true}} \\
&= (\mathbf{X}^T\mathbf{X} + \lambda_{\text{wd}}\mathbf{I})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}_{\text{true}} - \boldsymbol{\beta}_{\text{true}} \\
&= -\lambda_{\text{wd}}(\mathbf{X}^T\mathbf{X} + \lambda_{\text{wd}}\mathbf{I})^{-1}\boldsymbol{\beta}_{\text{true}} \\
&= -\lambda_{\text{wd}}\mathbf{Q}(\boldsymbol{\Lambda} + \lambda\mathbf{I})^{-1}\mathbf{Q}^T\boldsymbol{\beta}_{\text{true}}
\end{aligned}$$

下面进行比较　　比较总方差相同、但 $\boldsymbol{X}$ 分别具有各向同性和各向异性协方差时的偏差：

$$\frac{\lambda_1 + \lambda_2 + \cdots + \lambda_p}{p} = \bar{\lambda}. \tag{10}$$

对于 $\boldsymbol{X}$ 的任意各向异性协方差矩阵，将最小特征值对应的特征向量记为 $\boldsymbol{q}_1$，并用 $\kappa > 0$ 表示正常数。下面定义

$$\boldsymbol{\beta}_{\text{true}} = \kappa \cdot \mathbf{q}_p, \tag{11}$$

从而得到

$$\begin{aligned}
\|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{isotropic}} &= \frac{\lambda_{\text{wd}}}{\bar{\lambda} + \lambda_{\text{wd}}}\|\boldsymbol{\beta}_{\text{true}}\|, \\
\|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{non-isotropic}} &= \frac{\lambda_{\text{wd}}}{\lambda_p + \lambda_{\text{wd}}}\|\boldsymbol{\beta}_{\text{true}}\|
\end{aligned}$$

由于 $\lambda_p < \bar{\lambda}$（非各向同性时为严格不等式）：

$$\frac{\lambda_{\text{wd}}}{\lambda_p + \lambda_{\text{wd}}} > \frac{\lambda_{\text{wd}}}{\bar{\lambda} + \lambda_{\text{wd}}}$$

得到

$$\|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{non-isotropic}} > \|\mathrm{Bias}(\hat{\boldsymbol{\beta}})\|_{\text{isotropic}}$$

因此，只要 $\boldsymbol{X}$ 的协方差矩阵是各向异性的，就存在某些下游任务，其估计量偏差比各向同性协方差矩阵下更大。当真实参数向量与数据协方差结构的对齐方式不利时，各向异性协方差结构会放大正则化偏差。□

B.2 Proof of lemma. 2

Proof. We use the same formula as in Section B.1 with $\lambda_{\text{wd}} = 0$. We first see that the estimator is unbiased. We will now leverage that result to compute the covariance matrix of the estimator

$$\begin{aligned}\text{Var}(\hat{\boldsymbol{\beta}}|\mathbf{X}) &= \mathbb{E}[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T|\mathbf{X}] \\ &= \mathbb{E}[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}|\mathbf{X}] \\ &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T|\mathbf{X}]\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} \\ &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\sigma^2\mathbf{I}_n)\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\end{aligned}$$

leading to the total variance

$$\text{tr}(\text{Var}(\hat{\boldsymbol{\beta}})) = \sigma^2\text{tr}(\mathbf{G}^{-1}) = \sigma^2\sum_{j=1}^{p}\frac{1}{\lambda_j}$$

where we used the eigendecomposition:

$$\mathbf{G} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T$$

The function $f(x) = \frac{1}{x}$ is strictly convex on $(0, \infty)$ allowing us to leverage Jensen's Inequality:

$$\begin{aligned}&\frac{1}{K}\sum_{k=1}^{K}\frac{1}{\lambda_k} > \frac{1}{\frac{1}{K}\sum_{j=1}^{K}\lambda_k} \\ \iff &\frac{1}{K}\sum_{k=1}^{K}\frac{1}{\lambda_k} > \frac{1}{K}\sum_{k=1}^{K}\frac{1}{\frac{1}{K}\sum_{j=1}^{K}\lambda_k} \\ \iff &\sum_{k=1}^{K}\frac{1}{\lambda_k} > \sum_{k=1}^{K}\frac{1}{\frac{1}{K}\sum_{j=1}^{K}\lambda_k} \\ \iff &\text{tr}(\text{Var}(\hat{\boldsymbol{\beta}}))_{\text{aniso}} > \text{tr}(\text{Var}(\hat{\boldsymbol{\beta}}))_{\text{iso}}\end{aligned}$$

The inequality is strict whenever the eigenvalues $\{\lambda_j\}_{j=1}^{p}$ are not all equal. □

B.3 Proof of lemma. 4

Proof. Under PPP, conditional expectations of $\widehat{\eta}(x)$ coincide with the normalized ball average

$$\mathbb{E}\big[\widehat{\eta}(x)\big] \;=\; \frac{\int_{\mathrm{B}(0,r_0)}\eta(x+z)p(x+z)dz}{\int_{\mathrm{B}(0,r_0)}p(x+z)dz} \quad \text{to second order in } r_0,$$

which is the key surrogate used below. **Ball integrals.** For computations we use (by symmetry) for any $r > 0$:

$$\int_{\mathrm{B}(0,r)} z dz = 0, \qquad \int_{\mathrm{B}(0,r)} zz^\top dz = \frac{\mathrm{Vol}^{d+2}}{d+2}I_d, \qquad \int_{\mathrm{B}(0,r)}\|z\|^2 dz = \frac{d\mathrm{Vol}^{d+2}}{d+2}.$$

Fix $x \in \mathbb{R}^d$ and write $z \in \mathrm{B}(0, r_0)$ for local displacements. Assume $p \in C^3$, $\eta \in C^2$ with bounded derivatives on the region of interest, and expand a second-order Taylor expansion:

$$\begin{aligned}p(x+z) &= p(x) + \nabla p(x)^\top z + \tfrac{1}{2}z^\top Hp(x)z + O(\|z\|^3), \\ \eta(x+z) &= \eta(x) + \nabla\eta(x)^\top z + \tfrac{1}{2}z^\top H\eta(x)z + O(\|z\|^3),\end{aligned}$$

with remainders satisfying $|R_\eta(x;z)| \le C_\eta\|z\|^3$ and $|R_p(x;z)| \le C_p\|z\|^3$ uniformly for $\|z\| \le r_0$. Using the ball identities

B.2 引理 2 的证明

证明。使用与附录 B.1 相同的公式，并令 $\lambda_{\text{wd}} = 0$。首先可见估计量无偏。下面利用这一结果计算估计量的协方差矩阵：

$$\begin{aligned}\text{Var}(\hat{\boldsymbol{\beta}}|\mathbf{X}) &= \mathbb{E}[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^T|\mathbf{X}] \\ &= \mathbb{E}[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}|\mathbf{X}] \\ &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}[\boldsymbol{\varepsilon}\boldsymbol{\varepsilon}^T|\mathbf{X}]\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} \\ &= (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\sigma^2\mathbf{I}_n)\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\end{aligned}$$

从而得到总方差

$$\text{tr}(\text{Var}(\hat{\boldsymbol{\beta}})) = \sigma^2\text{tr}(\mathbf{G}^{-1}) = \sigma^2\sum_{j=1}^{p}\frac{1}{\lambda_j}$$

其中使用了特征分解：

$$\mathbf{G} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^T$$

函数 函数 $f(x) = \frac{1}{x}$ 在 $(0, \infty)$ 上严格凸，因此可应用 Jensen 不等式， :

$$\begin{aligned}&\frac{1}{K}\sum_{k=1}^{K}\frac{1}{\lambda_k} > \frac{1}{\frac{1}{K}\sum_{j=1}^{K}\lambda_k} \\ \iff &\frac{1}{K}\sum_{k=1}^{K}\frac{1}{\lambda_k} > \frac{1}{K}\sum_{k=1}^{K}\frac{1}{\frac{1}{K}\sum_{j=1}^{K}\lambda_k} \\ \iff &\sum_{k=1}^{K}\frac{1}{\lambda_k} > \sum_{k=1}^{K}\frac{1}{\frac{1}{K}\sum_{j=1}^{K}\lambda_k} \\ \iff &\text{tr}(\text{Var}(\hat{\boldsymbol{\beta}}))_{\text{aniso}} > \text{tr}(\text{Var}(\hat{\boldsymbol{\beta}}))_{\text{iso}}\end{aligned}$$

当特征值 $\{\lambda_j\}_{j=1}^{p}$ 不全相等时，不等式严格成立。 □

B.3 引理 4 的证明

证明。在 PPP 下， $\widehat{\eta}(x)$ 的条件期望等于归一化的球平均：

$$\mathbb{E}\big[\widehat{\eta}(x)\big] \;=\; \frac{\int_{\mathrm{B}(0,r_0)}\eta(x+z)p(x+z)dz}{\int_{\mathrm{B}(0,r_0)}p(x+z)dz} \quad \text{to second order in } r_0,$$

这是关键的替代量 以下将用到。球上的积分。计算中利用对称性，对任意 $r > 0$ 有 :

$$\int_{\mathrm{B}(0,r)} z dz = 0, \qquad \int_{\mathrm{B}(0,r)} zz^\top dz = \frac{\mathrm{Vol}^{d+2}}{d+2}I_d, \qquad \int_{\mathrm{B}(0,r)}\|z\|^2 dz = \frac{d\mathrm{Vol}^{d+2}}{d+2}.$$

固定 $x \in \mathbb{R}^d$ ，并用 $z \in \mathrm{B}(0, r_0)$ 表示局部位移。假设 $p \in C^3$, $\eta \in C^2$ 在所关注区域具有有界导数，作二阶泰勒展开：

$$\begin{aligned}p(x+z) &= p(x) + \nabla p(x)^\top z + \tfrac{1}{2}z^\top Hp(x)z + O(\|z\|^3), \\ \eta(x+z) &= \eta(x) + \nabla\eta(x)^\top z + \tfrac{1}{2}z^\top H\eta(x)z + O(\|z\|^3),\end{aligned}$$

其余项对 $\|z\| \le r_0$ 一致满足 $|R_\eta(x;\ z)| \le C_\eta\|z\|^3$ 和 $|R_p(x;z)| \le C_p\|z\|^3$。利用球上的恒等式，

$\int_{B(0,r)} z dz = 0$ and $\int_{B(0,r)} zz^\top dz = \frac{v_d r^{d+2}}{d+2} I_d$ and collecting terms up to order r_0^{d+2}, we simplify the denominator as

$$\begin{aligned}\mathcal{D}(x) &\triangleq \int_{\mathrm{B}(0,r_0)} p(x+z)dz \\ &= \int_{\mathrm{B}(0,r_0)} \Big[p(x) + \nabla p(x)^\top z + \tfrac{1}{2} z^\top H p(x) z + R_p(x;z)\Big] dz \\ &= \mathrm{Vol}_0^d p(x) \;+\; \frac{\mathrm{Vol}_0^{d+2}}{2(d+2)} \mathrm{tr}\big(Hp(x)\big) \;+\; O(r_0^{d+3}),\end{aligned}$$

since $\int z dz = 0$ and $\int z^\top H p z dz = \mathrm{tr}(Hp)\frac{v_d r_0^{d+2}}{d+2}$ and the denominator as

$$\begin{aligned}\mathcal{N}(x) &\triangleq \int_{\mathrm{B}(0,r_0)} \eta(x+z)p(x+z)dz \\ &= \int \Big[\eta(x) + \nabla\eta(x)^\top z + \tfrac{1}{2} z^\top H\eta(x) z\Big]\Big[p(x) + \nabla p(x)^\top z + \tfrac{1}{2} z^\top H p(x) z\Big] dz + O(r_0^{d+3}) \\ &= \eta(x)p(x) v_d r_0^d + \eta(x)\frac{v_d r_0^{d+2}}{2(d+2)}\mathrm{tr}\big(Hp(x)\big) + \frac{v_d r_0^{d+2}}{d+2}\nabla\eta(x)\cdot\nabla p(x) + \frac{v_d r_0^{d+2}}{2(d+2)} p(x)\mathrm{tr}\big(H\eta(x)\big) + O(r_0^{d+3}).\end{aligned}$$

Cubic terms vanish by symmetry, and quartic terms are $O(r_0^{d+4})$. Subtract $\eta(x)\mathcal{D}(x)$ to obtain the bias numerator:

$$\mathcal{N}(x) - \eta(x)\mathcal{D}(x) = \frac{v_d r_0^{d+2}}{d+2}\Big(\nabla\eta(x)\cdot\nabla p(x) + \tfrac{1}{2} p(x)\Delta\eta(x)\Big) + O(r_0^{d+3}).$$

Write $\mathcal{D}(x) = v_d r_0^d p(x)\big(1 + \alpha(x) r_0^2 + O(r_0^3)\big)$ where $\alpha(x) := \frac{1}{2(d+2)p(x)}\mathrm{tr}(Hp(x))$. Then

$$\begin{aligned}\frac{\mathcal{N}(x)}{\mathcal{D}(x)} - \eta(x) &= \frac{\frac{v_d r_0^{d+2}}{d+2}\big(\nabla\eta\cdot\nabla p + \frac{1}{2} p\Delta\eta\big) + O(r_0^{d+3})}{v_d r_0^d p\,\big(1 + \alpha r_0^2 + O(r_0^3)\big)} \\ &= \frac{r_0^2}{d+2}\left(\frac{\nabla\eta\cdot\nabla p}{p} + \frac{1}{2}\Delta\eta\right)\Big(1 - \alpha r_0^2 + O(r_0^3)\Big) \;+\; O(r_0^3) \\ &= \frac{r_0^2}{d+2}\Big(\nabla\eta(x)\cdot\nabla\log p(x) + \tfrac{1}{2}\Delta\eta(x)\Big) \;+\; o(r_0^2),\end{aligned}$$

uniformly on $\mathcal{K}$. This gives the bias formula

$$\mathbb{E}\big[\widehat{\eta}(x)\big] - \eta(x) = \frac{r_0^2}{d+2}\Big(\nabla\eta(x)\cdot\nabla\log p(x) + \tfrac{1}{2}\Delta\eta(x)\Big) \;+\; o(r_0^2),$$

completing the proof. □

B.4 Proof of thm. 7

Proof. Recall from Section B.3 that the bias term as sample $\boldsymbol{x}$ is given by

$$\begin{aligned}\mathrm{Bias}(\boldsymbol{x}) =& \frac{r_0^2}{d+2}\Big(\nabla\eta(x)\cdot\nabla\log p(x)\Big) \;+\; \frac{r_0^2}{2(d+2)}\Delta\eta(x) \;+\; o(r_0^2) \\ =& \frac{r_0^2}{d+2}\big(A(x) + C(x)\big) + o(r_0^2),\end{aligned}$$

$\int_{B(0,r)} z dz = 0$ 与 $\int_{B(0,r)} zz^\top dz = \frac{v_d r^{d+2}}{d+2} I_d$，并收集直到 r_0^{d+2} 阶的项，将分母化简为

$$\begin{aligned}\mathcal{D}(x) &\triangleq \int_{\mathrm{B}(0,r_0)} p(x+z)dz \\ &= \int_{\mathrm{B}(0,r_0)} \Big[p(x) + \nabla p(x)^\top z + \tfrac{1}{2} z^\top H p(x) z + R_p(x;z)\Big] dz \\ &= \mathrm{Vol}_0^d p(x) \;+\; \frac{\mathrm{Vol}_0^{d+2}}{2(d+2)} \mathrm{tr}\big(Hp(x)\big) \;+\; O(r_0^{d+3}),\end{aligned}$$

因为 $\int z dz = 0$ 和 $\int z^\top H p z dz = \mathrm{tr}(Hp)\frac{v_d r_0^{d+2}}{d+2}$，分母为

$$\begin{aligned}\mathcal{N}(x) &\triangleq \int_{\mathrm{B}(0,r_0)} \eta(x+z)p(x+z)dz \\ &= \int \Big[\eta(x) + \nabla\eta(x)^\top z + \tfrac{1}{2} z^\top H\eta(x) z\Big]\Big[p(x) + \nabla p(x)^\top z + \tfrac{1}{2} z^\top H p(x) z\Big] dz + O(r_0^{d+3}) \\ &= \eta(x)p(x) v_d r_0^d + \eta(x)\frac{v_d r_0^{d+2}}{2(d+2)}\mathrm{tr}\big(Hp(x)\big) + \frac{v_d r_0^{d+2}}{d+2}\nabla\eta(x)\cdot\nabla p(x) + \frac{v_d r_0^{d+2}}{2(d+2)} p(x)\mathrm{tr}\big(H\eta(x)\big) + O(r_0^{d+3}).\end{aligned}$$

三次项由于对称性消失　对称性，四次项为 $O(r_0^{d+4})$。减去 $\eta(x)\mathcal{D}(x)$，得到偏差的分子：　　：

$$\mathcal{N}(x) - \eta(x)\mathcal{D}(x) = \frac{v_d r_0^{d+2}}{d+2}\Big(\nabla\eta(x)\cdot\nabla p(x) + \tfrac{1}{2} p(x)\Delta\eta(x)\Big) + O(r_0^{d+3}).$$

写成 $\mathcal{D}(x) = v_d r_0^d p(x)\big(1 + \alpha(x) r_0^2 + O(r_0^3)\big)$，其中 $\alpha(x) := \frac{1}{2}\frac{1}{d+2}(\,)_{p(x)}\mathrm{tr}(Hp(x))$。于是

$$\begin{aligned}\frac{\mathcal{N}(x)}{\mathcal{D}(x)} - \eta(x) &= \frac{\frac{v_d r_0^{d+2}}{d+2}\big(\nabla\eta\cdot\nabla p + \frac{1}{2} p\Delta\eta\big) + O(r_0^{d+3})}{v_d r_0^d p\,\big(1 + \alpha r_0^2 + O(r_0^3)\big)} \\ &= \frac{r_0^2}{d+2}\left(\frac{\nabla\eta\cdot\nabla p}{p} + \frac{1}{2}\Delta\eta\right)\Big(1 - \alpha r_0^2 + O(r_0^3)\Big) \;+\; O(r_0^3) \\ &= \frac{r_0^2}{d+2}\Big(\nabla\eta(x)\cdot\nabla\log p(x) + \tfrac{1}{2}\Delta\eta(x)\Big) \;+\; o(r_0^2),\end{aligned}$$

在 $\mathcal{K}$ 上一致成立。由此得到偏差公式

$$\mathbb{E}\big[\widehat{\eta}(x)\big] - \eta(x) = \frac{r_0^2}{d+2}\Big(\nabla\eta(x)\cdot\nabla\log p(x) + \tfrac{1}{2}\Delta\eta(x)\Big) \;+\; o(r_0^2),$$

证明完毕。 □

B.4 定理 7 的证明

证明。回顾附录 B.3，样本 $\boldsymbol{x}$ 的偏差项为

$$\begin{aligned}\mathrm{Bias}(\boldsymbol{x}) =& \frac{r_0^2}{d+2}\Big(\nabla\eta(x)\cdot\nabla\log p(x)\Big) \;+\; \frac{r_0^2}{2(d+2)}\Delta\eta(x) \;+\; o(r_0^2) \\ =& \frac{r_0^2}{d+2}\big(A(x) + C(x)\big) + o(r_0^2),\end{aligned}$$

where we defined $A(x) \triangleq \nabla\eta(x) \cdot \nabla \log p(x)$ and $C(x) \triangleq \frac{1}{2}\Delta\eta(x)$. We now square and take expectation of $X \sim p$ and the isotropic gradient prior

$$\mathbb{E}\big[\text{Bias}(X)^2\big] = \mathbb{E}\Big[\left(\frac{r_0^2}{d+2}\right)^2 \big(A(x)^2 + 2A(x)C(x) + C(x)^2\big) + o(r_0^4)\Big] \tag{12}$$

$$= \left(\frac{r_0^2}{d+2}\right)^2 \Big\{ \underbrace{\mathbb{E}\big[A(X)^2\big]}_{\text{score-gradient term}} + 2\underbrace{\mathbb{E}\big[A(X)C(X)\big]}_{\text{cross term}} + \underbrace{\mathbb{E}\big[C(X)^2\big]}_{\text{curvature term}} \Big\} + o(r_0^4). \tag{13}$$

We will derive each term separately, recalling that we assume an isotropic gradient prior for η, i.e., $\mathbb{E}\big[\nabla\eta(x)\big] = 0$ and $\mathbb{E}\big[\nabla\eta(x)\nabla\eta(x)^\top\big] = \tau_g^2 I_d$, for some $\tau_g^2 \in (0, \infty)$.

1) The score-gradient term $\mathbb{E}[A(X)^2]$. Using $v(x) := \nabla \log p(x)$ for brevity:

$$\begin{aligned}
\mathbb{E}\big[A(X)^2\big] &= \mathbb{E}_X\big[\mathbb{E}_\eta[A(X)^2]\big] \\
&= \mathbb{E}_X\big[\mathbb{E}_\eta[\big(\nabla\eta(x)^\top v(x)\big)^2]\big] \\
&= \mathbb{E}_X\big[\mathbb{E}_\eta[\nabla\eta(x)^\top\Big(v(x)v(x)^\top\Big)\nabla\eta(x)]\big] \\
&= \mathbb{E}_X\big[\mathbb{E}_\eta[\text{tr}\Big(v(x)v(x)^\top\nabla\eta(x)\nabla\eta(x)^\top\Big)]\big] \\
&= \mathbb{E}_X\big[\text{tr}\Big(v(x)v(x)^\top\mathbb{E}_\eta[\nabla\eta(x)\nabla\eta(x)^\top]\Big)\big] \\
&= \mathbb{E}_X\big[\tau_g^2\|v(x)\|^2\big] \\
&= \tau_g^2\mathbb{E}_X\big[\|v(X)\|^2\big] \\
&= \tau_g^2\int_{\mathbb{R}^d}\|\nabla\log p(x)\|^2 p(x)dx
\end{aligned}$$

recovering the Fisher-information functional $J(p)$, scaled by τ_g^2

2) The cross term $2\mathbb{E}[A(X)C(X)]$. We have

$$A(x)C(x) = \frac{1}{2}\big(\nabla\eta(x)^\top v(x)\big)\Delta\eta(x).$$

Under the prior, $\nabla\eta$ is mean-zero and isotropic; if, additionally, $\Delta\eta$ is uncorrelated with $\nabla\eta$ and has zero mean (or is bounded and mean-zero after centering), then $\mathbb{E}_\eta[A(x)C(x)] = 0$. If one does *not* assume the orthogonality/vanishing covariance above, then $\mathbb{E}[A(X)C(X)]$ is a finite constant (depending on the joint law of derivatives of η), and the cross term contributes

$$\left(\frac{r_0^2}{d+2}\right)^2 \cdot 2\mathbb{E}[A(X)C(X)] = \ O(r_0^4),$$

not $o(r_0^4)$. In that general case, the leading p-dependent term of $\mathbb{E}[\text{Bias}(X)^2]$ is still the *score-gradient* $\tau_g^2 J(p)$.

3) The curvature term $\mathbb{E}[C(X)^2]$.

$$\begin{aligned}
\mathbb{E}\big[C(X)^2\big] &= \mathbb{E}_X\big[\mathbb{E}_\eta[C(X)^2]\big] \\
&= \frac{1}{4}\mathbb{E}_X\big[\mathbb{E}_\eta[(\Delta\eta(X))^2]\big]
\end{aligned}$$

which is independent of p, hence $\mathbb{E}\big[C(X)^2\big] = O(1)$

这里定义 $A(x) \triangleq \nabla\eta(x) \cdot \nabla \log p(x)$ 和 $C(x) \triangleq \frac{1}{2}\Delta\eta(x)$。下面取平方，并对 $X \sim p$ 及各向同性梯度先验求期望。

$$\mathbb{E}\big[\text{Bias}(X)^2\big] = \mathbb{E}\Big[\left(\frac{r_0^2}{d+2}\right)^2 \big(A(x)^2 + 2A(x)C(x) + C(x)^2\big) + o(r_0^4)\Big] \tag{12}$$

$$= \left(\frac{r_0^2}{d+2}\right)^2 \Big\{ \underbrace{\mathbb{E}\big[A(X)^2\big]}_{\text{score-gradient term}} + 2\underbrace{\mathbb{E}\big[A(X)C(X)\big]}_{\text{cross term}} + \underbrace{\mathbb{E}\big[C(X)^2\big]}_{\text{curvature term}} \Big\} + o(r_0^4). \tag{13}$$

我们分别推导各项，并回顾对 η 的假设：其梯度先验是各向同性的，即 $\mathbb{E}\big[\nabla\eta(x)\big] = 0$ 和 $\mathbb{E}\big[\nabla\eta(x)\nabla\eta(x)^\top\big] = \tau_g^2 I_d$，其中某个 $\tau_g^2 \in (0, \infty)$。

1）得分梯度项 $\mathbb{E}[A(X)^2]$。为简洁起见，使用 $v(x) := \nabla\log p(x)$：

$$\begin{aligned}
\mathbb{E}\big[A(X)^2\big] &= \mathbb{E}_X\big[\mathbb{E}_\eta[A(X)^2]\big] \\
&= \mathbb{E}_X\big[\mathbb{E}_\eta[\big(\nabla\eta(x)^\top v(x)\big)^2]\big] \\
&= \mathbb{E}_X\big[\mathbb{E}_\eta[\nabla\eta(x)^\top\Big(v(x)v(x)^\top\Big)\nabla\eta(x)]\big] \\
&= \mathbb{E}_X\big[\mathbb{E}_\eta[\text{tr}\Big(v(x)v(x)^\top\nabla\eta(x)\nabla\eta(x)^\top\Big)]\big] \\
&= \mathbb{E}_X\big[\text{tr}\Big(v(x)v(x)^\top\mathbb{E}_\eta[\nabla\eta(x)\nabla\eta(x)^\top]\Big)\big] \\
&= \mathbb{E}_X\big[\tau_g^2\|v(x)\|^2\big] \\
&= \tau_g^2\mathbb{E}_X\big[\|v(X)\|^2\big] \\
&= \tau_g^2\int_{\mathbb{R}^d}\|\nabla\log p(x)\|^2 p(x)dx
\end{aligned}$$

得到按 τ_g^2 缩放的 Fisher 信息泛函 $J(p)$。

2）交叉项 $2\mathbb{E}[A(X)C(X)]$。有

$$A(x)C(x) = \frac{1}{2}\big(\nabla\eta(x)^\top v(x)\big)\Delta\eta(x).$$

先验下， $\nabla\eta$ 均值零、各向同性；若 $\Delta\eta$ 与 $\nabla\eta$ 不相关且均值零（或中心化后有界且均值零），则 $\mathbb{E}_\eta[A(x)C(x)] = 0$。不假设正交／协方差为零，则 $\mathbb{E}[A(X)C(X)]$ 为依赖 η导数联合分布的有限常数，交叉项贡献：

$$\left(\frac{r_0^2}{d+2}\right)^2 \cdot 2\mathbb{E}[A(X)C(X)] = \ O(r_0^4),$$

而非 $o(r_0^4)$。在这一一般情形下， $\mathbb{E}[\text{Bias}(X)^2]$ 中依赖于 p 的主导项仍然是得分梯度 $\tau_g^2 J(p)$。

3）曲率项 $\mathbb{E}[C(X)^2]$。

$$\begin{aligned}
\mathbb{E}\big[C(X)^2\big] &= \mathbb{E}_X\big[\mathbb{E}_\eta[C(X)^2]\big] \\
&= \frac{1}{4}\mathbb{E}_X\big[\mathbb{E}_\eta[(\Delta\eta(X))^2]\big]
\end{aligned}$$

它与 p 无关，因此 $\mathbb{E}\big[C(X)^2\big] = O(1)$。

Putting it together. Substituting into (13):

$$\mathbb{E}\big[\mathrm{Bias}(X)^2\big] = \left(\frac{r_0^2}{d+2}\right)^2 \left\{\tau_g^2 J(p) + O(1)\right\} + o(r_0^4)$$
$$= \frac{r_0^4}{(d+2)^2}\tau_g^2 J(p) \;+\; O(r_0^4),$$

We show that, among all mean-zero distributions p on $\mathbb{R}^d$ with a given *scalar* constraint on the covariance (trace, determinant, Frobenius norm, or spectral radius), the density that minimizes the Fisher-information functional

$$J(p) \;:=\; \int_{\mathbb{R}^d} \|\nabla \log p(x)\|^2 p(x)dx$$

is the Gaussian with *isotropic* covariance satisfying the same scalar constraint. We proceed in two steps: (i) for fixed covariance matrix $\Sigma \succ 0$, $J(p)$ is minimized by the Gaussian $\mathcal{N}(0,\Sigma)$ and attains the value $\mathrm{tr}(\Sigma^{-1})$; (ii) for each scalar constraint, $\mathrm{tr}(\Sigma^{-1})$ is minimized by $\Sigma = sI_d$ for the appropriate scalar $s > 0$.

Lemma 6: Special case: Recovery of VCReg

Let p be a mean-zero probability density on $\mathbb{R}^d$ with covariance $\Sigma = \mathbb{E}[XX^\top] \succ 0$. Then

$$J(p) \;\geq\; \mathrm{tr}(\Sigma^{-1}),$$

with equality if and only if $p = \mathcal{N}(0,\Sigma)$.

Proof. Consider the location family $p_\theta(x) := p(x-\theta)$, $\theta \in \mathbb{R}^d$. Its Fisher-information matrix at θ is

$$\mathcal{I}(\theta) \;=\; \mathbb{E}\big[\nabla_\theta \log p_\theta(X) \nabla_\theta \log p_\theta(X)^\top\big] \;=\; \mathbb{E}\big[\nabla \log p(X) \nabla \log p(X)^\top\big],$$

so that $J(p) = \mathrm{tr}\mathcal{I}(\theta)$. The estimator $T(X) \equiv X$ is unbiased for θ under p_θ, with $\mathrm{Cov}(T) = \Sigma$. The matrix Cramér–Rao bound gives $\mathrm{Cov}(T) \succeq \mathcal{I}(\theta)^{-1}$, i.e., $\mathcal{I}(\theta) \succeq \Sigma^{-1}$. Taking traces yields $J(p) \geq \mathrm{tr}(\Sigma^{-1})$. Equality in the matrix Cramér–Rao bound holds if and only if the score is an *affine* function of $X-\theta$, i.e., $\nabla \log p_\theta(X) = A(X-\theta)$ a.s. for some matrix A; integrating this identity shows p_θ is Gaussian with precision matrix $-A$, hence $p = \mathcal{N}(0,\Sigma)$. □

Step 2: Optimizing over covariance shapes under scalar constraints

Write the eigenvalues of Σ as $\lambda_1, \ldots, \lambda_d > 0$. Then

$$\mathrm{tr}(\Sigma^{-1}) = \sum_{i=1}^d \frac{1}{\lambda_i}.$$

We now solve $\min \sum_i 1/\lambda_i$ under each scalar constraint; in every case the minimum is attained when all λ_i are equal, i.e., $\Sigma = sI_d$.

(a) Trace constraint. Given $\mathrm{tr}(\Sigma) = \sum_i \lambda_i = t > 0$, by Cauchy–Schwarz,

$$\left(\sum_{i=1}^d \frac{1}{\lambda_i}\right)\left(\sum_{i=1}^d \lambda_i\right) \;\geq\; \left(\sum_{i=1}^d 1\right)^2 = d^2,$$

with equality if and only if $\lambda_1 = \cdots = \lambda_d$. Hence

$$\min_{\Sigma \succ 0:\ \mathrm{tr}(\Sigma)=t} \mathrm{tr}(\Sigma^{-1}) \;=\; \frac{d^2}{t}, \quad \text{attained at} \quad \Sigma = \frac{t}{d} I_d.$$

综合各项。代入式（13）：

$$\mathbb{E}\big[\mathrm{Bias}(X)^2\big] = \left(\frac{r_0^2}{d+2}\right)^2 \left\{\tau_g^2 J(p) + O(1)\right\} + o(r_0^4)$$
$$= \frac{r_0^4}{(d+2)^2}\tau_g^2 J(p) \;+\; O(r_0^4),$$

我们证明，在 $\mathbb{R}^d$ 上所有均值为零、协方差满足给定标量约束（迹、行列式、Frobenius 范数或谱半径）的分布 p 中，使 Fisher 信息泛函最小的密度

$$J(p) \;:=\; \int_{\mathbb{R}^d} \|\nabla \log p(x)\|^2 p(x)dx$$

为满足同一标量约束的各向同性协方差高斯分布。我们分两步进行：（i）对固定的协方差矩阵 $\Sigma \succ 0$，$J(p)$ 在高斯分布 $\mathcal{N}(0,\Sigma)$ 下最小，且达到 $\mathrm{tr}(\Sigma^{-1})$；（ii）在每种标量约束下，$\mathrm{tr}(\Sigma^{-1})$ 均在 $\Sigma = sI_d$ 下最小，其中 $s > 0$ 为相应的标量。

引理 6：特殊情形：恢复 VCReg

设 p 为 $\mathbb{R}^d$ 上均值为零、协方差为 $\Sigma = \mathbb{E}[XX^\top] \succ 0$ 的概率密度。则

$$J(p) \;\geq\; \mathrm{tr}(\Sigma^{-1}),$$

等号成立当且仅当 $p = \mathcal{N}(0,\Sigma)$。

证明。考虑位置族 $p_\theta(x) := p(x-\theta)$，$\theta \in \mathbb{R}^d$。其在 θ 处的 Fisher 信息矩阵为

$$\mathcal{I}(\theta) \;=\; \mathbb{E}\big[\nabla_\theta \log p_\theta(X) \nabla_\theta \log p_\theta(X)^\top\big] \;=\; \mathbb{E}\big[\nabla \log p(X) \nabla \log p(X)^\top\big],$$

所以 $J(p) = \mathrm{trI}(\theta)$。估计器 $T(X) \equiv X$ 在 p_θ下对 θ 无偏，$\mathrm{Cov}(T) = \Sigma$。矩阵 Cramér–Rao 界给出 $\mathrm{Cov}(T) \succeq \mathrm{I}(\theta)^{-1}$，即 $\mathrm{I}(\theta) \succeq \Sigma^{-1}$。取迹得 $J(p) \geq \mathrm{tr}(\Sigma^{-1})$。等号当且仅当分数为 $X-\theta$的仿射函数，即某矩阵 A满足 $\nabla \log p_\theta(X) = A(X-\theta)$ 几乎处处成立。积分此恒等式得 p_θ 是精度矩阵 $-A$的高斯，故 $p = \mathcal{N}(0,\Sigma)$。□

步骤 2：在标量约束下优化协方差形状

将 Σ 的特征值写为 $\lambda_1, \ldots, \lambda_d > 0$。于是

$$\mathrm{tr}(\Sigma^{-1}) = \sum_{i=1}^d \frac{1}{\lambda_i}.$$

下面在每种标量约束下求解 $\min \sum_i 1/\lambda_i$；所有情形中，当全部 λ_i 相等，即 $\Sigma = sI_d$ 时，均达到最小值。

（a）迹约束。给定 $\mathrm{tr}(\Sigma) = \sum_i \lambda_i = t > 0$，根据 Cauchy–Schwarz 不等式，

$$\left(\sum_{i=1}^d \frac{1}{\lambda_i}\right)\left(\sum_{i=1}^d \lambda_i\right) \;\geq\; \left(\sum_{i=1}^d 1\right)^2 = d^2,$$

等号成立当且仅当 $\lambda_1 = \cdots = \lambda_d$。因此

$$\min_{\Sigma \succ 0:\ \mathrm{tr}(\Sigma)=t} \mathrm{tr}(\Sigma^{-1}) \;=\; \frac{d^2}{t}, \quad \text{attained at} \quad \Sigma = \frac{t}{d} I_d.$$

(b) Determinant constraint. Given $\det(\Sigma) = \prod_i \lambda_i = \delta > 0$, set $\mu_i := 1/\lambda_i$ so that $\prod_i \mu_i = \delta^{-1}$. By the AM–GM inequality,

$$\frac{1}{d}\sum_{i=1}^{d} \mu_i \;\geq\; \left(\prod_{i=1}^{d} \mu_i\right)^{1/d} = \delta^{-1/d},$$

with equality iff $\mu_1 = \cdots = \mu_d$, i.e., $\lambda_1 = \cdots = \lambda_d$. Thus

$$\min_{\Sigma\succ 0:\ \det(\Sigma)=\delta} \operatorname{tr}(\Sigma^{-1}) \;=\; d\delta^{-1/d}, \quad \text{attained at} \quad \Sigma = \delta^{1/d} I_d.$$

(c) Frobenius-norm constraint. Given $\|\Sigma\|_F^2 = \sum_i \lambda_i^2 = c^2 > 0$, minimize $f(\lambda) := \sum_i 1/\lambda_i$ over $\lambda_i > 0$ subject to $g(\lambda) := \sum_i \lambda_i^2 = c^2$. The Lagrangian

$$\mathcal{L}(\lambda, \nu) = \sum_{i=1}^{d} \frac{1}{\lambda_i} + \nu\left(\sum_{i=1}^{d} \lambda_i^2 - c^2\right)$$

has first-order conditions $-\lambda_i^{-2} + 2\nu\lambda_i = 0$ for all i, i.e., $\lambda_i^3 = \frac{1}{2\nu}$, so all λ_i are equal. Imposing $\sum \lambda_i^2 = c^2$ yields $\lambda_i = c/\sqrt{d}$, hence

$$\min_{\Sigma\succ 0:\ \|\Sigma\|_F=c} \operatorname{tr}(\Sigma^{-1}) \;=\; \sum_{i=1}^{d} \frac{1}{\lambda_i} \;=\; \frac{d^{3/2}}{c}, \quad \text{attained at} \quad \Sigma = \frac{c}{\sqrt{d}} I_d.$$

(d) Spectral-radius constraint. Let the spectral radius be constrained by $\rho(\Sigma) = \max_i \lambda_i \leq r$ for some $r > 0$. Since $x \mapsto 1/x$ is strictly decreasing on $(0, \infty)$,

$$\sum_{i=1}^{d} \frac{1}{\lambda_i} \;\geq\; \sum_{i=1}^{d} \frac{1}{r} \;=\; \frac{d}{r},$$

with equality if and only if $\lambda_i = r$ for all i. Therefore

$$\min_{\Sigma\succ 0:\ \rho(\Sigma)\leq r} \operatorname{tr}(\Sigma^{-1}) \;=\; \frac{d}{r}, \quad \text{attained at} \quad \Sigma = rI_d.$$

(The same conclusion holds if the constraint is $\rho(\Sigma) = r$, since one may take all eigenvalues equal to r.)

Conclusion: Isotropic Gaussian is optimal

Combining Lemma 6 with the solutions (a)–(d), we obtain:

（b）行列式约束。给定 $\det(\Sigma) = \prod_i \lambda_i = \delta > 0$，令 $\mu_i := 1/\lambda_i$，从而 $\prod_i \mu_i = \delta^{-1}$。根据算术平均数–几何平均数不等式，

$$\frac{1}{d}\sum_{i=1}^{d} \mu_i \;\geq\; \left(\prod_{i=1}^{d} \mu_i\right)^{1/d} = \delta^{-1/d},$$

等号成立当且仅当 $\mu_1 = \cdots = \mu_d$，即 $\lambda_1 = \cdots = \lambda_d$。因此

$$\min_{\Sigma\succ 0:\ \det(\Sigma)=\delta} \operatorname{tr}(\Sigma^{-1}) \;=\; d\delta^{-1/d}, \quad \text{attained at} \quad \Sigma = \delta^{1/d} I_d.$$

（c）Frobenius 范数约束。给定 $\|\Sigma\|_F^2 = \sum_i \lambda_i^2 = c^2 > 0$，在 $\lambda_i > 0$ 上最小化 $f(\lambda) := \sum_i 1/\lambda_i$，约束为 $g(\lambda) := \sum_i \lambda_i^2 = c^2$。拉格朗日函数为

$$\mathcal{L}(\lambda, \nu) = \sum_{i=1}^{d} \frac{1}{\lambda_i} + \nu\left(\sum_{i=1}^{d} \lambda_i^2 - c^2\right)$$

其一阶条件为：对所有 i，均有 $-\lambda_i^{-2} + 2\nu\lambda_i = 0$，即 $\lambda_i^3 = \frac{1}{2\nu}$，因此所有 λ_i 相等。施加 $\sum \lambda_i^2 = c^2$ 得到 $\lambda_i = c/\sqrt{d}$，因此

$$\min_{\Sigma\succ 0:\ \|\Sigma\|_F=c} \operatorname{tr}(\Sigma^{-1}) \;=\; \sum_{i=1}^{d} \frac{1}{\lambda_i} \;=\; \frac{d^{3/2}}{c}, \quad \text{attained at} \quad \Sigma = \frac{c}{\sqrt{d}} I_d.$$

（d）谱半径约束。设谱半径满足 $\rho(\Sigma) = \max_i \lambda_i \leq r$，其中某个 $r > 0$。由于 $x \mapsto 1/x$ 在 $(0, \infty)$ 上严格递减，

$$\sum_{i=1}^{d} \frac{1}{\lambda_i} \;\geq\; \sum_{i=1}^{d} \frac{1}{r} \;=\; \frac{d}{r},$$

等号成立当且仅当对所有 i 均有 $\lambda_i = r$。因此

$$\min_{\Sigma\succ 0:\ \rho(\Sigma)\leq r} \operatorname{tr}(\Sigma^{-1}) \;=\; \frac{d}{r}, \quad \text{attained at} \quad \Sigma = rI_d.$$

（若约束为 $\rho(\Sigma) = r$，结论也相同，因为可以令所有特征值均等于 r。）

结论：各向同性高斯分布最优

将引理 6 与（a）至（d）的解结合，可得：

Theorem 9: Special case: Recovery of VCReg

Fix one of the following scalar covariance constraints for a mean-zero distribution p on $\mathbb{R}^d$:

- trace: $\mathrm{tr}(\mathrm{Cov}(X)) = t$,
- determinant: $\det(\mathrm{Cov}(X)) = \delta$,
- Frobenius norm: $\|\mathrm{Cov}(X)\|_F = c$,
- spectral radius upper bound: $\rho(\mathrm{Cov}(X)) \le r$.

Then the Fisher-information functional $J(p)$ is minimized over all such p by the isotropic Gaussian $p_G = \mathcal{N}(0, sI_d)$ with s chosen to satisfy the constraint. The minimal values are:

$$\begin{aligned}
&\text{trace } t: && J_{\min} = \frac{d^2}{t}, \quad s = \frac{t}{d},\\
&\text{determinant } \delta: && J_{\min} = d\delta^{-1/d}, \quad s = \delta^{1/d},\\
&\text{Frobenius } c: && J_{\min} = \frac{d^{3/2}}{c}, \quad s = \frac{c}{\sqrt{d}},\\
&\text{spectral radius } r: && J_{\min} = \frac{d}{r}, \quad s = r.
\end{aligned}$$

In each case, p_G is the unique minimizer (up to null sets).

Proof. For any admissible p with covariance Σ, Lemma 6 gives $J(p) \ge \mathrm{tr}(\Sigma^{-1})$. Minimizing the right-hand side under the stated scalar constraint yields $\Sigma = sI_d$ by the calculations in (a)–(d). Equality in Lemma 6 holds if and only if p is Gaussian with that covariance, hence p_G uniquely attains the bound. □

□

B.5 Proof of lemma. 5

Proof. Write the numerator and denominator of $\widehat{m}(x)$ as

$$B_n(x) := \sum_{i=1}^{n} K_h(x - X_i)Y_i, \qquad A_n(x) := \sum_{i=1}^{n} K_h(x - X_i),$$

so that $\widehat{m}(x) = \frac{B_n(x)}{A_n(x)}$. *Bias.* Compute expectations using independence and change of variables. For the denominator,

$$\begin{aligned}
\mathbb{E}[A_n(x)] &= n\mathbb{E}\Big[K_h(x - X)\Big]\\
&= n\int_{\mathbb{R}^d} h^{-d}K\Big(\frac{x-u}{h}\Big)p(u)du\\
&= n\int_{\mathbb{R}^d} K(t)p(x - ht)dt \qquad (t := (x-u)/h)\\
&= n\int_{\mathbb{R}^d} K(t)\Big(p(x) - ht^\top\nabla p(x) + \frac{h^2}{2}t^\top\nabla^2 p(x)t + o(h^2)\Big)dt\\
&= n\Big(p(x) + \frac{h^2}{2}\underbrace{\int t^\top\nabla^2 p(x)tK(t)dt}_{=\mu_2(K)\Delta p(x)} + o(h^2)\Big),
\end{aligned}$$

定理 9：特殊情形：恢复 VCReg

对于 $\mathbb{R}^d$ 上均值为零的分布 p，固定以下一种标量协方差约束：

- 迹：$\mathrm{tr}(\mathrm{Cov}(X)) = t$,
- 行列式：$\det(\mathrm{Cov}(X)) = \delta$,
- Frobenius 范数：$\|\mathrm{Cov}(X)\|_F = c$,
- 谱半径上界：$\rho(\mathrm{Cov}(X)) \le r$。

则在所有这样的 p 中，Fisher 信息泛函 $J(p)$ 在各向同性高斯分布 $p_G = \mathcal{N}(0, sI_d)$ 下最小，其中选择 s 以满足约束。最小值为：

$$\begin{aligned}
&\text{trace } t: && J_{\min} = \frac{d^2}{t}, \quad s = \frac{t}{d},\\
&\text{determinant } \delta: && J_{\min} = d\delta^{-1/d}, \quad s = \delta^{1/d},\\
&\text{Frobenius } c: && J_{\min} = \frac{d^{3/2}}{c}, \quad s = \frac{c}{\sqrt{d}},\\
&\text{spectral radius } r: && J_{\min} = \frac{d}{r}, \quad s = r.
\end{aligned}$$

每种情形中，p_G 均为唯一最小化解（忽略零测集）。

证明。对于任意协方差为 Σ 的可行 p，引理 6 给出 $J(p) \ge \mathrm{tr}(\Sigma^{-1})$。根据（a）至（d）中的计算，在所述标量约束下最小化右侧可得 $\Sigma = sI_d$。引理 6 中等号成立，当且仅当 p 是具有该协方差的高斯分布，因此 p_G 唯一地达到该界。□

□

B.5 引理 5 的证明

证明。将 $\widehat{m}(x)$ 的分子和分母写为

$$B_n(x) := \sum_{i=1}^{n} K_h(x - X_i)Y_i, \qquad A_n(x) := \sum_{i=1}^{n} K_h(x - X_i),$$

从而 $\widehat{m}(x) = \frac{B_n(x)}{A_n(x)}$。偏差。利用独立性和变量代换计算期望。对于分母，

r,

$$\begin{aligned}
\mathbb{E}[A_n(x)] &= n\mathbb{E}\Big[K_h(x - X)\Big]\\
&= n\int_{\mathbb{R}^d} h^{-d}K\Big(\frac{x-u}{h}\Big)p(u)du\\
&= n\int_{\mathbb{R}^d} K(t)p(x - ht)dt \qquad (t := (x-u)/h)\\
&= n\int_{\mathbb{R}^d} K(t)\Big(p(x) - ht^\top\nabla p(x) + \frac{h^2}{2}t^\top\nabla^2 p(x)t + o(h^2)\Big)dt\\
&= n\Big(p(x) + \frac{h^2}{2}\underbrace{\int t^\top\nabla^2 p(x)tK(t)dt}_{=\mu_2(K)\Delta p(x)} + o(h^2)\Big),
\end{aligned}$$

where we used symmetry $\int tK(t)dt = 0$ and isotropy $\int tt^\top K(t)dt = \mu_2(K)I_d$, which implies $\int t^\top \nabla^2 p(x) t K(t)dt = \mu_2(K)\mathrm{tr}(\nabla^2 p(x)) = \mu_2(K)\Delta p(x)$. Similarly, for the numerator,

$$\begin{aligned}
\mathbb{E}[B_n(x)] &= n\mathbb{E}\big[K_h(x-X)Y\big] = n\int K(t)(mp)(x-ht)dt \\
&= n\int K(t)\Big((mp)(x) - ht^\top\nabla(mp)(x) + \frac{h^2}{2}t^\top\nabla^2(mp)(x)t + o(h^2)\Big)dt \\
&= n\Big(m(x)p(x) + \frac{h^2}{2}\mu_2(K)\mathrm{tr}\big(\nabla^2(mp)(x)\big) + o(h^2)\Big) \\
&= n\Big(m(x)p(x) + \frac{h^2\mu_2(K)}{2}\big(p\Delta m + m\Delta p + 2\nabla m^\top\nabla p\big)(x) + o(h^2)\Big),
\end{aligned}$$

where the last step uses the fact that $\mathrm{tr}\big(\nabla^2(mp)\big) = p\Delta m + m\Delta p + 2\nabla m^\top\nabla p$ by the product rule and symmetry of mixed derivatives.

Now expand the ratio $\frac{\mathbb{E}[B_n(x)]}{\mathbb{E}[A_n(x)]}$ using the identity

$$\frac{a_0 + h^2a_2 + o(h^2)}{b_0 + h^2b_2 + o(h^2)} = \frac{a_0}{b_0} + h^2\frac{a_2b_0 - a_0b_2}{b_0^2} + o(h^2),$$

with $a_0 = m(x)p(x)$, $a_2 = \frac{\mu_2(K)}{2}\big(p\Delta m + m\Delta p + 2\nabla m^\top\nabla p\big)(x)$, $b_0 = p(x)$, and $b_2 = \frac{\mu_2(K)}{2}\Delta p(x)$. This yields

$$\begin{aligned}
\frac{\mathbb{E}[B_n(x)]}{\mathbb{E}[A_n(x)]} &= m(x) + \frac{h^2\mu_2(K)}{2}\frac{\big(p\Delta m + m\Delta p + 2\nabla m^\top\nabla p\big)p - mp\Delta p}{p^2}\Big|_x + o(h^2) \\
&= m(x) + \frac{h^2\mu_2(K)}{2}\Big(\Delta m(x) + 2\nabla m(x)^\top\frac{\nabla p(x)}{p(x)}\Big) + o(h^2),
\end{aligned}$$

which recovers our statement. *Variance.* Linearize $\widehat{m}(x) = B_n(x)/A_n(x)$ around $(\mathbb{E}[B_n(x)], \mathbb{E}[A_n(x)])$ and use independence. To leading order,

$$\mathrm{Var}[\widehat{m}(x)] \approx \frac{\mathrm{Var}[B_n(x)]}{(\mathbb{E}[A_n(x)])^2}.$$

Compute

$$\begin{aligned}
\mathrm{Var}[B_n(x)] &= \sum_{i=1}^n \mathrm{Var}\big(K_h(x-X_i)Y_i\big) \quad \text{(independence)} \\
&= n\mathbb{E}\big[K_h(x-X)^2\mathrm{Var}(Y\mid X)\big] = n\mathbb{E}\big[K_h(x-X)^2 v(X)\big] \\
&= n\int h^{-2d}K\Big(\frac{x-u}{h}\Big)^2 v(u)p(u)du \\
&= nh^{-d}\int K(t)^2 v(x-ht)p(x-ht)dt = nh^{-d}\Big(R(K)v(x)p(x) + o(1)\Big),
\end{aligned}$$

while

$$\mathbb{E}[A_n(x)] = n\big(p(x) + o(1)\big).$$

Therefore,

$$\mathrm{Var}[\widehat{m}(x)] \approx \frac{nh^{-d}R(K)v(x)p(x)}{n^2p(x)^2} = \frac{R(K)}{nh^d}\frac{v(x)}{p(x)} + o\big((nh^d)^{-1}\big),$$

completing the proof. □

B.6 Proof of Equation (5) to Equation (6)

Proof. Let $\bar{\mathbf{z}} = \frac{1}{V_g}\sum_{v=1}^{V_g}\mathbf{z}_{n,v}$ denote the mean of the first V_g vectors.

这里使用了对称性 $\int tK\,t\,dt = 0$ $\int tt^\top K\,t\,dt = \mu_2\,K\,I_d$ $\int t^\top\nabla^2 p\,x\,tK\,t\,dt = ()$ $\mu_2\,K\quad\nabla^2 p\,x\;= \mu_2\,K\,\Delta p\,x$ 和各向同性 ( ) ( )，由此可得 ( ) ( ) ( )tr(( )) ( ) ( )。同理，对于分子，

$$\begin{aligned}
\mathbb{E}[B_n(x)] &= n\mathbb{E}\big[K_h(x-X)Y\big] = n\int K(t)(mp)(x-ht)dt \\
&= n\int K(t)\Big((mp)(x) - ht^\top\nabla(mp)(x) + \frac{h^2}{2}t^\top\nabla^2(mp)(x)t + o(h^2)\Big)dt \\
&= n\Big(m(x)p(x) + \frac{h^2}{2}\mu_2(K)\mathrm{tr}\big(\nabla^2(mp)(x)\big) + o(h^2)\Big) \\
&= n\Big(m(x)p(x) + \frac{h^2\mu_2(K)}{2}\big(p\Delta m + m\Delta p + 2\nabla m^\top\nabla p\big)(x) + o(h^2)\Big),
\end{aligned}$$

最后一步使用了 $\mathrm{tr}\big(\nabla^2(mp)\big) = p\Delta m + m\Delta p + 2\nabla m^\top\nabla p$，这来自乘积法则及混合导数的对称性。

现在使用以下恒等式展~~开比值~~ $\frac{\mathbb{E}[B_n\;x\;]}{\mathbb{E}[A_n\;x\;]}$ ( ) ( ):

$$\frac{a_0 + h^2a_2 + o(h^2)}{b_0 + h^2b_2 + o(h^2)} = \frac{a_0}{b_0} + h^2\frac{a_2b_0 - a_0b_2}{b_0^2} + o(h^2),$$

其中 $a_0 = m(x)p(x)$、 $a_2 = \frac{\mu_2(K)}{2}\big(p\Delta m + m\Delta p + 2\nabla m^\top\nabla p\big)(x)$、 $b_0 = p(x)$， 以及 $b_2 = \frac{\mu_2(K)}{2}\Delta p(x)$。由此可得

$$\begin{aligned}
\frac{\mathbb{E}[B_n(x)]}{\mathbb{E}[A_n(x)]} &= m(x) + \frac{h^2\mu_2(K)}{2}\frac{\big(p\Delta m + m\Delta p + 2\nabla m^\top\nabla p\big)p - mp\Delta p}{p^2}\Big|_x + o(h^2) \\
&= m(x) + \frac{h^2\mu_2(K)}{2}\Big(\Delta m(x) + 2\nabla m(x)^\top\frac{\nabla p(x)}{p(x)}\Big) + o(h^2),
\end{aligned}$$

由此得到所述结论。方差。在 $(\mathbb{E}[B_n(x)], \mathbb{E}[A_n(x)])$ 附近对 $\widehat{m}(x) = B_n(x)/A_n(x)$ 线性化，并利用独立性。保留主导阶项，

$$\mathrm{Var}[\widehat{m}(x)] \approx \frac{\mathrm{Var}[B_n(x)]}{(\mathbb{E}[A_n(x)])^2}.$$

计算量

$$\begin{aligned}
\mathrm{Var}[B_n(x)] &= \sum_{i=1}^n \mathrm{Var}\big(K_h(x-X_i)Y_i\big) \quad \text{(independence)} \\
&= n\mathbb{E}\big[K_h(x-X)^2\mathrm{Var}(Y\mid X)\big] = n\mathbb{E}\big[K_h(x-X)^2 v(X)\big] \\
&= n\int h^{-2d}K\Big(\frac{x-u}{h}\Big)^2 v(u)p(u)du \\
&= nh^{-d}\int K(t)^2 v(x-ht)p(x-ht)dt = nh^{-d}\Big(R(K)v(x)p(x) + o(1)\Big),
\end{aligned}$$

而

$$\mathbb{E}[A_n(x)] = n\big(p(x) + o(1)\big).$$

因此，

$$\mathrm{Var}[\widehat{m}(x)] \approx \frac{nh^{-d}R(K)v(x)p(x)}{n^2p(x)^2} = \frac{R(K)}{nh^d}\frac{v(x)}{p(x)} + o\big((nh^d)^{-1}\big),$$

证明完毕。 □

B.6 从式（5）到式（6）的推导证明

证明。令 $\bar{\mathrm{z}} = \frac{1}{V_g}\sum_{v=1}^{V_g}\mathrm{z}_{n,v}$ 表示前 V_g 个向量的均值。

We prove that:

$$\frac{1}{V_g}\sum_{v=1}^{V_g}\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v}-\mathbf{z}_{n,v'}\|_2^2=\frac{1}{V}\sum_{v'=1}^{V}\|\bar{\mathbf{z}}-\mathbf{z}_{n,v'}\|_2^2 \tag{14}$$

Expanding the left-hand side:

$$\begin{aligned}\text{LHS}&=\frac{1}{V_gV}\sum_{v=1}^{V_g}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v}-\mathbf{z}_{n,v'}\|_2^2 &(15)\\&=\frac{1}{V_gV}\sum_{v=1}^{V_g}\sum_{v'=1}^{V}\left(\|\mathbf{z}_{n,v}\|_2^2-2\mathbf{z}_{n,v}^T\mathbf{z}_{n,v'}+\|\mathbf{z}_{n,v'}\|_2^2\right) &(16)\\&=\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2-\frac{2}{V_gV}\sum_{v=1}^{V_g}\sum_{v'=1}^{V}\mathbf{z}_{n,v}^T\mathbf{z}_{n,v'}+\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v'}\|_2^2 &(17)\\&=\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2-\frac{2}{V}\bar{\mathbf{z}}^T\sum_{v'=1}^{V}\mathbf{z}_{n,v'}+\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v'}\|_2^2 &(18)\end{aligned}$$

Expanding the right-hand side:

$$\begin{aligned}\text{RHS}&=\frac{1}{V}\sum_{v'=1}^{V}\left(\|\bar{\mathbf{z}}\|_2^2-2\bar{\mathbf{z}}^T\mathbf{z}_{n,v'}+\|\mathbf{z}_{n,v'}\|_2^2\right) &(19)\\&=\|\bar{\mathbf{z}}\|_2^2-\frac{2}{V}\bar{\mathbf{z}}^T\sum_{v'=1}^{V}\mathbf{z}_{n,v'}+\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v'}\|_2^2 &(20)\end{aligned}$$

To complete the proof, we verify that:

$$\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2=\|\bar{\mathbf{z}}\|_2^2 \tag{21}$$

Expanding the right-hand side:

$$\begin{aligned}\|\bar{\mathbf{z}}\|_2^2&=\left\|\frac{1}{V_g}\sum_{v=1}^{V_g}\mathbf{z}_{n,v}\right\|_2^2 &(22)\\&=\frac{1}{V_g^2}\sum_{v=1}^{V_g}\sum_{v''=1}^{V_g}\mathbf{z}_{n,v}^T\mathbf{z}_{n,v''} &(23)\\&=\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2 &(24)\end{aligned}$$

Therefore, LHS = RHS, completing the proof. □

B.7 Proof of thm. 8

Proof. For each x,

$$\text{Bias}[\widehat{m}(x)]=\frac{h^2\mu_2(K)}{2}\Big(\Delta m(x)+2\nabla m(x)^\top\nabla\log p(x)\Big)+o(h^2).$$

我们证明：

$$\frac{1}{V_g}\sum_{v=1}^{V_g}\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v}-\mathbf{z}_{n,v'}\|_2^2=\frac{1}{V}\sum_{v'=1}^{V}\|\bar{\mathbf{z}}-\mathbf{z}_{n,v'}\|_2^2 \tag{14}$$

展开左侧：

$$\begin{aligned}\text{左侧}&=\frac{1}{V_gV}\sum_{v=1}^{V_g}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v}-\mathbf{z}_{n,v'}\|_2^2 &(15)\\&=\frac{1}{V_gV}\sum_{v=1}^{V_g}\sum_{v'=1}^{V}\left(\|\mathbf{z}_{n,v}\|_2^2-2\mathbf{z}_{n,v}^T\mathbf{z}_{n,v'}+\|\mathbf{z}_{n,v'}\|_2^2\right) &(16)\\&=\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2-\frac{2}{V_gV}\sum_{v=1}^{V_g}\sum_{v'=1}^{V}\mathbf{z}_{n,v}^T\mathbf{z}_{n,v'}+\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v'}\|_2^2 &(17)\\&=\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2-\frac{2}{V}\bar{\mathbf{z}}^T\sum_{v'=1}^{V}\mathbf{z}_{n,v'}+\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v'}\|_2^2 &(18)\end{aligned}$$

展开右侧：

$$\begin{aligned}\text{RHS}&=\frac{1}{V}\sum_{v'=1}^{V}\left(\|\bar{\mathbf{z}}\|_2^2-2\bar{\mathbf{z}}^T\mathbf{z}_{n,v'}+\|\mathbf{z}_{n,v'}\|_2^2\right) &(19)\\&=\|\bar{\mathbf{z}}\|_2^2-\frac{2}{V}\bar{\mathbf{z}}^T\sum_{v'=1}^{V}\mathbf{z}_{n,v'}+\frac{1}{V}\sum_{v'=1}^{V}\|\mathbf{z}_{n,v'}\|_2^2 &(20)\end{aligned}$$

为完成证明，我们验证：

$$\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2=\|\bar{\mathbf{z}}\|_2^2 \tag{21}$$

展开右侧：

$$\begin{aligned}\|\bar{\mathbf{z}}\|_2^2&=\left\|\frac{1}{V_g}\sum_{v=1}^{V_g}\mathbf{z}_{n,v}\right\|_2^2 &(22)\\&=\frac{1}{V_g^2}\sum_{v=1}^{V_g}\sum_{v''=1}^{V_g}\mathbf{z}_{n,v}^T\mathbf{z}_{n,v''} &(23)\\&=\frac{1}{V_g}\sum_{v=1}^{V_g}\|\mathbf{z}_{n,v}\|_2^2 &(24)\end{aligned}$$

因此，左侧 =右侧，证明完毕。 □

B.7 定理 8 的证明

证明。对每个 x,

$$\text{Bias}[\widehat{m}(x)]=\frac{h^2\mu_2(K)}{2}\Big(\Delta m(x)+2\nabla m(x)^\top\nabla\log p(x)\Big)+o(h^2).$$

Square and integrate against $p(x)$:

$$\begin{aligned}\mathcal{B}^2(h;p,m) &= \left(\frac{h^2\mu_2(K)}{2}\right)^2 \int \left(\Delta m(x) + 2\nabla m(x)^\top \nabla \log p(x)\right)^2 p(x)dx + o(h^4) \\ &\le \left(\frac{h^2\mu_2(K)}{2}\right)^2 \int \left(2(\Delta m(x))^2 + 2(2\nabla m(x)^\top \nabla \log p(x))^2\right) p(x)dx + o(h^4) \\ &= \left(\frac{h^2\mu_2(K)}{2}\right)^2 \left(2\int (\Delta m(x))^2 p(x)dx + 8\int (\nabla m(x)^\top \nabla \log p(x))^2 p(x)dx\right) + o(h^4),\end{aligned}$$

where we used $(a+b)^2 \le 2a^2 + 2b^2$ pointwise. Since $|\Delta m(x)| \le B$ for all x, we have

$$\int (\Delta m)^2 p \le \int B^2 p = B^2.$$

For the second term, first use Cauchy–Schwarz and then integrate against $p(x)$ to obtain

$$\begin{aligned}&(\nabla m(x)^\top \nabla \log p(x))^2 \le \|\nabla m(x)\|^2 \|\nabla \log p(x)\|^2 \le L^2 \|\nabla \log p(x)\|^2 \\ \Longrightarrow &\int (\nabla m(x)^\top \nabla \log p(x))^2 p(x)dx \le L^2 \int \|\nabla \log p(x)\|^2 p(x)dx = L^2 J(p).\end{aligned}$$

which can be combined with the bounds above to obtain the desired result. We similarly have for the integrated variance

$$\mathcal{V}(h;p) = \int \left(\frac{R(K)}{nh^d}\frac{v(x)}{p(x)} + o\big((nh^d)^{-1}\big)\right) p(x)dx = \frac{R(K)}{nh^d}\int v(x)dx + o\big((nh^d)^{-1}\big),$$

which is independent of p. □

B.8 Proof of lemma. 3

Proof. We first start by reminding the reader about the original Cramér-Wold theorem that is a function of all possible directions (not unit-norm ones).

Theorem 10: Cramér-Wold Cramér and Wold [1936]

Let X and Y be random vectors in $\mathbb{R}^D$:

$$X \overset{d}{=} Y \iff \langle X, a\rangle \overset{d}{=} \langle Y, a\rangle, \forall \boldsymbol{a} \in \mathbb{R}^D. \tag{25}$$

Our proof will follow the same proof as for thm. 10. Necessity is immediate: if $X \overset{d}{=} Y$, then every measurable function of X has the same distribution as the corresponding function of Y, from which the linear mapping $x \mapsto \langle u, x\rangle$ for $u \in \mathbb{S}^{d-1}$ is a special case. For sufficiency, assume $\langle u, X\rangle \overset{d}{=} \langle u, Y\rangle$ for all $u \in \mathbb{S}^{d-1}$. Let $\varphi_X(t) := \mathbb{E}\left[e^{i\langle t, X\rangle}\right]$ and $\varphi_Y(t) := \mathbb{E}\left[e^{i\langle t, Y\rangle}\right]$ denote the characteristic functions of X and Y. Fix an arbitrary $t \in \mathbb{R}^d$; if $t = 0$, then $\varphi_X(0) = \varphi_Y(0) = 1$. If $t \ne 0$, write $t = su$ with $s := \|t\| > 0$ and $u := t/\|t\| \in \mathbb{S}^{d-1}$. By the assumption, $\langle u, X\rangle \overset{d}{=} \langle u, Y\rangle$, hence for this u and s we have

$$\varphi_X(t) = \mathbb{E}\left[e^{i\langle t, X\rangle}\right] = \mathbb{E}\left[e^{is\langle u, X\rangle}\right] = \mathbb{E}\left[e^{is\langle u, Y\rangle}\right] = \mathbb{E}\left[e^{i\langle t, Y\rangle}\right] = \varphi_Y(t).$$

Thus $\varphi_X(t) = \varphi_Y(t)$ for all $t \in \mathbb{R}^d$, i.e., $\varphi_X \equiv \varphi_Y$ on $\mathbb{R}^d$. By the uniqueness theorem for characteristic functions, this implies $X \overset{d}{=} Y$. (ii) Define $\psi_{n,t} := \mathbb{E}\left[e^{i\langle t, X_n\rangle}\right]$ and $\psi_t := \mathbb{E}\left[e^{i\langle t, X\rangle}\right]$. Fix $t \in \mathbb{R}^d$ and decompose $t = su$ with $s := \|t\| \ge 0$ and $u \in \mathbb{S}^{d-1}$ (take, e.g., $u = t/\|t\|$ if $t \ne 0$, and any u if $t = 0$). The map $g_s : \mathbb{R} \to \mathbb{R}$, $g_s(x) = sx$, is continuous. By the continuous mapping theorem applied to the real-valued random variables $\langle u, X_n\rangle \xrightarrow{d} \langle u, X\rangle$, we obtain

$$\langle t, X_n\rangle = s\langle u, X_n\rangle \xrightarrow{d} s\langle u, X\rangle = \langle t, X\rangle.$$

取平方并对 $p(x)$ 积分：

$$\begin{aligned}\mathcal{B}^2(h;p,m) &= \left(\frac{h^2\mu_2(K)}{2}\right)^2 \int \left(\Delta m(x) + 2\nabla m(x)^\top \nabla \log p(x)\right)^2 p(x)dx + o(h^4) \\ &\le \left(\frac{h^2\mu_2(K)}{2}\right)^2 \int \left(2(\Delta m(x))^2 + 2(2\nabla m(x)^\top \nabla \log p(x))^2\right) p(x)dx + o(h^4) \\ &= \left(\frac{h^2\mu_2(K)}{2}\right)^2 \left(2\int (\Delta m(x))^2 p(x)dx + 8\int (\nabla m(x)^\top \nabla \log p(x))^2 p(x)dx\right) + o(h^4),\end{aligned}$$

其中逐点使用了 $(a+b)^2 \le 2a^2 + 2b^2$。由于对所有 x 均有 $|\Delta m(x)| \le B$，可得

$$\int (\Delta m)^2 p \le \int B^2 p = B^2.$$

对于第二项，先应用 Cauchy–Schwarz 不等式，再对 $p(x)$ 积分，得到 n

$$\begin{aligned}&(\nabla m(x)^\top \nabla \log p(x))^2 \le \|\nabla m(x)\|^2 \|\nabla \log p(x)\|^2 \le L^2 \|\nabla \log p(x)\|^2 \\ \Longrightarrow &\int (\nabla m(x)^\top \nabla \log p(x))^2 p(x)dx \le L^2 \int \|\nabla \log p(x)\|^2 p(x)dx = L^2 J(p).\end{aligned}$$

它可以与下列项结合 结合上述界得到所需结果。对积分方差，同样有 e

$$\mathcal{V}(h;p) = \int \left(\frac{R(K)}{nh^d}\frac{v(x)}{p(x)} + o\big((nh^d)^{-1}\big)\right) p(x)dx = \frac{R(K)}{nh^d}\int v(x)dx + o\big((nh^d)^{-1}\big),$$

它与 p 无关。 □

B.8 引理 3 的证明

证明。首先回顾原始 Cramér–Wold 定理，它涉及所有可能的方向（不限于单位范数方向）。

定理 10：Cramér–Wold 定理，Cramér and Wold [1936]

设 X 和 Y 是 $\mathbb{R}^D$ 中的随机向量：

$$X \overset{d}{=} Y \iff \langle X, a\rangle \overset{d}{=} \langle Y, a\rangle, \forall \boldsymbol{a} \in \mathbb{R}^D. \tag{25}$$

证明沿用定理 10。必要性显然：若 $X \overset{d}{=} Y$，X 的任意可测函数与 Y 对应函数同分布，线性映射 $x \mapsto u, x$（$u \in \mathbb{S}^{d-1}$）是特例。充分性：假设所有 $u \in \mathbb{S}^{d-1}$ 均有 $u, X \overset{d}{=} u, Y$。令 $\varphi_X(t) := \mathbb{E}\left[e^{i\langle t, X\rangle}\right]$ 和 $\varphi_Y(t) := \mathbb{E}\left[e^{i\langle t, Y\rangle}\right]$ 为 X 与 Y 特征函数。固定任意 $t \in \mathbb{R}^d$；若 $t = 0$，有 $\varphi_X(0) = \varphi_Y(0) = 1$；若 $t \ne 0$，写为 $t = su$，其中 $s := \|t\| > 0$、$u := t/\|t\| \in \mathbb{S}^{d-1}$。按假设，$u, X \overset{d}{=} u, Y$ 因此对此 u、s 有：

$$\varphi_X(t) = \mathbb{E}\left[e^{i\langle t, X\rangle}\right] = \mathbb{E}\left[e^{is\langle u, X\rangle}\right] = \mathbb{E}\left[e^{is\langle u, Y\rangle}\right] = \mathbb{E}\left[e^{i\langle t, Y\rangle}\right] = \varphi_Y(t).$$

因此所有 $t \in \mathbb{R}^d$ 均有 $\varphi_X(t) = \varphi_Y(t)$，即 $\varphi_X \equiv \varphi_Y$ 在 $\mathbb{R}^d$ 上相同。特征函数唯一性定理给出 $X \overset{d}{=} Y$。（ii）定义 $\psi_{n,t}$：$= \mathbb{E}\left[e^{i\langle t, X_n\rangle}\right]$ 以及 $\psi_t := \mathbb{E}\left[e^{i\langle t, X\rangle}\right]$。固定 $t \in \mathbb{R}^d$，分解 $t = su$，其中 $s := \|t\| \ge 0$、$u \in \mathbb{S}^{d-1}$（例如 $t \ne 0$ 时取 $u = t/\|t\|$，$t = 0$ 时任取 u）。映射 $g_s : \mathbb{R} \to \mathbb{R}$, $g_s(x) = sx$ 连续。对实随机变量 $u, X_n \xrightarrow{d} u, X$ 应用连续映射定理，得到：

$$\langle t, X_n\rangle = s\langle u, X_n\rangle \xrightarrow{d} s\langle u, X\rangle = \langle t, X\rangle.$$

Hence, for every fixed $t \in \mathbb{R}^d$, the one-dimensional projections satisfy $\langle t, X_n \rangle \xrightarrow{d} \langle t, X \rangle$, which in turn yields pointwise convergence of characteristic functions:

$$\psi_{n,t} \;=\; \mathbb{E}\big[e^{i\langle t, X_n\rangle}\big] \;\longrightarrow\; \mathbb{E}\big[e^{i\langle t, X\rangle}\big] \;=\; \psi_t, \qquad \text{for all } t \in \mathbb{R}^d.$$

Therefore, by Lévy's continuity theorem, $X_n \xrightarrow{d} X$. This completes the proof. □

B.9 Proof of thm. 2

Proof. We first formulate the following assumptions required for the proof–all of this are satisfied by typical univariate statistical tests.

$P = Q$ if and only if $P_a = Q_a$ for all $a \in S^{d-1}$ (population-level equivalence of laws).
A_n are finite sets with mesh $\Delta(A_n) := \sup_{u \in S^{d-1}} \min_{a \in A_n} \|u - a\| \to 0$ as $n \to \infty$.
If $P \neq Q$, there exists a separating direction $a^\star \in S^{d-1}$ and a neighborhood U of $a^\star$ such that

$$\inf_{a \in U} \lim_{n \to \infty} \Pr\big(T_{a,n} \geq u_n(\alpha)\big) = 1.$$

(Intuitively: near a truly separating direction, the 1D statistic eventually exceeds the global null threshold with probability $\to 1$.)

(i) Under $H_0 : P = Q$, our assumption implies no separating direction exists at the population level, and the calibration of $u_n(\alpha)$ ensures $\Pr(M_n \geq u_n(\alpha)) \leq \alpha$ for all n, hence $\limsup_{n\to\infty} \Pr(\Psi_n = 1) \leq \alpha$. (ii) Suppose $P \neq Q$. Our assumption guarantees that there exists at least one separating direction $a^\star$ with $P_{a^\star} \neq Q_{a^\star}$. Our assumption guarantees a neighborhood U of $a^\star$ in which the projection statistics exceed the global null threshold with probability tending to 1:

$$\inf_{a \in U} \lim_{n \to \infty} \Pr\big(T_{a,n} \geq u_n(\alpha)\big) \;=\; 1.$$

By assumption, for all large n the set A_n contains at least one direction $a_n \in U$ (dense coverage). Therefore,

$$\Pr(\Psi_n = 1) \;=\; \Pr\big(M_n \geq u_n(\alpha)\big) \;\geq\; \Pr\big(T_{a_n,n} \geq u_n(\alpha)\big) \;\longrightarrow\; 1,$$

which proves consistency. □

B.10 Proof of thm. 5

Proof. For each case, consider the function $g(a)$ on $\mathbb{S}^{D-1}$ defined by the quantity of interest (CF, CDF, or moment) at a fixed t or k. Since $f \in H^\alpha(\mathbb{R}^D)$, the mapping $a \mapsto g(a)$ is in $H^\alpha(\mathbb{S}^{D-1})$ for each fixed t or k.

Given M samples $\{a_i\}_{i=1}^M$ on the sphere, the best possible reconstruction of g from its values at these points is given by spherical interpolation. By classical results on Sobolev spaces and spherical harmonics (see, e.g., Narcowich et al. [2006]), the L^2 interpolation error for functions in $H^\alpha(\mathbb{S}^{D-1})$ using M points is bounded by

$$\mathbb{E}_b\left[|g(b) - g^*(b)|^2\right] \leq C(D, \alpha) M^{-2\alpha/(D-1)} \|g\|^2_{H^\alpha(\mathbb{S}^{D-1})},$$

where g^* is the interpolant matching g at the M sampled points. The interpolation error bound on the sphere follows from the theory of spherical harmonics and Marcinkiewicz–Zygmund (MZ) inequalities . Any $f \in H^\alpha(\mathbb{S}^d)$ admits a spherical harmonics expansion, and the best L^2 approximation by harmonics of degree at most L satisfies

$$\|f - P_L f\|_{L^2(\mathbb{S}^d)} \leq (1 + L^2)^{-\alpha/2} \|f\|_{H^\alpha(\mathbb{S}^d)},$$

where $P_L f$ is the projection onto harmonics of degree $\leq L$ [Narcowich et al., 2006, Lemma 2.1]. If M points are distributed quasi-uniformly on $\mathbb{S}^d$, then for $L \sim cM^{1/d}$, the set forms a Marcinkiewicz–Zygmund (MZ) set for degree L [Mhaskar et al., 2001, Theorem 1.1]. This allows reconstruction of any function in the space of harmonics of degree at most L from its values at these points, and the L^2 interpolation error for f is bounded by

$$\|f - I_M f\|_{L^2(\mathbb{S}^d)} \leq C(1 + L^2)^{-\alpha/2} \|f\|_{H^\alpha(\mathbb{S}^d)},$$

因此，对每个固定的 $t \in \mathbb{R}^d$，一维投影满足 $t, X_n \xrightarrow{d} t, X$，进而得到特征函数的逐点收敛：

$$\psi_{n,t} \;=\; \mathbb{E}\big[e^{i\langle t, X_n\rangle}\big] \;\longrightarrow\; \mathbb{E}\big[e^{i\langle t, X\rangle}\big] \;=\; \psi_t, \qquad \text{for all } t \in \mathbb{R}^d.$$

因此，由 Lévy 连续性定理， $X_n \xrightarrow{d} X$。证明完毕。 □

B.9 定理 2 的证明

证明。首先列出证明所需的假设；典型的一元统计检验均满足这些假设。

$P = Q$ 当且仅当对所有 $a \in S^{d-1}$ 均有 $P_a = Q_a$（总体层面的分布律等价性）。
A_n 为有限集，其网格尺度 $\Delta(A_n) := \sup_{u \in S^{d-1}} \min_{a \in A_n} \|u - a\| \to 0$ 随 $n \to \infty$。
若 $P \neq Q$，则存在一个区分方向 $a^\star \in S^{d-1}$ 以及 $a^\star$ 的邻域 U，使得

$$\inf_{a \in U} \lim_{n \to \infty} \Pr\big(T_{a,n} \geq u_n(\alpha)\big) = 1.$$

（直观上：在真正能区分分布的方向附近，一维统计量最终以概率 $\to 1$ 超过全局零假设阈值。）

（i） $H_0 : P = Q$下，假设意味着总体无分离方向。 $u_n(\alpha)$校准保证全部 n有 $\Pr(M_n \geq u_n(\alpha)) \leq \alpha$，因此 $\limsup_{n\to\infty} \Pr(\Psi_n = 1) \leq \alpha$。（ii）假设 $P \neq Q$，保证至少一分离方向 $a^\star$ 满足 $P_{a^\star} \neq Q_{a^\star}$；并存在 $a^\star$ 邻域 U，其投影统计以趋于 1 的概率超过全局零假设阈值：

$$\inf_{a \in U} \lim_{n \to \infty} \Pr\big(T_{a,n} \geq u_n(\alpha)\big) \;=\; 1.$$

由假设，对于一个 对所有充分大的 n，集合 A_n 至少包含一个方向 $a_n \in U$（稠密覆盖）。因此

$$\Pr(\Psi_n = 1) \;=\; \Pr\big(M_n \geq u_n(\alpha)\big) \;\geq\; \Pr\big(T_{a_n,n} \geq u_n(\alpha)\big) \;\longrightarrow\; 1,$$

由此证明一致性。 □

B.10 定理 5 的证明

证明。对每一种情形，考虑 $\mathbb{S}^{D-1}$ 上的函数 $g(a)$，它由固定 t 或 k 处所关注的量（特征函数、累积分布函数或矩）定义。由于 $f \in H^\alpha(\mathbb{R}^D)$，对于每个固定的 t 或 k，映射 $a \mapsto g(a)$ 属于 $H^\alpha(\mathbb{S}^{D-1})$。

给定球面上的 M 个样本 $\{a_i\}_{i=1}^M$，根据这些点上的函数值重建 g 的最佳方式是球面插值。依据 Sobolev 空间和球面调和函数的经典结果（例如参见 Narcowich et al. [2006]），对 $H^\alpha(\mathbb{S}^{D-1})$ 中的函数使用 M 个点进行插值时，其 L^2 插值误差满足以下界：

$$\mathbb{E}_b\left[|g(b) - g^*(b)|^2\right] \leq C(D, \alpha) M^{-2\alpha/(D-1)} \|g\|^2_{H^\alpha(\mathbb{S}^{D-1})},$$

其中 g^* 为在 M 个采样点处与 g 匹配的插值函数。球面上的插值误差界来自球面调和函数理论与 Marcinkiewicz–Zygmund（MZ）不等式。任意 $f \in H^\alpha(\mathbb{S}^d)$ 都可展开为球面调和函数，且使用次数不超过 L 的调和函数得到的最佳 L^2 近似满足

$$\|f - P_L f\|_{L^2(\mathbb{S}^d)} \leq (1 + L^2)^{-\alpha/2} \|f\|_{H^\alpha(\mathbb{S}^d)},$$

其中 $P_L f$ 为 $\leq L$ [次球谐的投影，Narcowich 等，2006，引理 2.1]。若 M 点准均匀分布于 $\mathbb{S}^d$，则 $L \sim cM^{1/d}$时构成 L [次 Marcinkiewicz–Zygmund（MZ）集，Mhaskar 等，2001，定理 1.1]。可从点值恢复次数不超过 L 的谐函数， f 的 L^2 插值误差有界：

$$\|f - I_M f\|_{L^2(\mathbb{S}^d)} \leq C(1 + L^2)^{-\alpha/2} \|f\|_{H^\alpha(\mathbb{S}^d)},$$

where $I_M f$ is any interpolant matching f at the M points [Narcowich et al., 2006, Theorem 3.1]. Substituting $L \sim cM^{1/d}$ yields the rate $M^{-\alpha/d}$, and thus

$$\mathbb{E}_\omega |f(\omega) - I_M f(\omega)|^2 \le C(d,\alpha) M^{-2\alpha/d} \|f\|^2_{H^\alpha(\mathbb{S}^d)},$$

with explicit $C(d,\alpha)$ as in the main theorem. Integrating (or summing) over t (for CF and CDF) or k (for moments, with weights w_k) yields the stated bounds. The explicit constant $C(D,\alpha)$ arises from the theory of spherical Sobolev spaces and is given above.

For the moment case, the sum over k is weighted to ensure convergence, as higher moments may grow rapidly. The weights w_k can be chosen, for example, as $w_k = 1/k!$.

This completes the proof. □

B.11 Proof of thm. 3

Pick distinct $x_0, \dots, x_{K+1} \in \mathbb{R}$ and consider the linear map $A : \mathbb{R}^{K+2} \to \mathbb{R}^{K+1}$, $(Ap)_r = \sum_{j=0}^{K+1} p_j x_j^r$ for $r = 0, \dots, K$. Then $\operatorname{rank}(A) \le K+1$, so $\ker(A) \neq \{0\}$. Let $v \in \ker(A) \setminus \{0\}$; from $(Ap)_0 = \sum_j p_j$, we get $\sum_j v_j = 0$, hence v has positive and negative entries. Choose a strictly positive probability vector p and $\varepsilon > 0$ small such that $p^\pm := p \pm \varepsilon v$ remain probability vectors. Then $Ap^+ = Ap^-$, so the distributions supported on $\{x_j\}$ with masses $p^\pm$ are distinct yet match moments up to order K.

B.12 Proof of thm. 4

Proof. Fix the Gaussian weight

$$w_s(t) = e^{-s^2 t^2}, \qquad s > 0,$$

and define the population CF distance

$$D(P,G) = \int_{\mathbb{R}} w_s(t) \left|\varphi_P(t) - \varphi_G(t)\right|^2 dt.$$

Let the empirical CF be

$$\widehat{\varphi}_N(t) = \frac{1}{N} \sum_{i=1}^{N} e^{itX_i},$$

and consider the V-statistic estimator

$$\widehat{D}_V = \int_{\mathbb{R}} w_s(t) \left|\widehat{\varphi}_N(t) - \varphi_G(t)\right|^2 dt.$$

We use only that $|e^{itX}| = 1$, $|\varphi_P(t)| \le 1$, $|\varphi_G(t)| \le 1$, and integrability of w_s. For each i differentiate under the integral (dominated convergence applies because the integrand and its derivative are bounded)

$$\begin{aligned}
\frac{\partial \widehat{D}_V}{\partial X_i} &= \int_{\mathbb{R}} w_s(t) 2\Re\Big(\big(\widehat{\varphi}_N(t) - \varphi_G(t)\big) \overline{\frac{\partial \widehat{\varphi}_N(t)}{\partial X_i}} \Big) dt, \\
\frac{\partial \widehat{\varphi}_N(t)}{\partial X_i} &= \frac{1}{N} it e^{itX_i},
\end{aligned}$$

since $|\widehat{\varphi}_N(t)| \le 1$ and $|\varphi_G(t)| \le 1$,

$$\begin{aligned}
\left|\frac{\partial \widehat{D}_V}{\partial X_i}\right| &\le \frac{2}{N} \int w_s(t)|t| \big(|\widehat{\varphi}_N(t)| + |\varphi_G(t)|\big) dt \\
&\le \frac{4}{N} \int w_s(t)|t| dt \\
&= \frac{4}{Ns^2},
\end{aligned}$$

其中 $I_M f$ 是在 M 个点处与 f 匹配的任意插值函数 [Narcowich et al., 2006, Theorem 3.1]。代入 $L \sim cM^{1/d}$ 可得速率 $M^{-\alpha/d}$，因此

$$\mathbb{E}_\omega |f(\omega) - I_M f(\omega)|^2 \le C(d,\alpha) M^{-2\alpha/d} \|f\|^2_{H^\alpha(\mathbb{S}^d)},$$

其中显式的 $C(d,\alpha)$ 与主定理相同。对 t（用于特征函数和累积分布函数）或 k（用于各阶矩，权重为 w_k）积分（或求和），即可得到所述界。显式常数 $C(D,\alpha)$ 来自球面 Sobolev 空间理论，已在上文给出。

对于矩的情形，对 k 的求和采用加权以确保收敛，因为高阶矩可能迅速增长。例如，可将权重 w_k 选为 $w_k = 1/k!$。

证明完毕。 □

B.11 定理 3 的证明

取不同的 $x_0, \dots, x_{K+1} \in \mathbb{R}$，考虑线性映射 $A : \mathbb{R}^{K+2} \to \mathbb{R}^{K+1}$, $(Ap)_r = \sum_{j=0}^{K+1} p_j x_j^r$，其中 $r = 0, \dots, K$。有 $\operatorname{rank}(A) \le K+1$，故 $\ker(A) \neq \{0\}$。取 $v \in \ker(A) \setminus \{0\}$；由$(Ap)_0 = \sum_j p_j$得到 $\sum_j v_j = 0$，所以 v 含正、负项。选择严格正概率向量 p 及足够小的 $\varepsilon > 0$，使 $p^\pm := p \pm \varepsilon v$ 仍为概率向量。于是 $Ap^+ = Ap^-$，支撑于$\{x_j\}$、质量为 $p^\pm$ 的分布虽不同，却匹配直到 K阶的矩。

B.12 定理 4 的证明

证明。固定高斯权重

$$w_s(t) = e^{-s^2 t^2}, \qquad s > 0,$$

并定义总体特征函数距离

$$D(P,G) = \int_{\mathbb{R}} w_s(t) \left|\varphi_P(t) - \varphi_G(t)\right|^2 dt.$$

设经验特征函数为

$$\widehat{\varphi}_N(t) = \frac{1}{N} \sum_{i=1}^{N} e^{itX_i},$$

并考虑 V 统计量估计器

$$\widehat{D}_V = \int_{\mathbb{R}} w_s(t) \left|\widehat{\varphi}_N(t) - \varphi_G(t)\right|^2 dt.$$

我们仅使用以下条件：$|e^{itX}| = 1$、$|\varphi_P(t)| \le 1$、$|\varphi_G(t)| \le 1$，以及 w_s 的可积性。对每个 i，在积分号下求导（由于被积函数及其导数有界，可应用控制收敛定理），

$$\begin{aligned}
\frac{\partial \widehat{D}_V}{\partial X_i} &= \int_{\mathbb{R}} w_s(t) 2\Re\Big(\big(\widehat{\varphi}_N(t) - \varphi_G(t)\big) \overline{\frac{\partial \widehat{\varphi}_N(t)}{\partial X_i}} \Big) dt, \\
\frac{\partial \widehat{\varphi}_N(t)}{\partial X_i} &= \frac{1}{N} it e^{itX_i},
\end{aligned}$$

因为 $|\widehat{\varphi}_N(t)| \le 1$ 且 $|\varphi_G(t)| \le 1$,

$$\begin{aligned}
\left|\frac{\partial \widehat{D}_V}{\partial X_i}\right| &\le \frac{2}{N} \int w_s(t)|t| \big(|\widehat{\varphi}_N(t)| + |\varphi_G(t)|\big) dt \\
&\le \frac{4}{N} \int w_s(t)|t| dt \\
&= \frac{4}{Ns^2},
\end{aligned}$$

using $\int_{\mathbb{R}} e^{-s^2t^2}|t|dt = 1/s^2$.

$$\left|\frac{\partial \widehat{D}_V}{\partial X_i}\right| \;\le\; \frac{4}{N}\int_{\mathbb{R}} w_s(t)|t|dt \;=\; \frac{4}{Ns^2}.$$

Moreover, differentiating once more in X_i and using $|\widehat{\varphi}_N(t)| \le 1$, $|\varphi_G(t)| \le 1$ gives a global Lipschitz bound

$$\left|\frac{\partial^2 \widehat{D}_V}{\partial X_i^2}\right| \;\le\; \frac{C}{N}\int_{\mathbb{R}} w_s(t)t^2dt \;=\; \frac{C}{N}\cdot\frac{\sqrt{\pi}}{2s^3},$$

for some absolute constant C arising from bounded factors and product rule. Hence ECF gradients are uniformly bounded and Lipschitz, with scale controlled only by (N, s).

(B) (Moment sample-gradients are polynomial in X_i and unbounded for $k \ge 2$.) Let $\widehat{D}_V$ be as above. Define the moment objective

$$\widehat{D}_k \;=\; (\bar{\phi}-\mu)^\top W(\bar{\phi}-\mu), \qquad \bar{\phi} := \frac{1}{N}\sum_{i=1}^{N}\phi(X_i), \quad \phi(x) = (x, x^2, \dots, x^k)^\top,$$

for a symmetric positive semidefinite $W \in \mathbb{R}^{k\times k}$ and Gaussian target moments $\mu = \mathbb{E}_G[\phi(Y)]$. For each i,

$$\begin{aligned}\frac{\partial \widehat{D}_k}{\partial X_i} &= \frac{2}{N}(\bar{\phi}-\mu)^\top W\frac{\partial \phi(X_i)}{\partial X_i},\\ \frac{\partial \phi(X)}{\partial X} &= \big(1, 2X, 3X^2, \dots, kX^{k-1}\big)^\top.\end{aligned}$$

The gradient formula follows by the chain rule and linearity of $\bar{\phi}$. Let $c := W(\bar{\phi}-\mu)$ and write c_r for its r-th coordinate. Then

$$\frac{\partial \widehat{D}_k}{\partial X_i} = \frac{2}{N}\sum_{r=1}^{k} c_r r X_i^{r-1},$$

which is a polynomial in X_i of degree $\deg = \max\{r-1 : c_r \ne 0\} \le k-1$. In particular, if $c_k \ne 0$ (the generic case when the top-weighted deviation is nonzero), then

$$\left|\frac{\partial \widehat{D}_k}{\partial X_i}\right| \xrightarrow[|X_i|\to\infty]{} \infty \quad \text{as} \quad |X_i|^{k-1}.$$

The expression is a nonconstant polynomial in X_i of degree $\deg \le k-1$ whenever some $c_r \ne 0$ with $r \ge 2$. Thus the gradient cannot be uniformly bounded on $\mathbb{R}$. If $c_k \ne 0$, the leading term dominates and the magnitude grows like $|X_i|^{k-1}$, proving unboundedness for $k \ge 2$. □

B.13 Proof of thm. 6

Proof. A direct calculation shows Fix $t \in \mathbb{R}^d$ and abbreviate $Z_j := e^{it^\top X_j}$, so that $\phi_n(t) = \frac{1}{n}\sum_{j=1}^{n} Z_j$. Note that $|Z_j| = 1$ almost surely (since $t^\top X_j \in \mathbb{R}$), and $\mathbb{E}[Z_j] = \phi_\theta(t)$ for all j. We start from the algebraic identity

$$\big|\phi_n(t) - \psi(t)\big|^2 = \phi_n(t)\overline{\phi_n(t)} - \psi(t)\overline{\phi_n(t)} - \overline{\psi(t)}\phi_n(t) + \big|\psi(t)\big|^2.$$

使用 $\int_{\mathbb{R}} e^{-s^2t^2}|t|dt = 1/s^2$。

$$\left|\frac{\partial \widehat{D}_V}{\partial X_i}\right| \;\le\; \frac{4}{N}\int_{\mathbb{R}} w_s(t)|t|dt \;=\; \frac{4}{Ns^2}.$$

此外，求导　　再对 X_i 求一次导数，并使用 $|\widehat{\varphi}_N(t)| \le 1$、$|\varphi_G(t)| \le 1$，可得全局 Lipschitz 界：

$$\left|\frac{\partial^2 \widehat{D}_V}{\partial X_i^2}\right| \;\le\; \frac{C}{N}\int_{\mathbb{R}} w_s(t)t^2dt \;=\; \frac{C}{N}\cdot\frac{\sqrt{\pi}}{2s^3},$$

其中某个绝对常数 C 来自有界因子和乘积法则。因此，ECF 梯度一致有界且满足 Lipschitz 条件，其尺度仅由 (N, s) 控制。

（B）（矩的样本梯度是关于 X_i 的多项式，且在 $k \ge 2$ 时无界。）设 D_V 如上。定义矩目标：

$$\widehat{D}_k \;=\; (\bar{\phi}-\mu)^\top W(\bar{\phi}-\mu), \qquad \bar{\phi} := \frac{1}{N}\sum_{i=1}^{N}\phi(X_i), \quad \phi(x) = (x, x^2, \dots, x^k)^\top,$$

其中 $W \in \mathbb{R}^{k\times k}$ 为对称半正定矩阵，$\mu = \mathbb{E}_G[\phi(Y)]$ 为高斯目标矩。对每个 i,

$$\begin{aligned}\frac{\partial \widehat{D}_k}{\partial X_i} &= \frac{2}{N}(\bar{\phi}-\mu)^\top W\frac{\partial \phi(X_i)}{\partial X_i},\\ \frac{\partial \phi(X)}{\partial X} &= \big(1, 2X, 3X^2, \dots, kX^{k-1}\big)^\top.\end{aligned}$$

梯度公式由链式法则和 ϕ 的线性性得出。令 $c := W(\bar{\phi}-\mu)$，并用 c_r 表示其第 r 个坐标。则

$$\frac{\partial \widehat{D}_k}{\partial X_i} = \frac{2}{N}\sum_{r=1}^{k} c_r r X_i^{r-1},$$

这是关于 X_i 的多项式，其次数为 $\deg = \max\{r-1 : c_r \ne 0\} \le k-1$。特别地，若 $c_k \ne 0$（最高权重对应的偏差非零时的一般情形），则

$$\left|\frac{\partial \widehat{D}_k}{\partial X_i}\right| \xrightarrow[|X_i|\to\infty]{} \infty \quad \text{as} \quad |X_i|^{k-1}.$$

当存在某个 $c_r \ne 0$ 满足 $r \ge 2$ 时，该表达式是关于 X_i 的非常数多项式，其次数为 $\deg \le k-1$。因此，梯度不可能在 $\mathbb{R}$ 上一致有界。若 $c_k \ne 0$，最高次项占主导，幅值按 $|X_i|^{k-1}$ 增长，从而证明在 $k \ge 2$ 情况下无界。□

B.13 定理 6 的证明

证明。直接计算可得。固定 $t \in \mathbb{R}^d$，并简记为 $Z_j := e^{it^\top X_j}$，于是 $\phi_n(t) = \frac{1}{n}\sum_{j=1}^{n} Z_j$。注意，$|Z_j| = 1$ 几乎处处成立（因为 $t^\top X_j \in \mathbb{R}$），且对所有 j，均有 $\mathbb{E}[Z_j] = \phi_\theta(t)$。我们从以下代数恒等式出发：

$$\big|\phi_n(t) - \psi(t)\big|^2 = \phi_n(t)\overline{\phi_n(t)} - \psi(t)\overline{\phi_n(t)} - \overline{\psi(t)}\phi_n(t) + \big|\psi(t)\big|^2.$$

Taking expectations term by term gives

$$\begin{aligned}
\mathbb{E}\left[\left|\phi_n-\psi\right|^2\right] =& \mathbb{E}\left[|\phi_n|^2\right]-\psi\mathbb{E}\left[\overline{\phi_n}\right]-\overline{\psi}\mathbb{E}\left[\phi_n\right]+|\psi|^2, && (26)\\
=& \mathbb{E}\left[|\phi_n|^2\right]-\psi\overline{\mathbb{E}[\phi_n]}-\overline{\psi}\frac{1}{n}\sum_{j=1}^{n}\mathbb{E}[Z_j]+|\psi|^2, && (27)\\
=& \mathbb{E}\left[|\phi_n|^2\right]-\psi\overline{\phi_\theta}-\overline{\psi}\phi_\theta+|\psi|^2, && (28)\\
=& \mathbb{E}\left[|\phi_n|^2\right]-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2, && (29)\\
=& \mathbb{E}\left[\left|\frac{1}{n}\sum_{j=1}^{n}Z_j\right|^2\right]-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2, && (30)\\
=& \frac{1}{n^2}\sum_{j=1}^{n}\sum_{l=1}^{n}\mathbb{E}\left[Z_j\overline{Z_l}\right]-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2, && (31)\\
& && (32)
\end{aligned}$$

Since the Z_j are i.i.d.,

$$\mathbb{E}\left[Z_j\overline{Z_l}\right]=\begin{cases}\mathbb{E}\left[|Z_1|^2\right]=1, & \text{if } j=l,\\ \mathbb{E}[Z_j]\overline{\mathbb{E}[Z_l]}=\phi_\theta\overline{\phi_\theta}=|\phi_\theta|^2, & \text{if } j\neq l,\end{cases}$$

hence

$$\begin{aligned}
\mathbb{E}\left[|\phi_n|^2\right] =& \frac{1}{n^2}\left(n+n(n-1)|\phi_\theta|^2\right)\\
=& \frac{1}{n}+\left(1-\frac{1}{n}\right)|\phi_\theta|^2\\
=& |\phi_\theta|^2+\frac{1-|\phi_\theta|^2}{n}
\end{aligned}$$

Plugging these, we obtain

$$\begin{aligned}
\mathbb{E}\left[\left|\phi_n-\psi\right|^2\right] &= \left(|\phi_\theta|^2+\frac{1-|\phi_\theta|^2}{n}\right)-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2\\
&= \left(|\phi_\theta|^2-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2\right)+\frac{1-|\phi_\theta|^2}{n}\\
&= \left|\phi_\theta-\psi\right|^2+\frac{1-|\phi_\theta|^2}{n}.
\end{aligned}$$

Under Dominated convergence, $\mathbb{E}[\nabla_\theta D_n(t)]=\nabla_\theta\mathbb{E}[D_n(t)]$, hence

$$\mathbb{E}\left[\nabla_\theta D_n(t)\right]=\nabla_\theta\left|\phi_\theta(t)-\psi(t)\right|^2+\nabla_\theta\frac{1-|\phi_\theta(t)|^2}{n},$$

concluding the proof.

In practice one replaces $\int_{\mathbb{R}} w(t)(\cdot)dt$ by a deterministic quadrature on a uniform grid $t_k\in[-T,T]$ with weights ω_k (e.g. trapezoidal rule) and a Gaussian window $w(t)=e^{-\alpha t^2}$. All statements above remain valid with the integral replaced by $\sum_k \omega_k(\cdot)$:

$$L(\theta)\approx\sum_k \omega_k\left|\phi_\theta(t_k)-\psi(t_k)\right|^2,\quad \widehat{L}_n(\theta)\approx\sum_k \omega_k\left|\phi_n(t_k)-\psi(t_k)\right|^2,$$

逐项求期望，得到

$$\begin{aligned}
\mathbb{E}\left[\left|\phi_n-\psi\right|^2\right] =& \mathbb{E}\left[|\phi_n|^2\right]-\psi\mathbb{E}\left[\overline{\phi_n}\right]-\overline{\psi}\mathbb{E}\left[\phi_n\right]+|\psi|^2, && (26)\\
=& \mathbb{E}\left[|\phi_n|^2\right]-\psi\overline{\mathbb{E}[\phi_n]}-\overline{\psi}\frac{1}{n}\sum_{j=1}^{n}\mathbb{E}[Z_j]+|\psi|^2, && (27)\\
=& \mathbb{E}\left[|\phi_n|^2\right]-\psi\overline{\phi_\theta}-\overline{\psi}\phi_\theta+|\psi|^2, && (28)\\
=& \mathbb{E}\left[|\phi_n|^2\right]-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2, && (29)\\
=& \mathbb{E}\left[\left|\frac{1}{n}\sum_{j=1}^{n}Z_j\right|^2\right]-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2, && (30)\\
=& \frac{1}{n^2}\sum_{j=1}^{n}\sum_{l=1}^{n}\mathbb{E}\left[Z_j\overline{Z_l}\right]-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2, && (31)\\
& && (32)
\end{aligned}$$

由于 Z_j 独立同分布，

$$\mathbb{E}\left[Z_j\overline{Z_l}\right]=\begin{cases}\mathbb{E}\left[|Z_1|^2\right]=1, & \text{if } j=l,\\ \mathbb{E}[Z_j]\overline{\mathbb{E}[Z_l]}=\phi_\theta\overline{\phi_\theta}=|\phi_\theta|^2, & \text{if } j\neq l,\end{cases}$$

因此

$$\begin{aligned}
\mathbb{E}\left[|\phi_n|^2\right] =& \frac{1}{n^2}\left(n+n(n-1)|\phi_\theta|^2\right)\\
=& \frac{1}{n}+\left(1-\frac{1}{n}\right)|\phi_\theta|^2\\
=& |\phi_\theta|^2+\frac{1-|\phi_\theta|^2}{n}
\end{aligned}$$

代入上述各项，得到

$$\begin{aligned}
\mathbb{E}\left[\left|\phi_n-\psi\right|^2\right] &= \left(|\phi_\theta|^2+\frac{1-|\phi_\theta|^2}{n}\right)-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2\\
&= \left(|\phi_\theta|^2-2\mathrm{Re}\left(\overline{\psi}\phi_\theta\right)+|\psi|^2\right)+\frac{1-|\phi_\theta|^2}{n}\\
&= \left|\phi_\theta-\psi\right|^2+\frac{1-|\phi_\theta|^2}{n}.
\end{aligned}$$

根据控制收敛定理， $\mathbb{E}[\nabla_\theta D_n(t)]=\nabla_\theta\mathbb{E}[D_n(t)]$, ，因此

$$\mathbb{E}\left[\nabla_\theta D_n(t)\right]=\nabla_\theta\left|\phi_\theta(t)-\psi(t)\right|^2+\nabla_\theta\frac{1-|\phi_\theta(t)|^2}{n},$$

证明完毕。

在实践中，用均匀网格 $t_k\in[-T,T]$ 上带权重 ω_k （例如梯形法则）的确定性求积，以及高斯窗 $w(t)=e^{-\alpha t^2}$，替代 $\int_{\mathbb{R}} w(t)(\cdot)dt$ 。将积分替换为 $\sum_k \omega_k(\cdot)$ 后，上述所有结论仍然成立：

$$L(\theta)\approx\sum_k \omega_k\left|\phi_\theta(t_k)-\psi(t_k)\right|^2,\quad \widehat{L}_n(\theta)\approx\sum_k \omega_k\left|\phi_n(t_k)-\psi(t_k)\right|^2,$$

and the bias term becomes

$$\text{Bias}(\theta) = -\frac{1}{n}\sum_k \omega_k \nabla_\theta \left|\phi_\theta(t_k)\right|^2.$$

Since the grid and weights are deterministic, they do not affect unbiasedness with respect to sampling; they only introduce a deterministic approximation error to the target functional $L(\theta)$.

□

B.14 Proof of VICReg's Recovery

Proof. We prove this result in two parts.

Part I: $\mathbb{E}[\mathbf{X}] = \mathbf{0}$ Given that $\mathbb{E}[\langle \mathbf{X}, \mathbf{a}\rangle] = 0$ for all unit vectors $\mathbf{a}$, and noting that $\langle \mathbf{X}, \mathbf{a}\rangle = \mathbf{a}^T\mathbf{X}$, we have:

$$\mathbb{E}[\mathbf{a}^T\mathbf{X}] = 0 \quad \text{for all } \mathbf{a} \in \mathbb{R}^d \text{ with } \|\mathbf{a}\| = 1 \tag{33}$$

By linearity of expectation:

$$\mathbf{a}^T\mathbb{E}[\mathbf{X}] = 0 \quad \text{for all unit vectors } \mathbf{a} \tag{34}$$

Let $\boldsymbol{\mu} = \mathbb{E}[\mathbf{X}]$. We claim that $\boldsymbol{\mu} = \mathbf{0}$. Suppose, for the sake of contradiction, that $\boldsymbol{\mu} \neq \mathbf{0}$. Then $\|\boldsymbol{\mu}\|_2 > 0$. Define the unit vector:

$$\mathbf{a}^* = \frac{\boldsymbol{\mu}}{\|\boldsymbol{\mu}\|_2} \tag{35}$$

Since $\mathbf{a}^*$ is a unit vector, equation (33) implies:

$$(\mathbf{a}^*)^T\boldsymbol{\mu} = 0 \tag{36}$$

However, substituting the definition of $\mathbf{a}^*$:

$$(\mathbf{a}^*)^T\boldsymbol{\mu} = \left(\frac{\boldsymbol{\mu}}{\|\boldsymbol{\mu}\|_2}\right)^T\boldsymbol{\mu} = \frac{\boldsymbol{\mu}^T\boldsymbol{\mu}}{\|\boldsymbol{\mu}\|_2} = \frac{\|\boldsymbol{\mu}\|_2^2}{\|\boldsymbol{\mu}\|_2} = \|\boldsymbol{\mu}\|_2 > 0 \tag{37}$$

This contradiction establishes that $\boldsymbol{\mu} = \mathbf{0}$.

Part II: $\text{Cov}(\mathbf{X}) = \mathbf{I}_d$ Since $\mathbb{E}[\mathbf{X}] = \mathbf{0}$, we have:

$$\text{Var}(\langle \mathbf{X}, \mathbf{a}\rangle) = \mathbb{E}[(\langle \mathbf{X}, \mathbf{a}\rangle)^2] = \mathbb{E}[(\mathbf{a}^T\mathbf{X})^2] \tag{38}$$

Expanding the quadratic form:

$$\mathbb{E}[(\mathbf{a}^T\mathbf{X})^2] = \mathbb{E}[\mathbf{a}^T\mathbf{X}\mathbf{X}^T\mathbf{a}] = \mathbf{a}^T\mathbb{E}[\mathbf{X}\mathbf{X}^T]\mathbf{a} \tag{39}$$

Since $\mathbb{E}[\mathbf{X}] = \mathbf{0}$, the covariance matrix is $\text{Cov}(\mathbf{X}) = \mathbb{E}[\mathbf{X}\mathbf{X}^T]$. Let $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{X})$. The variance condition gives us:

$$\mathbf{a}^T\boldsymbol{\Sigma}\mathbf{a} = 1 \quad \text{for all unit vectors } \mathbf{a} \tag{40}$$

We now show that $\boldsymbol{\Sigma} = \mathbf{I}_d$. *Step 1: Diagonal entries.* For $i \in \{1, 2, \ldots, d\}$, let $\mathbf{e}_i$ denote the i-th standard basis vector. Setting $\mathbf{a} = \mathbf{e}_i$ in equation (40):

$$\mathbf{e}_i^T\boldsymbol{\Sigma}\mathbf{e}_i = \Sigma_{ii} = 1 \tag{41}$$

Therefore, all diagonal entries of $\boldsymbol{\Sigma}$ equal 1. *Step 2: Off-diagonal entries.* For distinct indices $i, j \in \{1, 2, \ldots, d\}$, consider the unit vector:

$$\mathbf{a} = \frac{\mathbf{e}_i + \mathbf{e}_j}{\|\mathbf{e}_i + \mathbf{e}_j\|_2} = \frac{\mathbf{e}_i + \mathbf{e}_j}{\sqrt{2}} \tag{42}$$

Applying equation (40):

$$\mathbf{a}^T\boldsymbol{\Sigma}\mathbf{a} = \frac{1}{2}(\mathbf{e}_i + \mathbf{e}_j)^T\boldsymbol{\Sigma}(\mathbf{e}_i + \mathbf{e}_j) = 1 \tag{43}$$

偏差项变为

$$\text{Bias}(\theta) = -\frac{1}{n}\sum_k \omega_k \nabla_\theta \left|\phi_\theta(t_k)\right|^2.$$

由于网格和权重是确定性的，它们不会影响关于采样的无偏性；只会对目标泛函 $L(\theta)$ 引入确定性的近似误差。

□

B.14 恢复 VICReg 的证明

证明。我们分两部分证明此结果。

第一部分：$\mathbb{E}[\mathrm{X}] = \mathbf{0}$。已知对所有单位向量 a 均有 $\mathbb{E}[\mathrm{X}, \mathrm{a}] = 0$，并注意到 $\mathrm{X}, \mathrm{a} = \mathrm{a}^T\mathrm{X}$，可得：

$$\mathbb{E}[\mathbf{a}^T\mathbf{X}] = 0 \quad \text{for all } \mathbf{a} \in \mathbb{R}^d \text{ with } \|\mathbf{a}\| = 1 \tag{33}$$

由期望的线性性：

$$\mathbf{a}^T\mathbb{E}[\mathbf{X}] = 0 \quad \text{for all unit vectors } \mathbf{a} \tag{34}$$

令 $\boldsymbol{\mu} = \mathbb{E}[\mathrm{X}]$。我们断言 $\boldsymbol{\mu} = \mathbf{0}$。为反证，假设 $\boldsymbol{\mu} \neq \mathbf{0}$。于是 $\|\boldsymbol{\mu}\|_2 > 0$。定义单位向量：

$$\mathbf{a}^* = \frac{\boldsymbol{\mu}}{\|\boldsymbol{\mu}\|_2} \tag{35}$$

由于 a^* 是单位向量，式（33）意味着：

$$(\mathbf{a}^*)^T\boldsymbol{\mu} = 0 \tag{36}$$

然而，代入 a^* 的定义：

$$(\mathbf{a}^*)^T\boldsymbol{\mu} = \left(\frac{\boldsymbol{\mu}}{\|\boldsymbol{\mu}\|_2}\right)^T\boldsymbol{\mu} = \frac{\boldsymbol{\mu}^T\boldsymbol{\mu}}{\|\boldsymbol{\mu}\|_2} = \frac{\|\boldsymbol{\mu}\|_2^2}{\|\boldsymbol{\mu}\|_2} = \|\boldsymbol{\mu}\|_2 > 0 \tag{37}$$

这一矛盾证明了 $\boldsymbol{\mu} = \mathbf{0}$。

第二部分：$\mathrm{Cov}(\mathrm{X}) = \mathrm{I}_d$。由于 $\mathbb{E}[\mathrm{X}] = \mathbf{0}$，可得：

$$\text{Var}(\langle \mathbf{X}, \mathbf{a}\rangle) = \mathbb{E}[(\langle \mathbf{X}, \mathbf{a}\rangle)^2] = \mathbb{E}[(\mathbf{a}^T\mathbf{X})^2] \tag{38}$$

展开二次型：

$$\mathbb{E}[(\mathbf{a}^T\mathbf{X})^2] = \mathbb{E}[\mathbf{a}^T\mathbf{X}\mathbf{X}^T\mathbf{a}] = \mathbf{a}^T\mathbb{E}[\mathbf{X}\mathbf{X}^T]\mathbf{a} \tag{39}$$

由于 $\mathbb{E}[\mathrm{X}] = \mathbf{0}$，协方差矩阵为 $\mathrm{Cov}(\mathrm{X}) = \mathbb{E}[\mathrm{X}\mathrm{X}^T]$。令 $\boldsymbol{\Sigma} = \mathrm{Cov}(\mathrm{X})$。方差条件给出：

$$\mathbf{a}^T\boldsymbol{\Sigma}\mathbf{a} = 1 \quad \text{for all unit vectors } \mathbf{a} \tag{40}$$

下面证明 $\boldsymbol{\Sigma} = \mathrm{I}_d$。步骤 1：对角元素。对 $i \in \{1, 2, \ldots, d\}$，令 e_i 表示第 i 个标准基向量。在式（40）中令 $\mathrm{a} = \mathrm{e}_i$：

$$\mathbf{e}_i^T\boldsymbol{\Sigma}\mathbf{e}_i = \Sigma_{ii} = 1 \tag{41}$$

因此，$\boldsymbol{\Sigma}$ 的所有对角元素均为 1。步骤 2：非对角元素。对于不同的索引 $i, j \in \{1, 2, \ldots, d\}$，考虑单位向量：

$$\mathbf{a} = \frac{\mathbf{e}_i + \mathbf{e}_j}{\|\mathbf{e}_i + \mathbf{e}_j\|_2} = \frac{\mathbf{e}_i + \mathbf{e}_j}{\sqrt{2}} \tag{42}$$

应用式（40）：

$$\mathbf{a}^T\boldsymbol{\Sigma}\mathbf{a} = \frac{1}{2}(\mathbf{e}_i + \mathbf{e}_j)^T\boldsymbol{\Sigma}(\mathbf{e}_i + \mathbf{e}_j) = 1 \tag{43}$$

Expanding the quadratic form and using the symmetry of $\mathbf{\Sigma}$:

$$\frac{1}{2}(\mathbf{e}_i^T\mathbf{\Sigma}\mathbf{e}_i + 2\mathbf{e}_i^T\mathbf{\Sigma}\mathbf{e}_j + \mathbf{e}_j^T\mathbf{\Sigma}\mathbf{e}_j) = 1 \tag{44}$$

$$\frac{1}{2}(\Sigma_{ii} + 2\Sigma_{ij} + \Sigma_{jj}) = 1 \tag{45}$$

$$\frac{1}{2}(1 + 2\Sigma_{ij} + 1) = 1 \tag{46}$$

$$1 + \Sigma_{ij} = 1 \tag{47}$$

$$\Sigma_{ij} = 0 \tag{48}$$

Therefore, all off-diagonal entries of $\mathbf{\Sigma}$ equal zero, establishing that $\mathbf{\Sigma} = \mathbf{I}_d$. □

C Background

Foundation: The Linear Regression Model We start with the standard linear regression model:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

where:

- $\mathbf{y} = [y_1, y_2, \ldots, y_n]^T \in \mathbb{R}^n$ is the response vector
- $\mathbf{X} \in \mathbb{R}^{n\times p}$ is the design matrix with $\mathbf{X}_{ij} = x_{ij}$
- $\boldsymbol{\beta} = [\beta_1, \beta_2, \ldots, \beta_p]^T \in \mathbb{R}^p$ is the parameter vector
- $\boldsymbol{\varepsilon} = [\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n]^T \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbf{I}_n)$ is the error vector

The error assumption means:

$$\mathbb{E}[\varepsilon_i] = 0, \quad \mathrm{Var}(\varepsilon_i) = \sigma^2, \quad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j$$

Step 1: Deriving the OLS Estimator To find the OLS estimator, we minimize the sum of squared residuals:

$$\mathrm{SSR}(\boldsymbol{\beta}) = \sum_{i=1}^{n}(y_i - \mathbf{x}_i^T\boldsymbol{\beta})^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

Expanding this quadratic form:

$$\mathrm{SSR}(\boldsymbol{\beta}) = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} \tag{49}$$

Taking the derivative with respect to $\boldsymbol{\beta}$:

$$\frac{\partial \mathrm{SSR}}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

Setting equal to zero and solving:

$$-2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$$

$$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$$

Assuming $\mathbf{X}^T\mathbf{X}$ is invertible:

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}}$$

D Details on Low-Discrepancy Sequences

Quasi-Monte Carlo (QMC) methods, such as the Sobol sequence, are widely used to generate low-discrepancy samples in the unit hypercube, providing improved uniformity over purely random sampling. To obtain samples uniformly distributed on the hypersphere, each QMC point is mapped to a standard normal vector via the inverse cumulative

展开二次型并利用 $\mathbf{\Sigma}$ 的对称性：

$$\frac{1}{2}(\mathbf{e}_i^T\mathbf{\Sigma}\mathbf{e}_i + 2\mathbf{e}_i^T\mathbf{\Sigma}\mathbf{e}_j + \mathbf{e}_j^T\mathbf{\Sigma}\mathbf{e}_j) = 1 \tag{44}$$

$$\frac{1}{2}(\Sigma_{ii} + 2\Sigma_{ij} + \Sigma_{jj}) = 1 \tag{45}$$

$$\frac{1}{2}(1 + 2\Sigma_{ij} + 1) = 1 \tag{46}$$

$$1 + \Sigma_{ij} = 1 \tag{47}$$

$$\Sigma_{ij} = 0 \tag{48}$$

因此， $\mathbf{\Sigma}$ 的所有非对角元素均为零，从而证明 $\mathbf{\Sigma} = \mathrm{I}_d$。 □

C 背景

基础 线性回归模型。我们从标准线性回归模型出发，

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$$

其中：

- $\mathrm{y} = [y_1, y_2, \ldots, y_n]^T \in \mathbb{R}^n$ 是响应向量
- $\mathrm{X} \in \mathbb{R}^{n\times p}$ 是设计矩阵，其中 $\mathrm{X}^{ij} = x_{ij}$
- $\boldsymbol{\beta} = [\beta_1, \beta_2, \ldots, \beta_p]^T \in \mathbb{R}^p$ 是参数向量
- $\boldsymbol{\varepsilon} = [\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n]^T \sim \mathcal{N}(0, \sigma^2\mathrm{I}_n)$ 是误差向量

误差假设意味着：

$$\mathbb{E}[\varepsilon_i] = 0, \quad \mathrm{Var}(\varepsilon_i) = \sigma^2, \quad \mathrm{Cov}(\varepsilon_i, \varepsilon_j) = 0 \text{ for } i \neq j$$

步骤 1：推导 OLS 估计量。为求出普通最小二乘估计量，我们最小化残差平方和：

$$\mathrm{SSR}(\boldsymbol{\beta}) = \sum_{i=1}^{n}(y_i - \mathbf{x}_i^T\boldsymbol{\beta})^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

展开该二次型：

$$\mathrm{SSR}(\boldsymbol{\beta}) = \mathbf{y}^T\mathbf{y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} \tag{49}$$

对 $\boldsymbol{\beta}$ 求导：

$$\frac{\partial \mathrm{SSR}}{\partial \boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}$$

令其等于零并求解：

$$-2\mathbf{X}^T\mathbf{y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$$

$$\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{X}^T\mathbf{y}$$

假设 $\mathrm{X}^T\mathrm{X}$ 可逆：

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}}$$

D 低差异序列的细节

准蒙特卡洛（QMC）方法，例如 Sobol 序列，广泛用于在单位超立方体中生成低差异样本，相比纯随机采样具有更好的均匀性。为获得超球面上的均匀分布样本，通过逆累积函数将每个 QMC 点映射为标准正态向量，

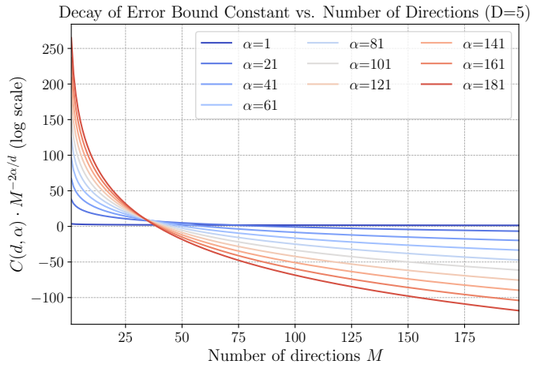


Figure 15. Depiction of the expected BCS loss upper bound (thm. 5) for various smoothness values α. We clearly see that as the smoothness increases (**blue to red**), as the upper bound decreases more and more rapidly with M.

Table 3. Performance metrics across different sample sizes from Figure 12

Freeze Backbone	Model Name	Samples per Class						
		All	**1**	**2**	**5**	**10**	**100**	**1000**
No	*LeJEPA (Ours)*							
	ConvNeXt-V2 Nano	82.72	**29.42**	**36.65**	**50.94**	**59.85**	**75.34**	81.97
	LeViT-128	79.41	18.45	24.08	33.11	41.76	64.59	77.59
	ResNet-18	82.15	23.34	31.56	43.82	54.64	73.53	81.41
	ResNet-34	**83.28**	24.27	31.51	44.23	53.95	74.93	**82.32**
	Baselines							
	DINOv2 Small	78.34	21.05	21.71	30.33	36.23	60.81	75.55
	DINOv3 ViT-S/16	81.60	24.71	29.43	37.71	44.71	69.87	80.54
Yes	*LeJEPA (Ours)*							
	ConvNeXt-V2 Nano	76.52	28.74	36.65	50.60	59.50	72.62	77.24
	LeViT-128	69.00	25.85	33.30	45.52	52.43	64.37	69.39
	ResNet-18	75.95	30.48	38.22	50.85	58.86	72.70	76.39
	ResNet-34	**78.17**	**31.08**	**38.33**	**52.26**	**60.63**	**74.77**	**78.62**
	Baselines							
	DINOv2 Small	67.62	27.68	32.22	40.72	47.72	62.49	67.89
	DINOv3 ViT-S/16	71.38	30.17	36.65	45.74	51.51	65.90	71.35

Table 4. Top 1 accuracy (in %) with LeJEPA pretraining on Imagenet-100 for 400 epochs (All values are percentages)

	backbone	resnet50			vit_small_patch8_224			vit_tiny_patch8_224		
	Projector	1-layer	2-layer	3-layer	1-layer	2-layer	3-layer	1-layer	2-layer	3-layer
w/ predictor	w/ SWA									
False	False	79.71	82.44	83.93	76.59	80.77	81.07	71.79	76.87	80.37
	True	79.79	82.69	83.50	79.96	83.63	84.12	75.86	82.36	80.50
True	False	79.41	82.44	83.57	77.58	79.41	81.91	67.74	77.64	80.73
	True	78.87	82.04	82.82	77.11	81.77	82.58	69.53	78.27	79.77

distribution function (CDF), and then projected onto the sphere by normalization. This approach leverages the rotational invariance of the multivariate normal distribution, ensuring that the resulting directions are uniformly distributed on

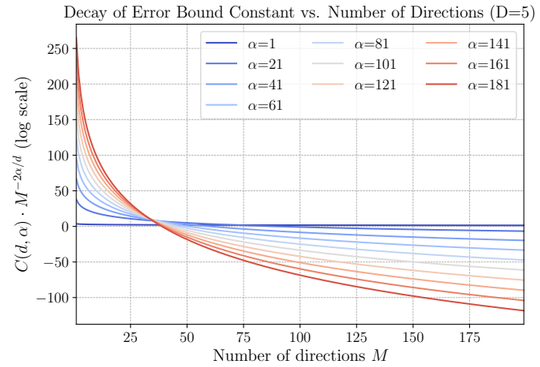


图 15. 不同光滑度 α 下的期望 BCS 损失上界（定理 5）。可以清楚看到，随着光滑度增加（从蓝色到红色），上界随 M 下降得越来越快。

表 3. 图 12 中不同样本量下的性能指标。

冻结骨干网络	模型名称	每类样本数						
		All	**1**	**2**	**5**	**10**	**100**	**1000**
No	*LeJEPA*（我们的方法）							
	ConvNeXt-V2 Nano	82.72	**29.42**	**36.65**	**50.94**	**59.85**	**75.34**	81.97
	LeViT-128	79.41	18.45	24.08	33.11	41.76	64.59	77.59
	ResNet-18	82.15	23.34	31.56	43.82	54.64	73.53	81.41
	ResNet-34	**83.28**	24.27	31.51	44.23	53.95	74.93	**82.32**
	基线							
	DINOv2 Small	78.34	21.05	21.71	30.33	36.23	60.81	75.55
	DINOv3 ViT-S/16	81.60	24.71	29.43	37.71	44.71	69.87	80.54
Yes	*LeJEPA*（我们的方法）							
	ConvNeXt-V2 Nano	76.52	28.74	36.65	50.60	59.50	72.62	77.24
	LeViT-128	69.00	25.85	33.30	45.52	52.43	64.37	69.39
	ResNet-18	75.95	30.48	38.22	50.85	58.86	72.70	76.39
	ResNet-34	**78.17**	**31.08**	**38.33**	**52.26**	**60.63**	**74.77**	**78.62**
	基线							
	DINOv2 Small	67.62	27.68	32.22	40.72	47.72	62.49	67.89
	DINOv3 ViT-S/16	71.38	30.17	36.65	45.74	51.51	65.90	71.35

表 4. 使用 LeJEPA 在 ImageNet-100 上预训练 400 个轮次后的 top-1 准确率（%，所有数值均为百分比）。

	骨干网络	resnet50			vit_small_patch8_224			vit_tiny_patch8_224		
	投影器	1 层	2 层	3 层	1 层	2 层	3 层	1 层	2 层	3 层
使用预测器／使用 SWA										
否	否	79.71	82.44	83.93	76.59	80.77	81.07	71.79	76.87	80.37
	True	79.79	82.69	83.50	79.96	83.63	84.12	75.86	82.36	80.50
True	否	79.41	82.44	83.57	77.58	79.41	81.91	67.74	77.64	80.73
	True	78.87	82.04	82.82	77.11	81.77	82.58	69.53	78.27	79.77

分布函数（CDF），随后通过归一化投影到球面上。这种方法利用多元正态分布的旋转不变性，保证所得方向均匀分布于

Table 5. Small architecture in-domain LeJEPA pretraining from random initialization across datasets and architectures, with frozen backbone linear evaluation. First, **LeJEPA is able to produce near state-of-the-art performances on tiny dataset with only a thousand samples**, e.g., flowers102. Second, **on non-natural image data, LeJEPA clearly outperforms the latest frontier vision models**, e.g., Galaxy10. See Figure 12 for additional experiments with varying number of training samples and with full finetuning.

	Pretraining # train. samples	flowers102 1020	cifar100 50000	food101 75750	inet10 13000	cifar10 50000	galaxy10 11008
LeJEPA (convnextv2_nano) 14M	in-domain	64.34	69.26	69.59	90.81	92.22	76.05
LeJEPA (resnet18) 11M	in-domain	74.57	69.94	73.57	92.36	92.51	75.32
LeJEPA (resnet34) 21M	in-domain	71.85	70.44	74.95	92.80	93.16	77.29
LeJEPA (resnext26ts) 8M	in-domain	82.19	69.10	76.77	92.82	91.59	73.78
LeJEPA (swin_tiny) 27M	in-domain	63.94	65.08	78.40	92.87	92.67	74.89
IJEPA-inet22k (ViT-H/14) 630M	inet1k	85.76	86.93	81.06	98.65	97.77	62.93

Table 6. Time (in millisecond) to compute the proposed SIGReg loss from algorithm 1 on a Tesla V100-SXM2-16GB for varying mini-batch size (N), number of slices (M), integration points. Results are computed over 10 runs.

N	M	# integration points	mean (ms)	std (ms)
512	512	16	0.465236	0.011642
512	512	64	0.461317	0.003894
512	512	256	0.627644	0.003337
2048	512	16	1.406441	0.002415
8192	512	16	6.188304	0.007226
8192	8192	16	8.685009	0.038829
32768	512	16	26.373118	0.012732
512	2048	16	0.465614	0.005274
512	8192	16	0.670379	0.006854

Table 7. Number of Figure 8.

	resnet50											
λ #views	0.001	0.005	0.010	0.020	0.025	0.050	0.100	0.150	0.200	0.300	0.400	0.500
2	81.41	82.73	83.49	82.99	82.23	-	-	-	-	-	-	-
4	79.88	83.04	84.36	84.68	84.33	83.00	82.91	81.05	78.58	-	-	-
8	76.67	81.58	83.59	83.49	83.76	84.32	83.66	83.07	82.16	81.00	79.25	77.72

the sphere's surface. While the low-discrepancy property is not strictly preserved under this nonlinear mapping, the resulting samples are empirically more uniform than random samples and are standard in high-dimensional applications Marsaglia [1972], Dick and Pillichshammer [2010], Caflisch [1998].

Require: Number of points N, dimension d
Ensure: Points $\{\mathbf{y}_i\}_{i=1}^{N}$ quasi-uniformly distributed on $\mathbb{S}^{d-1}$
1: **for** $i = 1$ to N **do**
2: Generate $\mathbf{x}_i \in [0,1]^d$ as the i-th point of a Sobol sequence
3: Transform each component: $z_{i,j} = \Phi^{-1}(x_{i,j})$ for $j = 1, \ldots, d$ ▷ Φ^{-1} is the inverse CDF of the standard normal
4: Normalize: $\mathbf{y}_i = \mathbf{z}_i / \|\mathbf{z}_i\|_2$
5: **end for**

E Shapiro-Wilk Test

Let X1 < X2 < . . . < Xn denote an ordered random sample of size n from a standard normal distribution. Also, let mÂ 5 (m1,m2,...,mn) be the vector of expected values of standard normal order statistics, and let V 5 (vij) be the corresponding

表 5. 多数据、架构从随机初始化域内预训练，冻结线性评估。仅千样本的 flowers102 也接近先进性能；非自然图像 Galaxy10 上明显超过最新模型。改变训练样本、完整微调见图 12。

	预训练 训练样本数	flowers102 1020	cifar100 50000	food101 75750	inet10 13000	cifar10 50000	galaxy10 11008
LeJEPA (convnextv2_nano) 14M	领域内	64.34	69.26	69.59	90.81	92.22	76.05
LeJEPA (resnet18) 11M	领域内	74.57	69.94	73.57	92.36	92.51	75.32
LeJEPA (resnet34) 21M	领域内	71.85	70.44	74.95	92.80	93.16	77.29
LeJEPA (resnext26ts) 8M	领域内	82.19	69.10	76.77	92.82	91.59	73.78
LeJEPA (swin_tiny) 27M	领域内	63.94	65.08	78.40	92.87	92.67	74.89
IJEPA-inet22k (ViT-H/14) 630M	inet1k	85.76	86.93	81.06	98.65	97.77	62.93

表 6. 计算耗时（毫秒）：
切片数（M）、积分点数

在 Tesla V100-SXM2-16GB 上，对于不同的小批量大小（N），计算算法 1 提出的 SIGReg 损失，
结果基于 10 次运行计算。

N	M	积分数量 点数	均值（毫秒）	标准差（毫秒）
512	512	16	0.465236	0.011642
512	512	64	0.461317	0.003894
512	512	256	0.627644	0.003337
2048	512	16	1.406441	0.002415
8192	512	16	6.188304	0.007226
8192	8192	16	8.685009	0.038829
32768	512	16	26.373118	0.012732
512	2048	16	0.465614	0.005274
512	8192	16	0.670379	0.006854

表 7. 图 8 中的数值。

	resnet50											
λ 视图数	0.001	0.005	0.010	0.020	0.025	0.050	0.100	0.150	0.200	0.300	0.400	0.500
2	81.41	82.73	83.49	82.99	82.23	-	-	-	-	-	-	-
4	79.88	83.04	84.36	84.68	84.33	83.00	82.91	81.05	78.58	-	-	-
8	76.67	81.58	83.59	83.49	83.76	84.32	83.66	83.07	82.16	81.00	79.25	77.72

球面的表面。虽然这种非线性映射不能严格保持低差异性质，但所得样本在经验上比随机样本更均匀，也是高维应用中的标准做法，参见 Marsaglia [1972]、Dick and Pillichshammer [2010]、Caflisch [1998]。

输入：点数 N，维度 d
保证：点集 $\{\mathbf{y}_i\}_{i=1}^{N}$ 在 $\mathbb{S}^{d-1}$ 上准均匀分布 1: 对 $i = 1$ 到 N 循环

2: 将 $\mathbf{x}_i \in [0,1]^d$ 生成为 Sobol 序列的第 i 个点 3: 变换每个分量：$z_{i,j} = \Phi^{-1}(x_{i,j})$，其中 $j = 1, \ldots, d$ ▷ Φ^{-1} 为标准正态分布的逆累积分布函数 4: 归一化：$\mathbf{y}_i = \mathbf{z}_i / \|\mathbf{z}_i\|_2$ 5: 结束循环

E Shapiro–Wilk 检验

令 X1 < X2 < . . . <Xn 表示来自标准正态分布、大小为 n 的有序随机样本。另令 mÂ 5 (m1,m2,...,mn) 为标准正态顺序统计量期望值组成的向量，V 5 (vij) 为对应的

n 3 n covariance matrix, so that

$$E\left(X_i\right) = m_i \quad \text{and} \quad \operatorname{cov}\left(X_i, X_j\right) = v_{ij}, \quad i, j = 1, 2, \ldots, n \tag{50}$$

The W test statistic Shapiro and Wilk [1965] for normality is then denoted by

$$\begin{aligned} W &= \frac{\left(\sum_{i=1}^{n} a_i Y_i\right)}{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2} = \frac{(\mathbf{aY})}{S^2} \\ \mathbf{a}' &= \left(a_1, a_2, \ldots, a_n\right) = \mathbf{m}\mathbf{V}^{-1}\left(\mathbf{m}\mathbf{V}^{-1}\mathbf{V}^{-1}\mathbf{m}\right)^{-1/2} \\ \mathrm{S}^2 &= \sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2 \end{aligned} \tag{51}$$

Shapiro and Francia [1972] suggested replacing the covariance matrix V by the identity matrix I, because for large samples, the observations Yi may be treated as if they are independent (see Gupta [1952]). Another asymptotic extension was suggested by Weisburg and Binham [1975]

$$E\left(X_i\right) = m_i \approx \Phi^{-1}\left(\frac{i - \frac{3}{8}}{n + \frac{1}{4}}\right) \quad i = 1, 2, \ldots, n \tag{52}$$

building atop Elfving [1947]'s approximation but using 3/8 instead of $\pi/8$.

Rahman and Govindarajulu [1997] proposed another variation using the approximation for the expected values of order statistics given by Blom [1958] and the approximations for the elements of the variance± covariance matrix given by Blom [1958], Mosteller [2006]. These approximations are

$$E\left(X_i\right) = m_i \approx \Phi^{-1}\left(\frac{i}{N+1}\right), \quad i = 1, 2, \ldots, n \tag{53}$$

$$\operatorname{cov}\left(X_i, X_j\right) = v_{ij} \approx \frac{p_i p_j}{(n+2) f\left(m_i\right) f\left(m_j\right)}, \quad i, j = 1, 2, \ldots, n \tag{54}$$

$$p_i = \frac{i}{n+1} \tag{55}$$

We know (see Hammersley and Morton [1954], Plackett [1958])

$$\begin{aligned} \mathbf{V}^{-1} &= (n+1)(n+2) \\ &\times \begin{pmatrix} 2\phi^2\left(m_1\right) & -\phi\left(m_1\right)\phi\left(m_2\right) & 0 & 0 & \ldots & 0 \\ -\phi\left(m_1\right)\phi\left(m_2\right) & 2\phi^2\left(m_2\right) & -\phi\left(m_2\right)\phi\left(m_3\right) & 0 & \ldots & 0 \\ 0 & -\phi\left(m_2\right)\phi\left(m_3\right) & 2\phi^2\left(m_3\right) & -\phi\left(m_3\right)\phi\left(m_4\right) & \ldots & 0 \\ \vdots & & & & & \\ 0 & 0 & 0 & 0 & \ldots & 2\phi^2\left(m_n\right) \end{pmatrix} \end{aligned} \tag{56}$$

F Multivariate Statistics

We ideally would like to compare the distributions. One slight variation is to compare the Characteristic function of the distributions. Given samples $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N$, the Empirical Characteristic Function (ECF) is defined as

$$\hat{\psi}_N(\boldsymbol{t}) = \frac{1}{N}\sum_{n=1}^{N} e^{-i\boldsymbol{t}^\top \boldsymbol{y}_n}.$$

We can now compare our ECF to the one of the target distribution and build the statistic

$$N\int |\hat{\psi}_N(\boldsymbol{t}) - \psi_0(\boldsymbol{t})|^2 \omega(\boldsymbol{t}) d\boldsymbol{t} = N\int |\hat{\psi}_N(\boldsymbol{t}) - e^{-\|\boldsymbol{t}\|_2/2}|^2 \omega(\boldsymbol{t}) d\boldsymbol{t},$$

n×n 协方差矩阵，从而

$$E\left(X_i\right) = m_i \quad \text{and} \quad \operatorname{cov}\left(X_i, X_j\right) = v_{ij}, \quad i, j = 1, 2, \ldots, n \tag{50}$$

Shapiro and Wilk [1965] 的正态性 W 检验统计量记为

$$\begin{aligned} W &= \frac{\left(\sum_{i=1}^{n} a_i Y_i\right)}{\sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2} = \frac{(\mathbf{aY})}{S^2} \\ \mathbf{a}' &= \left(a_1, a_2, \ldots, a_n\right) = \mathbf{m}\mathbf{V}^{-1}\left(\mathbf{m}\mathbf{V}^{-1}\mathbf{V}^{-1}\mathbf{m}\right)^{-1/2} \\ \mathrm{S}^2 &= \sum_{i=1}^{n}\left(Y_i - \bar{Y}\right)^2 \end{aligned} \tag{51}$$

Shapiro and Francia [1972] 建议用单位矩阵 I 替代协方差矩阵 V，因为对于大样本，可将观测 Yi 视为相互独立（参见 Gupta [1952]）。Weisburg and Binham [1975] 提出了另一种渐近推广。

$$E\left(X_i\right) = m_i \approx \Phi^{-1}\left(\frac{i - \frac{3}{8}}{n + \frac{1}{4}}\right) \quad i = 1, 2, \ldots, n \tag{52}$$

基于 Elfving [1947] 的近似，但使用 3/8 而非 $\pi/8$。

Rahman and Govindarajulu [1997] 提出另一种变体，采用 Blom [1958] 给出的顺序统计量期望值近似，以及 Blom [1958], 和 Mosteller [2006] 给出的方差± 协方差矩阵元素近似。这些近似为

$$E\left(X_i\right) = m_i \approx \Phi^{-1}\left(\frac{i}{N+1}\right), \quad i = 1, 2, \ldots, n \tag{53}$$

$$\operatorname{cov}\left(X_i, X_j\right) = v_{ij} \approx \frac{p_i p_j}{(n+2) f\left(m_i\right) f\left(m_j\right)}, \quad i, j = 1, 2, \ldots, n \tag{54}$$

$$p_i = \frac{i}{n+1} \tag{55}$$

已知（参见 Hammersley and Morton [1954]、Plackett [1958]），

$$\begin{aligned} \mathbf{V}^{-1} &= (n+1)(n+2) \\ &\times \begin{pmatrix} 2\phi^2\left(m_1\right) & -\phi\left(m_1\right)\phi\left(m_2\right) & 0 & 0 & \ldots & 0 \\ -\phi\left(m_1\right)\phi\left(m_2\right) & 2\phi^2\left(m_2\right) & -\phi\left(m_2\right)\phi\left(m_3\right) & 0 & \ldots & 0 \\ 0 & -\phi\left(m_2\right)\phi\left(m_3\right) & 2\phi^2\left(m_3\right) & -\phi\left(m_3\right)\phi\left(m_4\right) & \ldots & 0 \\ \vdots & & & & & \\ 0 & 0 & 0 & 0 & \ldots & 2\phi^2\left(m_n\right) \end{pmatrix} \end{aligned} \tag{56}$$

F 多元统计量

理想情况下，我们希望比较两个分布。一种稍有变化的做法是比较分布的特征函数。给定样本 $\boldsymbol{x}_1, \ldots, \boldsymbol{x}_N$，经验特征函数（ECF）定义为

$$\hat{\psi}_N(\boldsymbol{t}) = \frac{1}{N}\sum_{n=1}^{N} e^{-i\boldsymbol{t}^\top \boldsymbol{y}_n}.$$

现在可以比较我们的 ECF 与目标分布的 ECF，并构造统计量。

$$N\int |\hat{\psi}_N(\boldsymbol{t}) - \psi_0(\boldsymbol{t})|^2 \omega(\boldsymbol{t}) d\boldsymbol{t} = N\int |\hat{\psi}_N(\boldsymbol{t}) - e^{-\|\boldsymbol{t}\|_2/2}|^2 \omega(\boldsymbol{t}) d\boldsymbol{t},$$

if the weighting function is given by $\omega(\boldsymbol{t}) = (2\pi\beta^2)^{-d/2} e^{-\frac{\|t\|_2^2}{2}}$ then the following simplification can be made

$$\begin{aligned}\mathrm{BHEP}_{n,\beta} =& \frac{1}{n}\sum_{j,k=1}^{n}\exp\left(-\frac{\beta^2\left\|Y_{n,j}-Y_{n,k}\right\|^2}{2}\right)\\ &-\frac{2}{\left(1+\beta^2\right)^{d/2}}\sum_{j=1}^{n}\exp\left(-\frac{\beta^2\left\|Y_{n,j}\right\|^2}{2\left(1+\beta^2\right)}\right)+\frac{n}{\left(1+2\beta^2\right)^{d/2}}.\end{aligned}$$

with $\beta > 0$, Baringhaus-Henze-Epps-Pulley. From [1] leading to the HZ test [2] uses

$$\beta_n = 2^{-1/2}((2d+1)n/4)^{1/(d+4)} \tag{57}$$

the same can be done with the moment generating function [3]

$$\begin{aligned}T_{n,\beta} = \pi^{d/2}\Bigg(&\frac{1}{n}\sum_{i,j=1}^{n}\frac{1}{\beta^{d/2}}\exp\left(\frac{\left\|Y_{n,i}+Y_{n,j}\right\|^2}{4\beta}\right)+\frac{n}{(\beta-1)^{d/2}}\\ &-2\sum_{j=1}^{n}\frac{1}{(\beta-1/2)^{d/2}}\exp\left(\frac{\left\|Y_{n,j}\right\|^2}{4\beta-2}\right)\Bigg),\end{aligned}$$

here with $\beta > 2$

There is also one combining both[4]!

$$\begin{aligned}&T_{n,\gamma} := \textstyle\int_{\mathbb{R}^d} U_n^2(t) w_\gamma(t)\mathrm{d}t\\ &U_n(t) := \sqrt{n}\left(R_n(t)M_n(t) - 1\right)\end{aligned} \tag{58}$$

$$\begin{aligned}T_{n,\gamma} = \left(\frac{\pi}{\gamma}\right)^{d/2}\Bigg\{&\frac{1}{2n^3}\sum_{j,k,l,m=1}^{n}\left[\exp\left(\frac{\left\|Y_{jk}^{+}\right\|^2-\left\|Y_{\ell m}^{-}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{jk}^{+\top}Y_{\ell m}^{-}}{2\gamma}\right)\right.\\ &\left.+\exp\left(\frac{\left\|Y_{jk}^{+}\right\|^2-\left\|Y_{\ell m}^{+}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{jk}^{+\top}Y_{\ell m}^{+}}{2\gamma}\right)\right]\\ &-\frac{2}{n}\sum_{j,k=1}^{n}\exp\left(\frac{\left\|Y_{n,j}\right\|^2-\left\|Y_{n,k}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{n,j}^{\top}Y_{n,k}}{2\gamma}\right)+n\Bigg\},\end{aligned} \tag{59}$$

and its simplified version

$$\widetilde{T}_{n,\gamma} := \int_{\mathbb{R}^d} U_n(t) w_\gamma(t)\mathrm{d}t. \tag{60}$$

$$\widetilde{T}_{n,\gamma} = \left(\frac{\pi}{\gamma}\right)^{d/2}\sqrt{n}\left(\frac{1}{n^2}\sum_{j,k=1}^{n}\exp\left(\frac{\left\|Y_{n,j}\right\|^2-\left\|Y_{n,k}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{n,j}^{\top}Y_{n,k}}{2\gamma}\right)-1\right) \tag{61}$$

Also one testing the derivative [5]

$$\mathrm{HV}_{n,\gamma} := n\int\left\|\nabla M_n(t) - tM_n(t)\right\|^2\widetilde{w}_\gamma(t)\mathrm{d}t \tag{62}$$

[1] https://www.routledge.com/Density-Estimation-for-Statistics-and-Data-Analysis/Silverman/p/book/9780412246203?srsltid=AfmBOoodlL-CtlqL0JVC-LcP6mOWw6VTt51_YstdZOW4W3iuicu1VFyg
[2] https://www.tandfonline.com/doi/abs/10.1080/03610929008830400
[3] https://arxiv.org/pdf/1711.07199
[4] https://arxiv.org/pdf/1706.03029
[5] https://arxiv.org/pdf/1901.03986

若权重函数为 $\omega(\boldsymbol{t}) = (2\pi\beta^2)^{-d/2} e^{-\frac{\|t\|_2^2}{2}}$，则可以作如下化简：

$$\begin{aligned}\mathrm{BHEP}_{n,\beta} =& \frac{1}{n}\sum_{j,k=1}^{n}\exp\left(-\frac{\beta^2\left\|Y_{n,j}-Y_{n,k}\right\|^2}{2}\right)\\ &-\frac{2}{\left(1+\beta^2\right)^{d/2}}\sum_{j=1}^{n}\exp\left(-\frac{\beta^2\left\|Y_{n,j}\right\|^2}{2\left(1+\beta^2\right)}\right)+\frac{n}{\left(1+2\beta^2\right)^{d/2}}.\end{aligned}$$

其中 $\beta > 0$，即 Baringhaus–Henze–Epps–Pulley。由 [1] 得到的 HZ 检验 [2] 使用

$$\beta_n = 2^{-1/2}((2d+1)n/4)^{1/(d+4)} \tag{57}$$

对于矩生成函数 [3]，也可作相同处理。

$$\begin{aligned}T_{n,\beta} = \pi^{d/2}\Bigg(&\frac{1}{n}\sum_{i,j=1}^{n}\frac{1}{\beta^{d/2}}\exp\left(\frac{\left\|Y_{n,i}+Y_{n,j}\right\|^2}{4\beta}\right)+\frac{n}{(\beta-1)^{d/2}}\\ &-2\sum_{j=1}^{n}\frac{1}{(\beta-1/2)^{d/2}}\exp\left(\frac{\left\|Y_{n,j}\right\|^2}{4\beta-2}\right)\Bigg),\end{aligned}$$

这里取 $\beta > 2$

还有一种将两者结合的检验 [4]!

$$\begin{aligned}&T_{n,\gamma} := \textstyle\int_{\mathbb{R}^d} U_n^2(t) w_\gamma(t)\mathrm{d}t\\ &U_n(t) := \sqrt{n}\left(R_n(t)M_n(t) - 1\right)\end{aligned} \tag{58}$$

$$\begin{aligned}T_{n,\gamma} = \left(\frac{\pi}{\gamma}\right)^{d/2}\Bigg\{&\frac{1}{2n^3}\sum_{j,k,l,m=1}^{n}\left[\exp\left(\frac{\left\|Y_{jk}^{+}\right\|^2-\left\|Y_{\ell m}^{-}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{jk}^{+\top}Y_{\ell m}^{-}}{2\gamma}\right)\right.\\ &\left.+\exp\left(\frac{\left\|Y_{jk}^{+}\right\|^2-\left\|Y_{\ell m}^{+}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{jk}^{+\top}Y_{\ell m}^{+}}{2\gamma}\right)\right]\\ &-\frac{2}{n}\sum_{j,k=1}^{n}\exp\left(\frac{\left\|Y_{n,j}\right\|^2-\left\|Y_{n,k}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{n,j}^{\top}Y_{n,k}}{2\gamma}\right)+n\Bigg\},\end{aligned} \tag{59}$$

及其简化版本

$$\widetilde{T}_{n,\gamma} := \int_{\mathbb{R}^d} U_n(t) w_\gamma(t)\mathrm{d}t. \tag{60}$$

$$\widetilde{T}_{n,\gamma} = \left(\frac{\pi}{\gamma}\right)^{d/2}\sqrt{n}\left(\frac{1}{n^2}\sum_{j,k=1}^{n}\exp\left(\frac{\left\|Y_{n,j}\right\|^2-\left\|Y_{n,k}\right\|^2}{4\gamma}\right)\cos\left(\frac{Y_{n,j}^{\top}Y_{n,k}}{2\gamma}\right)-1\right) \tag{61}$$

还有一种检验导数的方法 [5]

$$\mathrm{HV}_{n,\gamma} := n\int\left\|\nabla M_n(t) - tM_n(t)\right\|^2\widetilde{w}_\gamma(t)\mathrm{d}t \tag{62}$$

[1] https://www.routledge.com/Density-Estimation-for-Statistics-and-Data-Analysis/Silverman/p/book/9780412246203?srsltid=AfmBOoodlL-CtlqL0JVC-LcP6mOWw6VTt51_YstdZOW4W3iuicu1VFyg [2] https://www.tandfonline.com/doi/abs/10.1080/03610929008830400 [3] https://arxiv.org/pdf/1711.07199 [4] https://arxiv.org/pdf/1706.03029 [5] https://arxiv.org/pdf/1901.03986

$$\mathrm{HV}_{n,\gamma} = \frac{1}{n}\left(\frac{\pi}{\gamma}\right)^{d/2} \sum_{j,k=1}^{n} \exp\left(\frac{\left\|Y^{+}_{n,j,k}\right\|^2}{4\gamma}\right)\left(Y_{n,j}^{\top} Y_{n,k} - \frac{\left\|Y^{+}_{n,j,k}\right\|^2}{2\gamma} + \frac{d}{2\gamma} + \frac{\left\|Y^{+}_{n,j,k}\right\|^2}{4\gamma^2}\right). \tag{63}$$

skewness [6]:

$$b_{1,d} = \frac{1}{n^2} \sum_{j,k=1}^{n} \left(Y_{n,j}^{\top} Y_{n,k}\right)^3 \tag{64}$$

skewness [7]:

$$\widetilde{b}_{1,d} = \frac{1}{n^2} \sum_{j,k=1}^{n} Y_{n,j}^{\top} Y_{n,k} \left\|Y_{n,j}\right\|^2 \left\|Y_{n,k}\right\|^2 \tag{65}$$

which should be 0 for Gaussian and Kurtosis which should be d(d+2)

$$b_{2,d} = \frac{1}{n} \sum_{j=1}^{n} \left\|Y_{n,j}\right\|^4 \tag{66}$$

[6] https://www.jstor.org/stable/2334770
[7] https://link.springer.com/article/10.1007/s13171-020-00211-6

$$\mathrm{HV}_{n,\gamma} = \frac{1}{n}\left(\frac{\pi}{\gamma}\right)^{d/2} \sum_{j,k=1}^{n} \exp\left(\frac{\left\|Y^{+}_{n,j,k}\right\|^2}{4\gamma}\right)\left(Y_{n,j}^{\top} Y_{n,k} - \frac{\left\|Y^{+}_{n,j,k}\right\|^2}{2\gamma} + \frac{d}{2\gamma} + \frac{\left\|Y^{+}_{n,j,k}\right\|^2}{4\gamma^2}\right). \tag{63}$$

偏度 [6]:

$$b_{1,d} = \frac{1}{n^2} \sum_{j,k=1}^{n} \left(Y_{n,j}^{\top} Y_{n,k}\right)^3 \tag{64}$$

偏度 [7]:

$$\widetilde{b}_{1,d} = \frac{1}{n^2} \sum_{j,k=1}^{n} Y_{n,j}^{\top} Y_{n,k} \left\|Y_{n,j}\right\|^2 \left\|Y_{n,k}\right\|^2 \tag{65}$$

对于高斯分布，应为 0；峰度应为 d(d+2)。

$$b_{2,d} = \frac{1}{n} \sum_{j=1}^{n} \left\|Y_{n,j}\right\|^4 \tag{66}$$

[6] https://www.jstor.org/stable/2334770[7]
https://link.springer.com/article/10.1007/s13171-020-00211-6

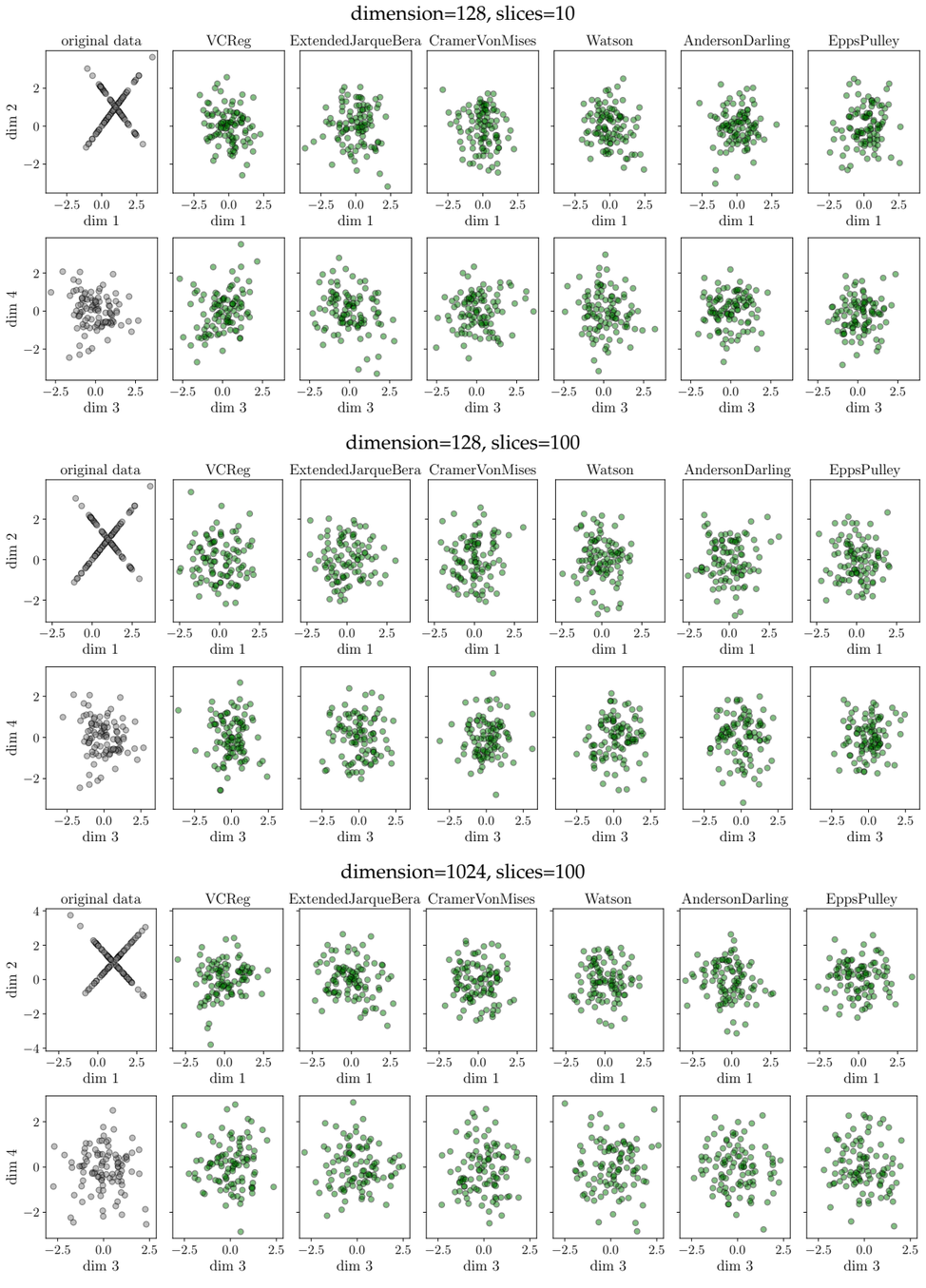


Figure 16. Reprise of Figure 6 for additional dimensions and number of 1d projections.

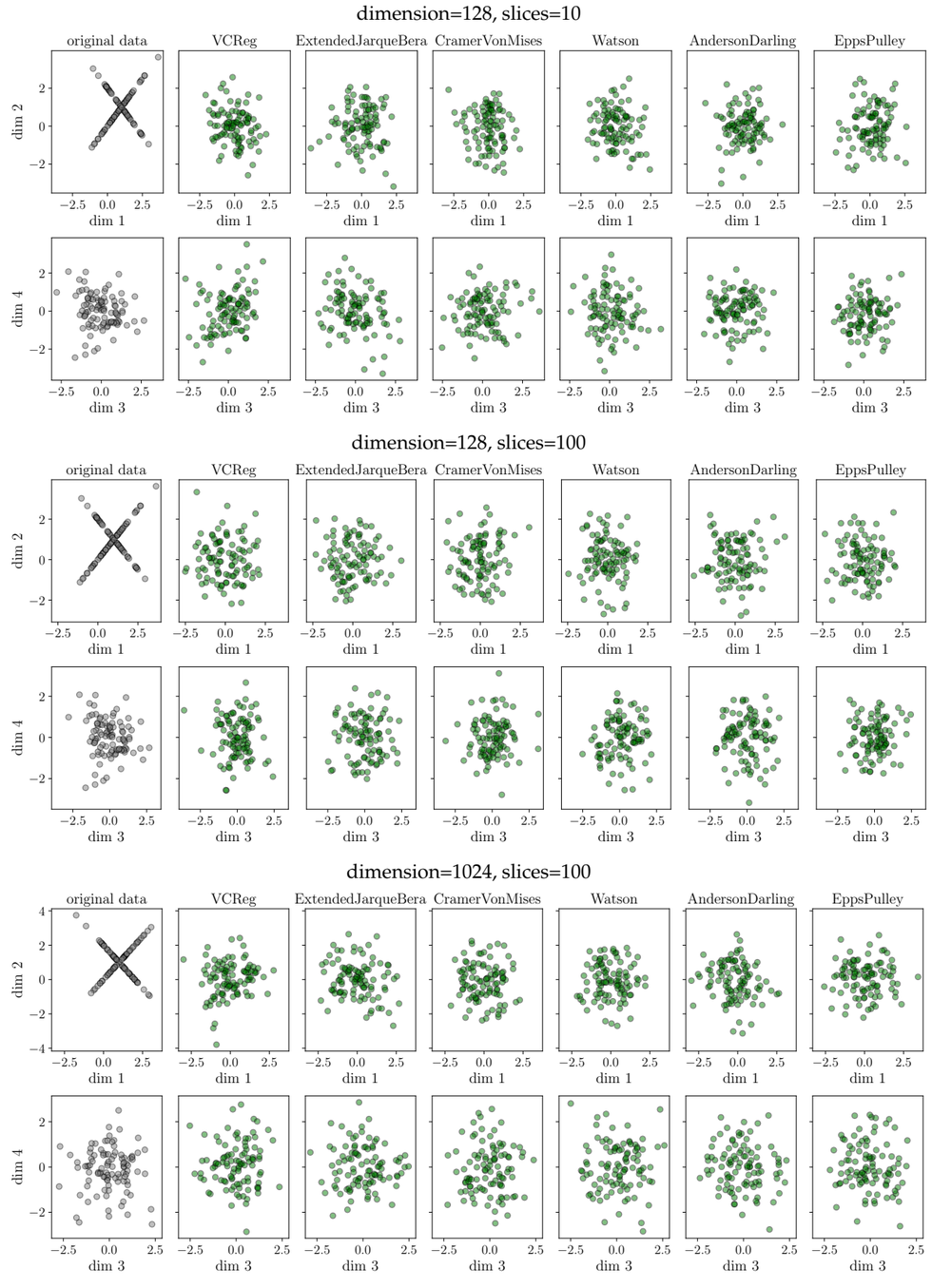


图 16. 在更多维度和更多一维投影数下复现图 6。

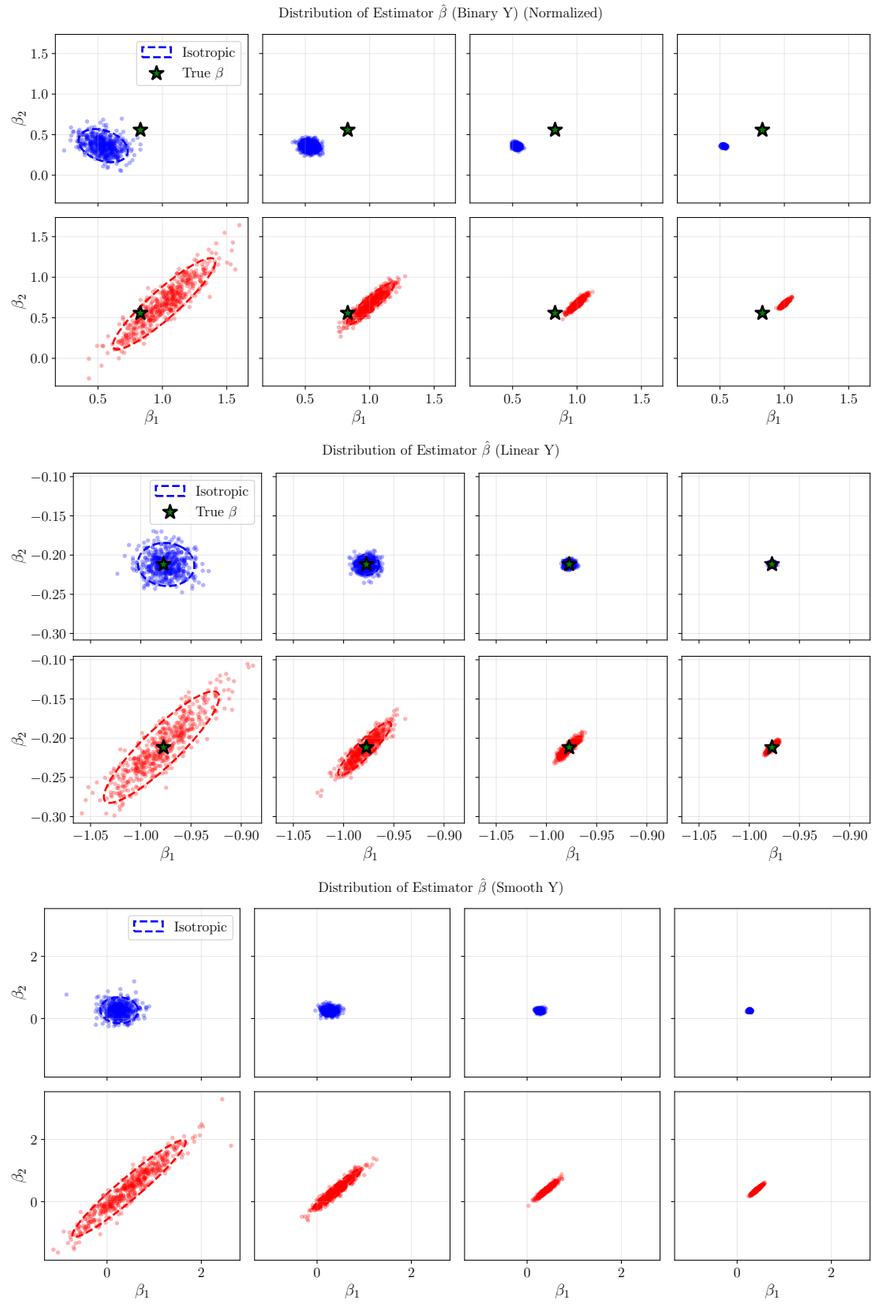


Figure 17. Depiction of the distribution of optimized β values from OLS when comparing $\boldsymbol{Z}_{\text{iso}}$ and $\boldsymbol{Z}_{\text{aniso}}$ from lemmas. 1 and 2. We clearly observe that the anisotropic version (**blue**) provides much lower variance compared to the isotropic case (**red**). We consider a binary classification (linear separable class) (**top row**), a linear regression task (**middle row**), and a nonlinear regression task with smooth targets (**bottom row**). For each case, we resample the training samples numerous times and produce an estimate for β each time. Because the data is 2-dimensional, we can visualize the β distribution directly.

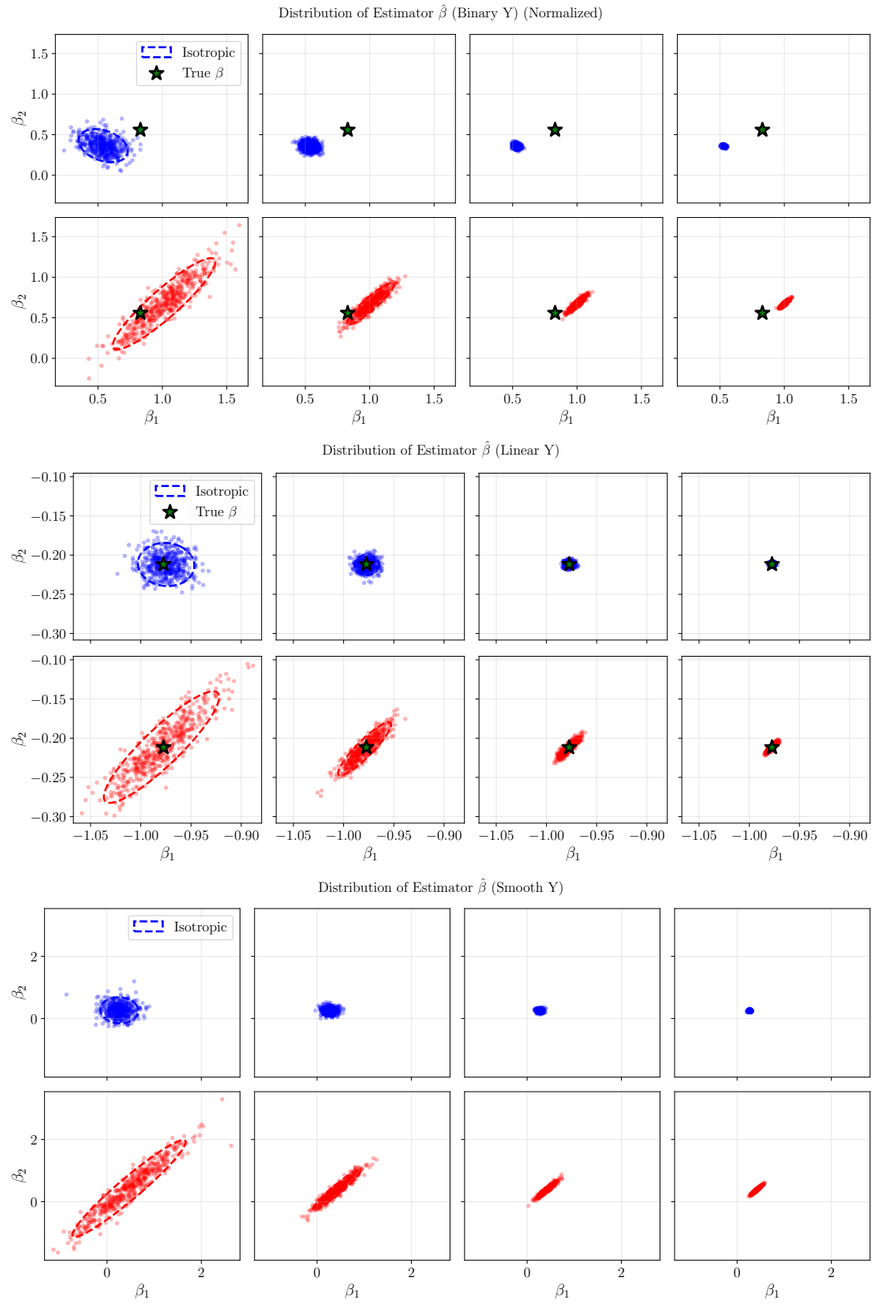


图 17. 比较引理 1、2 的 $\boldsymbol{Z}_{\text{iso}}$ 、 $\boldsymbol{Z}_{\text{aniso}}$ 时，OLS 优化 β 值的分布。各向异性（蓝）比各向同性（红）方差低得多。上：二分类（线性可分）；中：线性回归；下：平滑目标非线性回归。反复重采训练集，每次估计 β；数据二维，可直接可视化 β 分布。

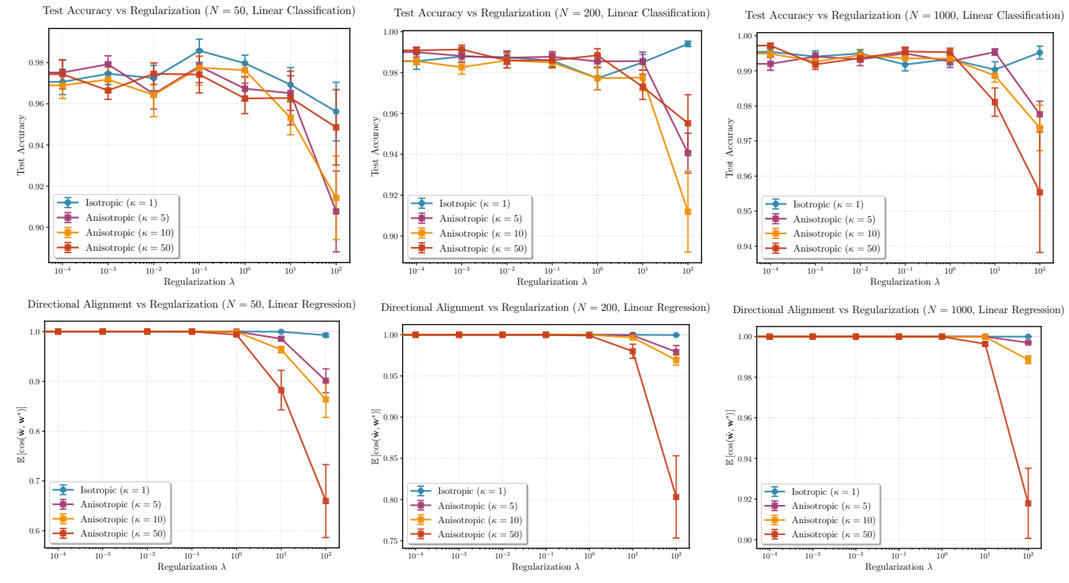


Figure 18. Depiction of accuracy (**top**) and cosine similarity between estimated and true estimator (**bottom**) for the OLS setting with varying strength of Tikhonov regularization (**x-axis**) comparing isotropic and anisotropic embeddings. As per thm. 6, the anisotropic distribution creates a bias in the OLS estimation for nonzero regularization.

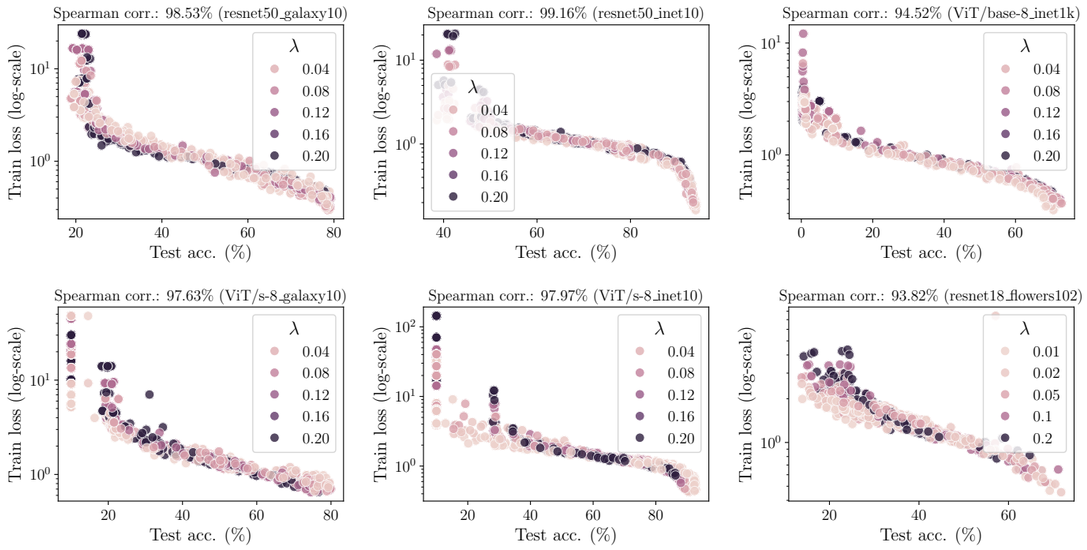


Figure 19. Additional figures provides in Figure 19

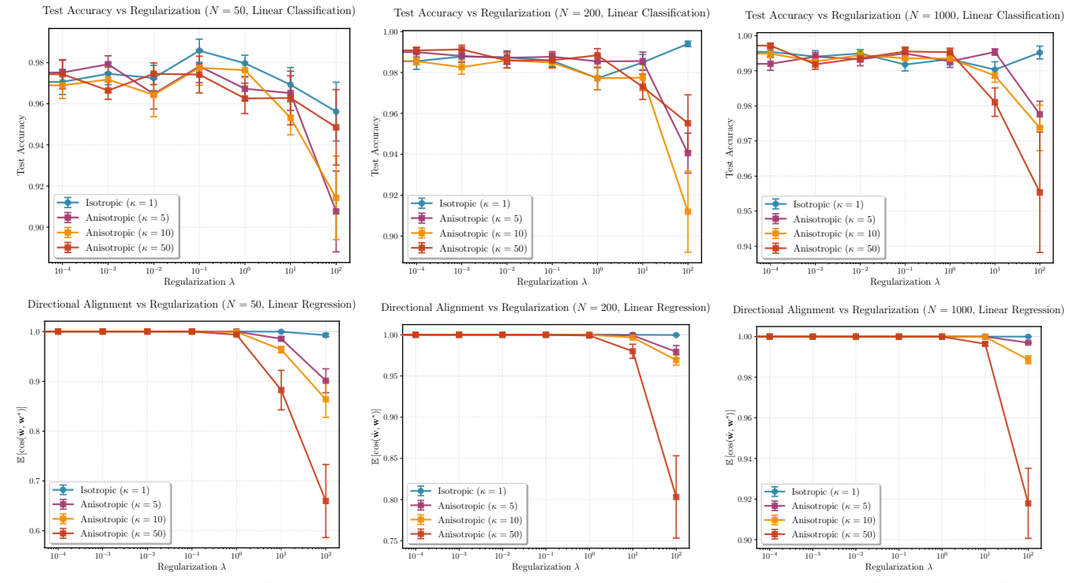


图 18. 在普通最小二乘（OLS）设置下，比较各向同性与各向异性嵌入，绘制准确率（上）以及估计量与真实估计量之间的余弦相似度（下）随 Tikhonov 正则化强度（横轴）的变化。依据定理 6，当正则化非零时，各向异性分布会使 OLS 估计产生偏差。

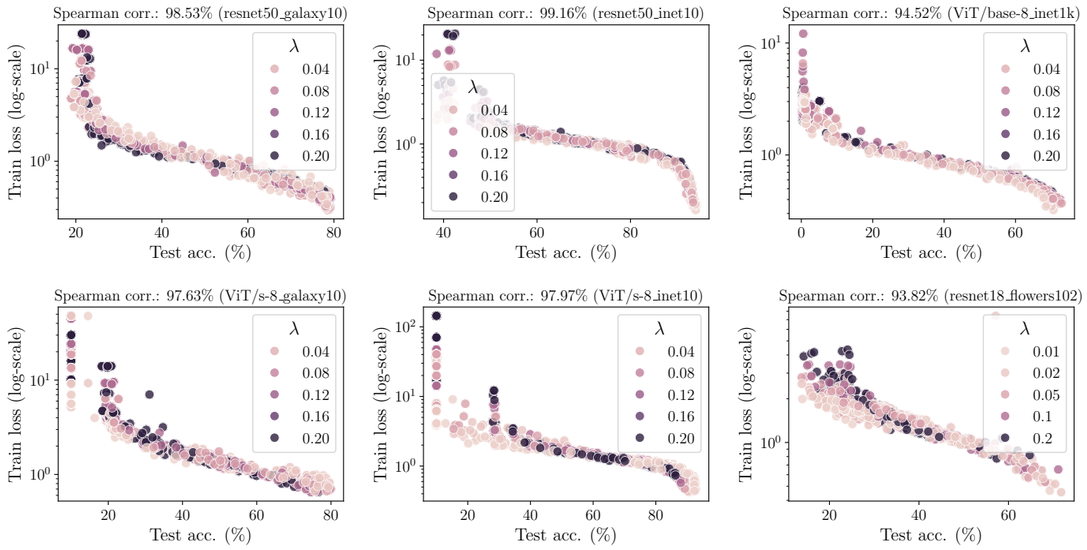


图 19. 图 19 中提供的补充图。

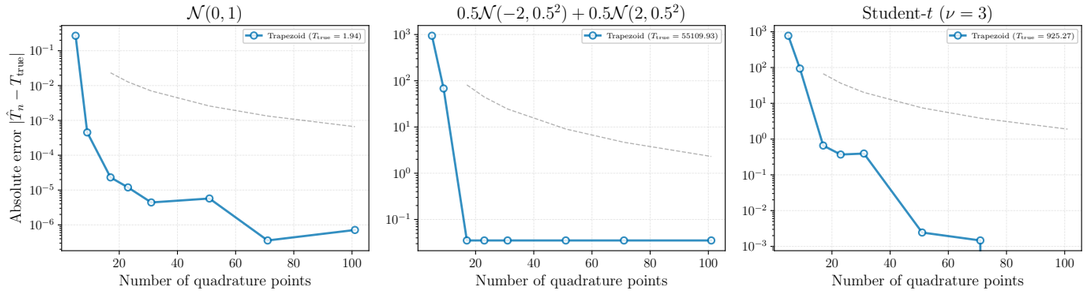


Figure 20. Proposed trapezoid quadrature for the Epps-Pulley statistic as implemented in algorithm 1. We depict the approximation error of the integral for various distributions, demonstrate rapid convergence (faster than quadratic show in **grey line**) across possible embedding distributions.

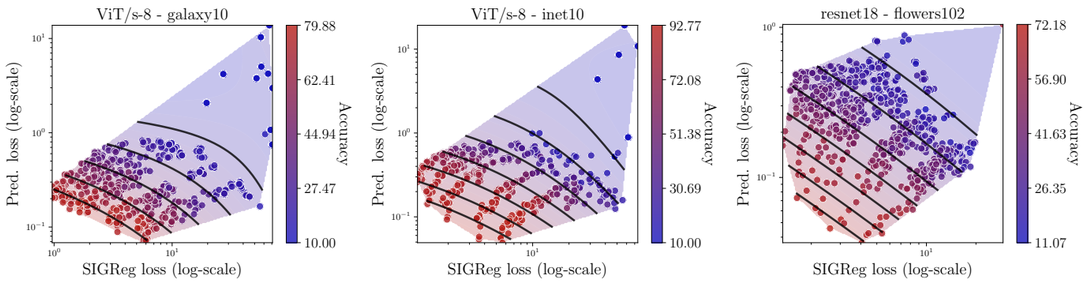


Figure 21. Additional figures for Figure 10.

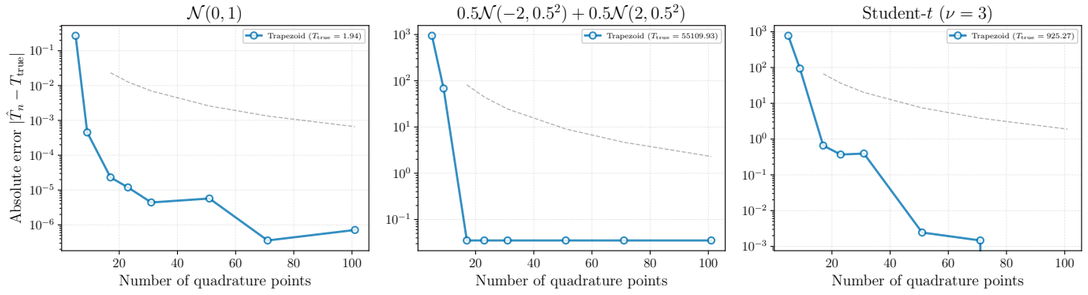


图 20. 为 Epps–Pulley 统计量提出的梯形求积法，具体实现见算法 1。图中绘制了各种分布下的积分近似误差，说明对于不同的可能嵌入分布，该方法均快速收敛（快于灰线表示的二次收敛速率）。

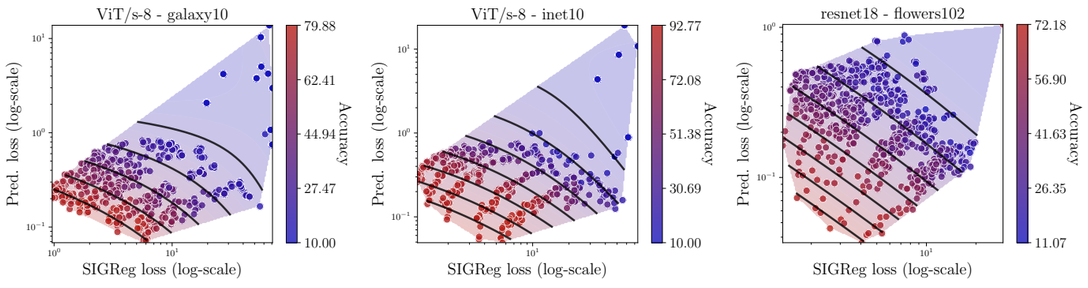


图 21. 图 10 的补充图。